DESIGN RESEARCH

# Where Nature Remains

## Historical space, nature and dreamlike immersion

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AN IMMERSIVE VR EXPERIENCE

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01

# INITIAL IDEA

WHERE NATURE REMAINS

*An immersive exploration of historical spaces, nature and human perception.*

# 1.1 Research Background

## 1.1.1 Personal Motivation and Initial Question

The initial motivation for this project grew out of nearly two years of living in Shenzhen, China, and Milan, Italy.

Shenzhen is a highly modern city with a fast pace of life. High-rise buildings, technology companies and dense traffic form my most immediate impression of the city. People heading to interviews, commuting to work or delivering food often move hurriedly, and urban space seems to revolve around work, efficiency and the economy.

Along my everyday routes, natural landscapes mostly appear as planned and maintained greenery or landscaped areas. In busy districts, these landscapes tend to be carefully composed, which to some extent reduces the unpredictability and wildness of nature. Elsewhere, planting more often separates roads, organises traffic or decorates space. Although Shenzhen is a coastal city, I still find it difficult in daily life to encounter places where I can stay for a long time and attend closely to nature. When I describe Shenzhen to friends, the first word that comes to mind is often still “work”.

Figure 1.1-1. Shenzhen Bay.

Motorcycle journey outside Shenzhen.

While living in this environment, I made a special trip to nearby Huizhou to buy a motorcycle. Because motorcycle access is restricted in urban Shenzhen, I spent many subsequent weekends travelling to surrounding cities in search of mountains, woodland and more open natural environments, riding through these places. For me, riding became both a means of transport and a way to feel wind, temperature, terrain and plants again.

Yet whenever I returned to Shenzhen on a weekday and faced its familiar cityscape, I felt a brief sense of disorientation.

As the city improves efficiency and convenience, does it also change how people perceive their surroundings and, to some extent, increase their distance from nature? I sometimes felt that the closer I came to the city, the further I became removed from it. Surrounded by high-rise buildings, were we also becoming less aware of the histories through which these buildings and landscapes had come into being? This question arose from personal experience and could not support a general conclusion on its own, but it became the starting point for this project.

Milan Cathedral.

Figure 1.1-2. Colonne di San Lorenzo, Milan. Photograph: FlavMi. Wikimedia Commons. CC BY-SA 3.0.

Later, I moved to Milan. Milan is also a modern city, yet its urban spaces felt different from those of Shenzhen. Modern buildings coexist with historical architecture from different periods. Homes, churches, ruins and murals along its streets make the accumulated traces of time perceptible in everyday life.

Milan's high-rise buildings are concentrated in certain central districts, while many other areas retain housing of a relatively modest height. Parks, squares and public facilities are interwoven with residential neighbourhoods, giving me a sense of an urban atmosphere more closely connected to everyday life.

Even so, when I wanted a more direct experience of nature, I still needed to travel to nearby towns, mountain areas or other natural environments.

Through living and travelling in Milan and elsewhere in Italy, I gradually realised that a sense of belonging to a city may arise from the accumulation of personal and historical memory as well as from present-day functions and convenience. Ruins, historical buildings, murals and weathered architectural surfaces allow past human activities to remain present in the contemporary city, providing a medium for emotional connections between people and places.

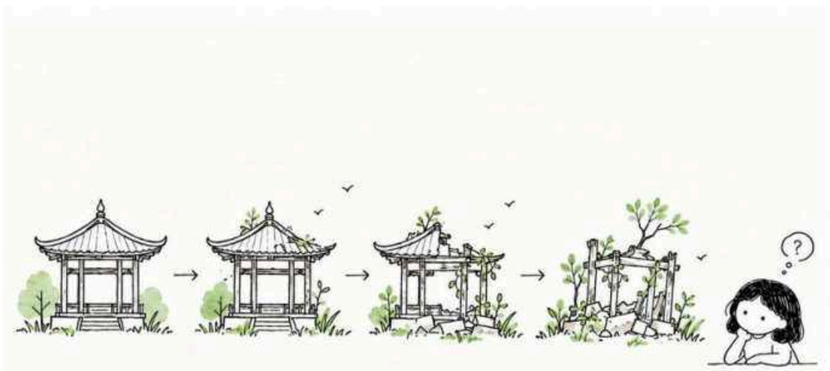




This feeling has something in common with my experience of natural environments. Both nature and historical remains reveal the accumulation of time, material change and conditions that cannot be fully controlled. Plant growth, weathered stone, surface erosion and traces of human activity turn a space into a place that holds time and memory alongside its immediate functions. In my own experience, modern high-rise environments organised primarily around work and efficiency are less likely to evoke the same feeling.

During visits to historical sites across Italy, I also observed that ancient buildings do not exist in isolation. They remain in ongoing relationships with surrounding terrain, plants, climate and changing materials. This led me to ask whether the relationship between nature and historical remains could provide a starting point for immersive design: could it help reimagine the connections between ancient architecture, natural environments and human perception, and allow others to enter and experience these relationships?

This question formed the project's initial design proposition and led to further research into human relationships with nature, historical spaces, immersive experience and methods of visual reconstruction.



# 1.1.2 Research Context

Urban landscape in Shenhe District, Shenyang, Liaoning, China. Photograph: Michael Myers.

Contemporary urban life increasingly takes place within dense built landscapes, transport systems, commercial spaces, digital interfaces and artificial environments. Streets, towers, screens, advertisements and streams of information continually organise everyday life. Cities offer efficiency, convenience and dense social relationships, while also creating perceptual environments that are highly constructed and functionally organised.

Nature may still be present in such environments. Street trees, urban parks, green belts, the sky, water and seasonal changes remain part of city life. However, they are often planned within designated areas or treated as scenery, leisure settings and destinations for brief escapes from everyday routines. Nature gradually becomes something that must be deliberately sought out and explored, with fewer opportunities for continuous contact in daily life.

Research often describes this issue using the terms “disconnection from nature” or “extinction of experience”. The central concern is a reduction in opportunities for direct, sustained and meaningful contact with natural environments. Relationships between people and nature continue to exist, but built landscapes and everyday rhythms can obscure them, making these connections less readily perceived.[1]

1. Soga and Gaston, “Extinction of Experience” (2016). Full notes: p. 195.





This study therefore avoids a simple opposition between cities and nature, and understands nature as an environmental system comprising landforms, plants, climate, time and living processes. Connections with nature can also become perceptible in everyday life through space, materials and memory, as well as in forests, mountains and other natural settings.

The study takes E. O. Wilson’s concept of biophilia as a theoretical starting point. The biophilia hypothesis suggests that humans may have an inherent tendency to attend to life and life-like processes. From this perspective, the relationship between people and nature can be understood as an ongoing perceptual relationship that may become less clearly apparent in contemporary urban environments.[2]

2. Wilson, Biophilia (1984). Full notes: p. 195.

Figure 1.1-3. Temple of Concordia and its surroundings, Agrigento, Sicily. Photograph: Berthold Werner. Wikimedia Commons. CC BY-SA 3.0.

Against this background, historical spaces become an important visual point of entry for the project. Many historical buildings and sites preserve natural materials, landform conditions, plant environments, traces of time and human cultural activity alongside their constructed forms. They make it possible to see how nature participates in human creation and how people build spaces, memories and cultures within natural environments.

Historical spaces thus serve as a visual bridge between natural environments and human cultural experience. Situated between nature and construction, landforms and architecture, time and memory, they provide a concrete and perceptible basis for reconsidering the relationship between people and nature.

On this basis, the project explores whether immersive environments can bring historical forms, natural surroundings, plants, landforms and cultural traces together within a dreamlike space. By temporarily stepping outside the perceptual order of everyday urban life, the experience seeks to redirect attention towards the self, space and essential connections with the natural world that may otherwise go unnoticed.





# 1.2 Research Aim and Significance

This research aims to create an immersive dreamlike experience grounded visually in specific historical spaces and incorporating natural environments and traces of human activity. Through spatial reorganisation, environmental atmosphere and a first-person perspective, the project invites viewers to step temporarily outside familiar rhythms of urban viewing. It opens a slower, more open space with perceptual continuity, bringing easily overlooked connections between people and nature back into awareness.

In this study, “reconnection” is defined as a temporary perceptual condition created through design. It concerns renewed attention to natural environments, one’s own position and relationships with the surrounding world during a specific experience. Its scope is limited to short-term experience and subjective perception.

The research does not examine digital environments as substitutes for real experiences of nature, nor does it establish therapeutic effects, attention restoration or long-term changes in nature connectedness.

At the level of experience design, the study seeks to translate a temporary departure from the perceptual order of everyday urban life into an experiential goal achievable through spatial design. The concepts of “being away” and “fascination” in attention restoration theory inform the experience structure. Being away helps explain a temporary disengagement from everyday environments and habitual patterns of attention. Fascination informs how the environment can sustain attention without forcing it. Spatial rhythm, environmental change, visual focal points and bodily movement are used to create these perceptual conditions and explore how immersive experience can evoke awareness of human relationships with nature.[3]

3. Kaplan and Kaplan, The Experience of Nature (1989). Full notes: p. 195.

At the level of visual design, the study investigates how architectural forms, natural systems and traces of human activity in specific historical environments can form an internally connected visual structure. Historical spaces carry evidence of landform conditions, plant growth, temporal change and cultural activities. Through source extraction, formal analysis, three-dimensional reconstruction and dreamlike transformation, these relationships are translated into a unified spatial language. Dreaming functions here as a method of visual organisation. Forms from different periods and regions establish new relationships through displacement, merging, suspension, growth and dissolution. This method expands expression beyond realistic reconstruction while retaining the regional grounding and identifying features of historical subjects.

At the level of media practice, the study explores the potential of immersive VR equipment to represent spaces combining history, nature and dreaming.

First-person perspective, bodily scale, viewing direction, spatial distance and movement all contribute to the experience, extending visual expression from two-dimensional images into spatial perception. The research focuses on the capacity of immersive media to support spatial layers and nonlinear visual narratives, while documenting the limits imposed on visual exploration by equipment performance, interaction methods, sensory scope and user comfort.

The central aim is to construct an immersive experience, using the visual translation of architecture, nature and human traces as the main design method and VR as the medium of implementation. Through concrete design practice, the study explores ways for viewers to shift their habitual perception temporarily and reconsider their relationship with the natural world. It also provides a practical case through which the immersive expression of combined historical and natural spaces can be analysed and reflected upon.





# 1.3 Research Question

The research selects site evidence, develops dreamlike animation prototypes and prepares a route and test plan for later headset viewing.

*How can a dreamlike animated route help viewers see and understand how natural processes shape historic sites?*

Three subsidiary questions develop this central research question.

### 1.3.1 How can visual elements in historical spaces be selected and analysed?

Specific historical spaces contain multiple layers of information, including architectural structures, plant forms, landform features, constructed remains and changes over time. The research needs explicit criteria for selecting sources, analysing the formal characteristics, spatial relationships and cultural contexts of visual elements, and determining their role in design translation. This question concerns how visual reorganisation can retain the necessary regional grounding and historical recognisability of the subject, establishing a reliable visual foundation for subsequent design.

### 1.3.2 How can dreamlike visual methods organise relationships between architecture, nature and human traces?

Architecture, nature and human traces differ in scale, form and temporal character. A relatively coherent visual logic is needed to bring them into the same immersive space. The research explores how changes in scale, formal merging, spatial displacement, repetition, proliferation and dissolution contribute to visual translation and establish continuity between spaces. This question seeks to clarify the organising principles of dreamlike expression so that it can construct spatial relationships and support experiential narrative.

### 1.3.3 How does VR support and constrain the intended immersive experience?

Through first-person perspective, bodily scale, movement and a sense of spatial enclosure, VR allows viewers to enter combined historical and natural spaces that have undergone visual translation. The study examines how viewing direction, spatial distance, movement rhythm, the sound environment and interaction methods jointly influence the experience. It also analyses the constraints imposed by equipment performance, sensory scope, operational stability and user comfort. This discussion is limited to the technical conditions and experiential context adopted in this project.

Together, these questions establish the research pathway. The first concerns the organisation of historical sources and visual analysis; the second concerns the development of a dreamlike visual language and spatial design; the third concerns VR prototyping and reflection on the experience. Viewer feedback will help assess how the design intention is communicated, identifying effective aspects, differences in understanding and media limitations.

Figure 1.3-1. Left to right, selecting historic details, dreamlike transformation and the field of view in VR.





# 1.4 Research Approach and Thesis Structure

This study adopts a predominantly qualitative, practice-oriented approach, with Research through Design (RtD) as its overall methodology. Source analysis, design experiments, prototyping, experience testing and reflective documentation produce research insights that can be discussed and traced back to practice. Zimmerman, Forlizzi and Evenson argue that the value of RtD depends on the transparency of the process, the relevance of the problem addressed and whether the resulting methods can be understood and used by subsequent research, alongside the final artefact.[4]

### 1.4.1 Literature Review

A literature review examines research on human relationships with nature, changes in experiences of nature under urbanisation, nature connectedness, attention restoration, virtual nature experiences, the representation of historical spaces and dreamlike visual language. It defines key concepts, establishes the theoretical framework and informs visual analysis, dreamlike translation, the construction of the VR experience and subsequent evaluation.

The principal theoretical framework comprises the biophilia hypothesis, nature connectedness and attention restoration theory. Wilson's biophilia hypothesis proposes a potential human tendency to attend to life and living processes. It provides the starting point for understanding human relationships with nature and a conceptual basis for incorporating natural forms, living processes and environmental change into an immersive experience.[5]

4. Zimmerman, Forlizzi and Evenson, "Research through Design" (2007).
5. Wilson, Biophilia (1984). Full notes: p. 195.

Nature connectedness defines the conceptual scope of “reconnection” in this study. It primarily concerns an individual’s understanding of their relationship with the natural world, emotional connection and sense of belonging. The research focuses on short-term shifts in perception and awareness that may occur during the VR experience, including renewed attention to one’s position, natural environments, human activities and their relationships. Later visitor feedback can show which relationships viewers notice and how they explain them.[6]

Kaplan and Kaplan’s attention restoration theory provides a reference for organising the experiential environment, with particular emphasis on “being away” and “fascination”.

Being away describes a temporary disengagement from everyday environments, habitual activities and established patterns of attention. Fascination describes an environment’s ability to sustain attention with less reliance on deliberate control. These concepts guide the organisation of environmental experience in VR and support design reasoning and experience analysis.

The literature review establishes three connected theoretical levels. Biophilia explains a potential basis for human attention to life and natural processes. Nature connectedness defines the scope of discussion about awareness of human relationships with nature. Being away and fascination guide spatial and perceptual design. These levels correspond respectively to the relational foundation, conceptual boundaries and principles of experience construction.[7]

6. Schultz (2002); Mayer and Frantz (2004), nature connectedness.
7. Kaplan and Kaplan, The Experience of Nature (1989). Full notes: p. 195.





### 1.4.2 Site Research and Visual Source Analysis

For places I visited, I drew on my own observations and records. For sites I did not visit, I worked from photographs taken by others, published drawings and historical images, and archaeological or architectural research. I compared structure, materials, setting and signs of use. I chose details for their connection to the place, how readily they could be recognised and how well they could show change in the animation. Each case identifies its sources and distinguishes what they support from my design interpretation.

### 1.4.3 Dreamlike Visual Translation and Design Experiments

Visual translation proceeds from source extraction and formal analysis to deformation, combination and spatial development. Architectural elements, natural environments and traces of human activity undergo changes of scale, formal merging and spatial displacement to generate varied and distinctive visual proposals.

Experiments develop progressively through sketches, two-dimensional composites, three-dimensional models, animated sequences and partial spatial prototypes. Proposals are compared in terms of their historical recognisability, the intelligibility of relationships between architecture and nature, and the relative continuity of the viewing experience during spatial change. Visual sources, translation rules, revisions and reasons for selection are documented throughout, grounding the final discussion in concrete design evidence.

### 1.4.4 VR Prototyping and Iterative Testing

The planned headset prototype will give viewers room to look around and pause within a short route. Observing which details they notice, where they lose their bearings and how movement feels can guide adjustments to scale, navigation and pacing.

The planned tests will examine visual expression, operational stability and user experience. They will cover spatial layers, focal points and atmosphere, as well as frame rate, loading and interaction response. The user experience assessment will consider whether the controls are clear and whether movement causes discomfort or motion sickness.

Production revisions and their rationale are documented. The planned trial will help identify whether viewing difficulties arise from visual design, spatial organisation or technical implementation.

### 1.4.5 Experience Feedback and Qualitative Analysis

Formal visitor testing has yet to take place. Once the headset prototype is ready, the planned study will use observation, post-experience questionnaires and interviews. Instructions, procedures, duration and equipment will be kept as consistent as possible. Observations will cover movement, pauses, viewing direction, difficulty with controls and signs of discomfort. Questionnaires will ask about visual appeal, spatial coherence, understanding and comfort.





Interviews will ask what participants noticed, how they understood the links among architecture, nature and human traces, and whether the dreamlike changes felt coherent or prompted thoughts about their own relationship with nature. The evaluation will compare these accounts with the design intentions through open-ended questions. Interview responses will be analysed using Braun and Clarke's thematic analysis, from familiarisation and coding through theme development, review, naming and writing. The analysis will draw on the research questions and allow themes to emerge from the interviews. It will include negative experiences, contradictory accounts and alternative explanations.[8]

### 1.4.6 Research Trustworthiness, Ethics and Limits of the Conclusions

Historical sources, site records and production versions document how the design decisions were made. Planned visitor feedback will help assess which relationships can be understood from the experience. As the researcher also acts as designer and analyst, personal aesthetic judgement cannot be fully excluded.

Expectations, reasons for revision and evaluation procedures therefore need to be documented, and the discussion of results must reflect on potential selection bias arising from the researcher's position.

Before the planned trial, participants will receive information about the research purpose, procedure, intended use of data and potential discomfort. Participation will be voluntary, and personal information will be anonymised. Participants will be able to pause or withdraw if they experience motion sickness, visual fatigue or other discomfort.

The current conclusions draw on the project's historical sources and design practice. Any future trial findings will be limited to the equipment, participants and short-term conditions of that trial. They could show which visual and spatial methods viewers understood and where they encountered difficulties. Long-term nature connectedness, attention restoration, therapeutic effects and general causal claims would require further research.

Interviews could document whether participants report a sense of nature. Assessing a change in nature connectedness would require pre- and post-tests, validated scales, control conditions and a more rigorous statistical design.

8. Braun and Clarke, "Using Thematic Analysis in Psychology" (2006). Full notes: p. 195.

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# RESEARCH

## 2.1 THEORETICAL RESEARCH

### 2.1.1 Basic Concepts of Human–Nature Disconnection

As cities continue to expand, everyday life increasingly takes place within environments composed of buildings, roads, transport systems, commercial spaces and digital media. Nature remains present in urban life through street trees, parks, bodies of water, the sky and seasonal change. People commonly encounter nature through these relatively identifiable forms.

Human–nature connection has several dimensions. Ives et al. distinguish material, experiential, cognitive, emotional and philosophical connections. Material connection concerns human dependence on nature for food, energy and resources; experiential connection concerns direct contact with nature; cognitive connection concerns knowledge of ecosystems and natural processes; emotional connection includes attachment, care and emotional responses; and philosophical connection concerns how people understand their position within natural systems.[1]

Even when nature is present in everyday environments, attention and understanding can vary considerably. Someone may pass an urban park every day while rarely noticing changes in plants, weather or seasons. Another person may develop a connection to an unvisited natural environment through knowledge, memory or cultural identification. Research therefore needs to distinguish opportunities for contact, levels of attention, ways of understanding and emotional responses. Beery et al.'s wheel of disconnection further shows that personal, social and institutional factors shape disconnection. Interests, cultural ideas, social norms, urban planning and resource allocation all influence how people encounter and understand nature. [2]

In this thesis, disconnection refers to a weakening of relationships between people and their surroundings, or a condition in which these relationships are difficult to notice and understand.

It may involve fewer opportunities for direct contact, limited attention to natural change, or difficulty understanding the connections between nature and human life. In this thesis, relational disconnection describes what happens when historical sites are presented in ways that separate, simplify or overlook the connections among architecture, natural processes, human activities and historical time. The concept draws on research on nature disconnection and heritage. In this thesis, it serves as a tool for site analysis and design.





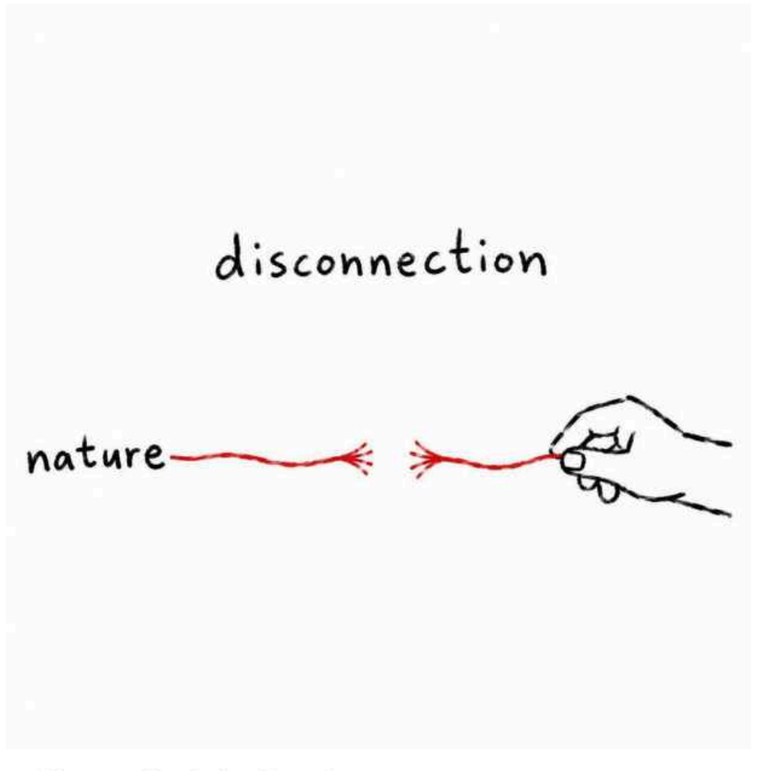


Figure 2.1-1. Relational disconnection. Project illustration.

### 2.1.2 Changes in Everyday Experiences of Nature

Soga and Gaston use the term extinction of experience to describe the gradual reduction of direct interactions with nature. Such changes may affect knowledge of nature, emotional attitudes, experiences of health and environmental behaviour. Reduced contact may also create a cycle: fewer opportunities can lower familiarity and interest, while lower interest may reduce the willingness to seek further contact.[3]

Interactions with nature are influenced chiefly by opportunity and orientation. Opportunity includes access to natural spaces, available time, distance and economic circumstances. Orientation includes personal interests, existing knowledge, emotional connections and willingness to approach nature. Together, they shape the frequency and forms of contact.[4] Urban experiences consequently vary. Parks, water, green belts and street trees offer everyday opportunities, while age, occupation, lifestyle, cultural background and personal interests influence whether people notice these elements.

Gaston and Soga further distinguish interaction from experience. Interaction refers to actual contact, such as seeing trees, hearing birds, feeling wind or walking beside water. Experience also includes the attention, understanding, emotions and memories that emerge during contact. The same environment can produce different responses in different people, so contact frequency and the experience formed through contact require separate discussion.[5] Current research cannot establish that all regions have experienced an equal decline. Cazalis et al. show that findings differ across countries, regions and indicators, while long-term data and cross-cultural comparisons remain limited.[6]

The analysis focuses on how opportunities to encounter nature and forms of contact are changing. Urban spatial structures, lifestyles and interests jointly shape these relationships, with urbanisation forming an important condition. Nature and history are commonly organised within designated spaces in modern cities. People visit parks, botanical gardens and nature reserves to encounter nature, and museums, archaeological sites and historical buildings to learn about history. These settings support protection, management and visitation, while also shaping ways of seeing.

Architectural appearance, construction dates, important events and cultural value are often immediately noticeable. Material ageing, climatic effects, everyday activities and histories of maintenance and repair usually require further explanation.

What makes it difficult to see how architecture, natural processes and human activity have shaped a site over time?

### 2.1.3 Nature–Culture Relationships at Historical Sites

Historical buildings are commonly classified as cultural heritage, while mountains, rivers, plants, climate and geological settings are more often classified as natural environments or natural heritage. These classifications help establish conservation systems and management rules, while also influencing public understanding.

Terrain influences location and layout, while material sources influence structural systems and construction methods. Wind, water, temperature, gravity, earthquakes, volcanoes and plant growth continually alter buildings.





Construction, habitation, religious practices, repair, maintenance, abandonment and tourism management also leave traces at different stages. The sites visible today have been formed through natural forces, human activities and long-term material change.

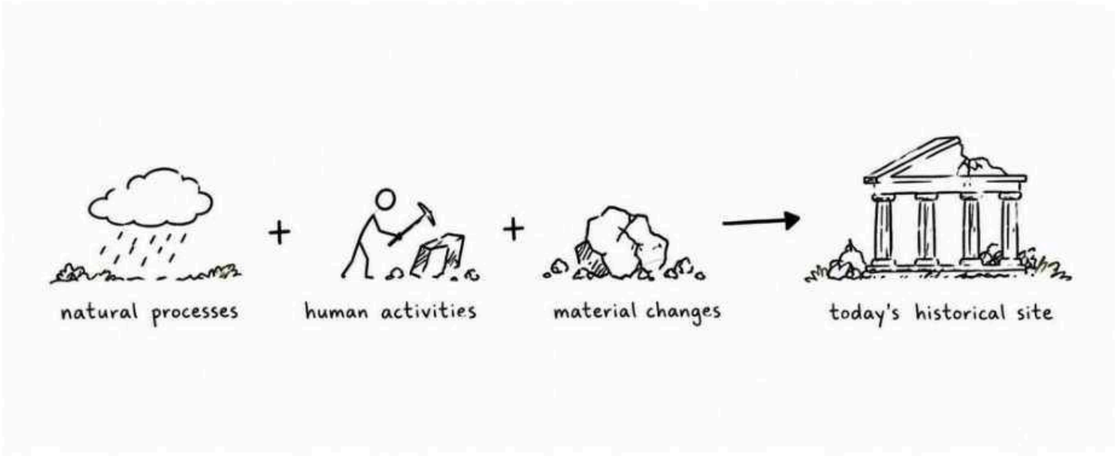


Figure 2.1-2. Natural processes, human activity and material change at a historic site. Project illustration.

Ingold's temporality of the landscape provides a theoretical basis for understanding this process. Landscapes continually take shape through dwelling, labour and environmental change. His concept of taskscape emphasises relationships among activity, time and space. Roads, buildings, land and traces record past activities and ongoing change.[7] This perspective also applies to historical architecture. Cracks, inclinations, wear, losses and repairs record natural processes and human actions.

Plants, dust, sediments and material ageing likewise belong to a site's history. Historical buildings continue to change with their environment and uses.

Harrison discusses the relationship between natural and cultural heritage. Real places contain people, materials, environments, organisms and technologies simultaneously. Heritage can be understood as an assemblage of these elements.[8]

Smith's Authorized Heritage Discourse explains how professional institutions, conservation systems and dominant historical narratives jointly establish heritage value. Such narratives place particular emphasis on age, aesthetic value, commemorative significance and material integrity, and on the role of expert knowledge in interpretation.[9]

This tendency also influences visual communication. Visual presentations often foreground façades, building forms, artistic styles and completed reconstructions. Weathering, disasters, everyday uses and long-term maintenance require explanation through archaeological evidence, architectural research and site observation.

The four cases examine whether a building's form is easier to see than the processes that shaped it.





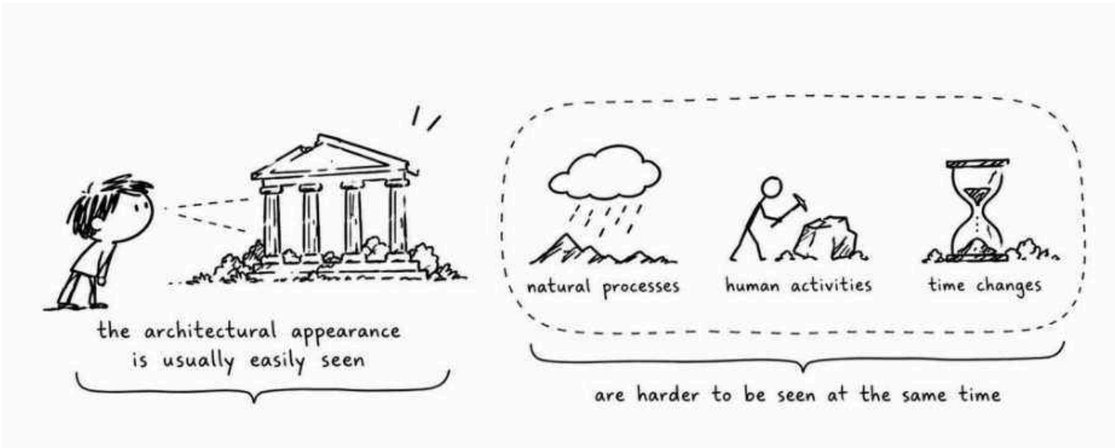


Figure 2.1-3. Visible architecture and less visible processes. Project illustration.

The four cases present different relationships:

Pompeii's condition relates to volcanic eruption, burial, excavation and ancient everyday life.

The material condition of ancient temples in Sicily relates to terrain, sunlight, wind and long-term weathering.

The continuity of Yingxian Wooden Pagoda relates to its timber system, vibration, religious activities and successive periods of maintenance.

The spatial form of the Hanging Temple relates to the mountain, gravity, timber supports and routes of human movement.

These cases provide a basis for comparison. Chapter 3 uses reliable site documentation to establish whether each relationship is supported by historical, architectural or material evidence.

## Relationships Shaping the Four Historical Sites

### Pompeii

FOCUS

Ruin condition

RELATIONSHIPS

Volcanic eruption; burial and preservation; archaeological excavation; ancient daily life.

### Ancient Temples of Sicily

FOCUS

Material condition

RELATIONSHIPS

Terrain; sunlight; wind; long-term weathering.

### Yingxian Wooden Pagoda

FOCUS

Architectural continuity

RELATIONSHIPS

Timber structural system; seismic movement; religious activities; maintenance across dynasties.

### Hanging Temple

FOCUS

Spatial form

RELATIONSHIPS

Mountain setting; gravity; timber support; human movement paths.

Figure 2.1-4. Relationships shaping the four historic sites. Project illustration.

### 2.1.4 Sensory Disconnection

Sensory disconnection describes the weakening of certain spatial information when historical sites are communicated through photographs, texts and plans.

Photographs and texts effectively present appearance, dates and background knowledge. They make it harder to directly experience architectural scale, bodily orientation, changing distance and spatial enclosure. The height of the pagoda, the narrow paths of the Hanging Temple, the rhythm of temple colonnades and the scale of Pompeii's streets and interiors all relate to movement and viewing position.

Wind, vibration, footsteps and environmental sounds can also reveal natural and spatial relationships. The VR experience uses sight, sound, movement and changes in viewing direction.





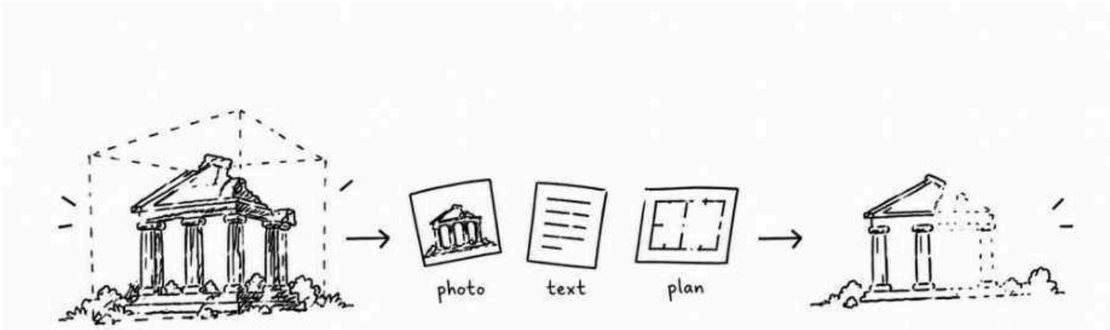


Figure 2.1-5. Static records and bodily experience of space. Project illustration.

### 2.1.5 From Relational Disconnection to Mediated Perceptual Reconnection

The four dimensions above converge on the study's design question:

How can VR help viewers notice connections among historical architecture, natural processes, human activities and temporal change?

VR can organise architecture, environments, sound, movement and temporal change within a three-dimensional space. Viewers can alter their position through head movements, bodily movement or virtual navigation. Cultural heritage applications already include digital reconstruction, virtual visits, education, preservation and public display. A review by Chong et al. examined 290 studies published between 2007 and 2020.

The review identifies spatial exploration, multimedia content, immersion and presence as important directions.[10]

VR expression can respond to four dimensions of relational disconnection:

Dimension of disconnection / VR translation

Spatial disconnection / Place buildings, terrain and environments in a shared space, and connect sites through transitions.

Temporal disconnection / Animate burial, weathering, vibration, erosion and maintenance.

Relational disconnection / Make connections among natural forces, architectural structures and traces of human activity visible within scenes.

Sensory disconnection / Enrich viewing through scale, direction, movement, sound and spatial enclosure.

## Dimensions of Relational Disconnection

### Spatial disconnection

CORE ISSUE

Weak connections between architecture and its surrounding environment.

MANIFESTATION

Architecture is viewed as an isolated object or tourist destination.

EVALUATION

Whether viewers can identify connections between architecture, topography, climate and the surrounding environment.

### Temporal disconnection

CORE ISSUE

Limited connection between the present condition and the processes that formed it.

MANIFESTATION

Viewers see the surviving result but gain limited understanding of destruction, change and maintenance.

EVALUATION

Whether viewers can describe the major transformations experienced by the site.

### Relational disconnection

CORE ISSUE

Architecture, natural forces and human activities are explained separately.

MANIFESTATION

The environment is treated as background, while human activity is supplementary information.

EVALUATION

Whether viewers can explain how different factors interact.

### Sensory disconnection

CORE ISSUE

The site is reduced to static images and textual descriptions.

MANIFESTATION

A limited sense of scale, orientation, movement, sound and spatial presence.

EVALUATION

Whether viewers can perceive scale, orientation, movement and environmental atmosphere.

The study refers to this process as mediated perceptual reconnection.

VR places the building and the natural processes, human activities and changes that shaped it within the same scene.





Here, reconnection primarily concerns visual communication and understanding. It includes:

Noticing site relationships that are easily overlooked.

Recognising interactions among different factors.

Understanding how architectural forms develop and change.

Making further associations among nature, architecture and history.

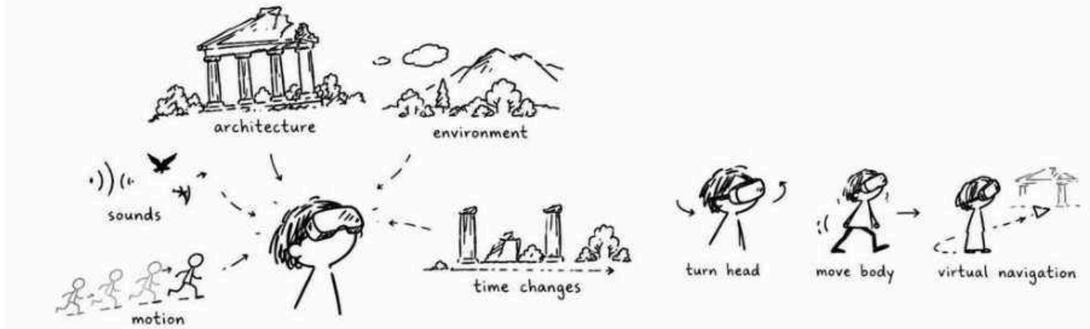


Figure 2.1-6. Attention, recognition, understanding and association. Project illustration.

Evaluation focuses on these immediate outcomes (see Section 1.4.6). Assessing lasting changes in emotional attachment to nature, environmental behaviour or nature connectedness would require longitudinal evidence. Connectedness scales and controlled experiments could help assess those changes.

VR experiences are also shaped by equipment and design. Image quality, interaction methods, navigation speed, prior device experience and existing knowledge may all affect evaluation. Visual novelty may increase interest without demonstrating understanding of the content.

Existing cultural heritage VR research indicates that participant backgrounds, sample sizes, interface design and evaluation methods influence results.[11] Cybersickness, disorientation and visual fatigue may also reduce experience quality.[12]

Audience evaluation in Chapter 5 will therefore examine:

Whether viewers can identify connections between architecture and natural processes.

Whether viewers can describe temporal changes shown in the scenes.

Whether viewers can recognise traces of human activities and maintenance.

Whether viewers can understand how different factors jointly affect architecture.

Whether feedback focuses primarily on visual appeal or also demonstrates understanding of core relationships.

Whether the experience prompts further reflection on relationships among nature, architecture and history.

The conclusions will consider what viewers notice and understand, and how the framework guides site analysis and visual design.

## 2.2 MARKET AND CASE STUDY RESEARCH

Immersive cultural heritage projects take many forms. Some overlay digital information at a site, some reconstruct vanished buildings, and others combine online exploration with physical visits. Bekele et al.'s review identifies cultural presentation, historical reconstruction and learning as major uses of augmented, virtual and mixed reality (Bekele et al., 2018). This section evaluates the explanation of historical evidence and the organisation of visits; technical sophistication is not an assessment criterion.

Five publicly exhibited projects are selected. Each concerns a real or vanished historical place and has verifiable official documentation. Comparing different media and visitor formats makes their approaches and difficulties clearer.

### 2.2.1 Comparative Immersive Cultural Experiences

#### 2.2.1.1 Ancient Olympia: Common Grounds: Cross-Platform Presentation

Ancient Olympia: Common Grounds was developed by the Hellenic Ministry of Culture and Sport with Microsoft.





Ground photography and drone imagery documented the site, providing the basis for reconstructing 27 buildings, including the temples of Zeus and Hera and the ancient stadium. Content was released through a website, a mobile application and a HoloLens 2 exhibition at the Athens Olympic Museum (Microsoft, 2021).

Figure 2.2-1. HoloLens presentation of Ancient Olympia: Common Grounds. Image: CFJ Productions / Claire Jackson.

The project selects media for different visiting situations. Websites and phones introduce the site to people unable to visit, while museum mixed reality presents vanished architectural elements and clarifies their original scale. All these formats draw on site documentation.

The project will likewise first establish where viewers encounter the work and what they need to understand, then select the medium. Headsets will handle content suited to immersive viewing.

As reconstruction becomes more convincing, projects need to distinguish evidence from conjecture.

Otherwise, viewers may interpret a complete image as certain history. The London Charter requires documentation of sources and disputed elements (Denard, 2009). Microsoft reports archaeological expert participation, but its public pages do not specify the evidential strength of each reconstructed part.

### 2.2.1.2 L'Ara com'era: Returning to the Original Object

L'Ara com'era is an onsite mixed reality experience at Rome's Ara Pacis Museum. Visitors wear headsets along a route with nine viewing points. Cameras track the real altar, overlaying conjectured ancient colours on its reliefs and sculptures. Visitors then see the ancient Campus Martius and ritual scenes before returning to observe the original object. The complete route lasts approximately 45 minutes, offers five languages and operates in small groups (Museo dell'Ara Pacis, 2016–2019).

Figure 2.2-2. L'Ara com'era combines virtual colour with the real altar. Image: Museo dell'Ara Pacis / Roma Capitale.

The experience centres on comparing virtual colour with the real altar. Visitors first observe today's white marble and then view its conjectured ancient appearance through the headset, maintaining a relationship with specific reliefs. The organisers report approximately 11,000 participants in the first three months and a later increase from 300 to 400 visitors per evening. Timed group admission also requires equipment and staffing arrangements.

Virtual imagery should help viewers inspect the altar's materials. After the added historical context, attention should return to the original object.

Headsets also introduce access constraints: children under 13 cannot participate, and group admission increases management requirements. Flights and ritual scenes add drama, but without clear reconstruction evidence visitors may remember only the sensation of travelling through time.

### 2.2.1.3 The Horizon of Khufu: Free-Roaming VR and the Risk of Spectacle

The Horizon of Khufu is a free-roaming VR experience produced by Excurio. In a venue covering several hundred square metres, visitors walk through the experience and see their companions in the virtual environment. The approximately 42-minute route is led by the virtual guide Mona.

It enters the Great Pyramid, moves onto the Giza plateau and presents ritual scenes from around 2500 BCE. The organisers report research collaboration with Egyptologists from the Harvard Giza Project (Excurio, 2022).

Figure 2.2-3. Giza scenes and shared free-roaming experience in The Horizon of Khufu. Image: Excurio / Fever Newsroom.

Free roaming allows visitors to move within the virtual pyramid and see companions, making the visit a shared experience. Reinstallation in different cities supports touring. However, the experience is detached from Giza: visitors cannot encounter the actual site, and the imagery loses direct contact with onsite materials.

Repeated dramatic scenes create multiple visual climaxes within a short period. This rhythm risks turning the pyramid into an adventure backdrop and relegating specific historical information to a secondary position.





### 2.2.1.4 Notre-Dame de Paris: The Augmented Exhibition: Exploring Historical Change by Tablet

Notre-Dame de Paris: The Augmented Exhibition provides HistoPad tablets in a physical exhibition. At each display point, visitors scan content to explore construction beginning in the twelfth century, the 2019 fire and subsequent restoration. Its account spans 850 years, extending history beyond a single reconstructed moment. Histovery developed the project with the public body responsible for Notre-Dame's conservation and restoration, and the exhibition subsequently toured several countries (Histovery, 2019).

Figure 2.2-4. Visitors use HistoPad to explore Notre-Dame's history and restoration. Image: Histovery / L'Oréal Groupe.

Tablets offer a lighter entry point to heritage experiences. Visitors can still see the gallery and communicate with companions. Touring also reaches people unable to travel to Paris. This format is more accessible to visitors for whom headsets are unsuitable.

Tablets also compete for attention. Constantly searching for activation points may keep visitors looking at screens and overlooking exhibits around them. This project can adopt the chronological approach while reducing fixed triggers and allowing viewers to decide when to continue.

### 2.2.1.5 The Lost Palace: Screen-Free Urban Storytelling

The Lost Palace was jointly developed by Historic Royal Palaces, Uninvited Guests, Chomko & Rosier and Calvium. It tells the story of London's Whitehall Palace, destroyed by fire more than 300 years ago. Starting at the surviving Banqueting House, visitors carry wooden controllers and headphones on an approximately one-mile walk through Whitehall. Location tracking triggers sound and vibration at specific points, with installations or live performances at some stops.





Vanished events thus return to the streets where they occurred (Chomko & Rosier, 2016; Calvium, 2018).

Figure 2.2-5. The Lost Palace's screen-free wooden interaction device and outdoor route. Image: Calvium.

The project turns existing streets into the visiting route. Stories and interactions activate only at relevant locations, bringing the contemporary city and its past together. The team also conducted months of onsite testing and used each trial to decide which interactions to retain. For a practice-based thesis, these decision records offer greater value than a presentation of finished work alone.

Calvium's published survey reports that 93% of respondents found the experience distinctive, 92% felt history had come alive and 90% felt more connected to the past.

Responses were broadly positive, but the published account does not specify sample size or questionnaire design. Reliance on Whitehall's streets also makes operation sensitive to weather, traffic and onsite management, and the complete route is difficult to transfer elsewhere.

### Comparative Summary

| Project | Experience format | Lessons and risks |
|---|---|---|
| Ancient Olympia | Cross-platform reconstruction; 27 buildings | Expand access; identify conjecture |
| L'Ara com'era | Onsite mixed reality; approximately 45 minutes | Return to the original object; access barriers |
| Horizon of Khufu | Shared free-roaming VR; approximately 42 minutes | Experience scale; avoid spectacle |
| Notre-Dame | Tablet augmented exhibition; 850 years of history | Present historical change; reduce screen distraction |
| The Lost Palace | Screen-free urban walk; approximately one mile | Connect to the original site; difficult to replicate |

Across these cases, experience depends on what viewers are guided to observe. L'Ara com'era returns virtual content to the original object; The Lost Palace keeps stories at their original locations; The Horizon of Khufu demonstrates risks associated with continuous spectacle. For this project, the question is where the technology directs attention and what it lets viewers notice about the remains.

### 2.2.2 Audience and Experience Positioning

Participation differs with location. L'Ara com'era serves museum visitors, so digital content needs to support renewed observation of the original object. Ancient Olympia and Notre-Dame reach people unable to visit onsite. The Horizon of Khufu can be installed in different cities but loses contact with Giza. The Lost Palace requires an actual neighbourhood, limiting participation and increasing sensitivity to local conditions.





Wider distribution does not guarantee ease of use. VR can present inaccessible sites, but headsets may cause dizziness and interactions may add effort. Chong et al.'s review identifies several usability barriers in heritage VR (Chong et al., 2021). The choice of format depends first on whether visitors can use the equipment and complete the experience comfortably.

Duration alone does not describe pacing. A 42-minute VR experience may continually push visitors forward, while an 80-minute walk may constrain them to a fixed route. What matters is whether visitors can pause, look back and decide when to continue. This project is better suited to viewers willing to attend closely, rather than aiming at maximum throughput or continuous stimulation. Duration and interaction methods will be established in Chapter 3 and assessed through prototype testing.

### 2.2.3 Project Differentiation

Where Nature Remains explores four historic sites. Pompeii looks back through time after volcanic burial. Concordia traces stone weathering in its landscape. At Yingxian, a larch gives way to the timber pagoda. At the Hanging Temple, beams set into the cliff support a platform as the halls emerge above it.

Where Nature Remains will distinguish historical evidence from design content. Production records will identify documented elements and research-based extrapolations, while dreamlike transformations will be explicitly treated as visual design. Realistic models preserve basic site characteristics; surreal changes express time and natural processes. This makes the work's sources understandable and responds to the transparency requirements of the London Charter and Seville Principles.

The case study yields four requirements. Modelling must document sources and identify uncertainties. Virtual content should guide observation of heritage materials. Viewers should control their pace, with pauses between scenes. Completed prototypes require testing on actual equipment. Each requirement has a case basis: Ancient Olympia demonstrates the need to explain reconstruction evidence; L'Ara com'era shows why digital content should return attention to the original object; The Lost Palace demonstrates onsite testing. Difficulties in The Horizon of Khufu and Notre-Dame also support controlling visual stimulation and screen distraction.

Section 2.3 turns these findings into design criteria.

Chapter 4 explains how the research informed the four scenes. Chapter 5 discusses how visitors may encounter the work and how their experience could be evaluated.

## 2.3 Research Synthesis and Design Criteria

The research suggests that the relationship between a building and its surroundings can be seen through change. Pompeii and Concordia show the effects of time and environment on vegetation, fragments and stone. Yingxian and the Hanging Temple approach the relationship through construction, from tree to timberwork and from rock sockets to supporting beams.

The case studies also shaped how viewers move through the scenes. Site evidence identifies the features of the buildings and terrain to retain. The sphere connects views at different scales, and pauses between wide and close views give viewers time to notice changes. Storyboards and process images show how those features informed the movements of the dream sequences.

These choices also guide evaluation. In a screen viewing, visitors first describe the changes they saw, then explain how they understood the relationship between the building and its surroundings. Headset trials can explore whether viewers stay oriented, follow the changes and feel comfortable when free to turn and pause.

# 03

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WHERE NATURE REMAINS / FINAL CONCEPT

# FINAL CONCEPT AND PROJECT IDEA

## 3.1 Project Overview and Experience

Figure 3.1-1. The two visual prototypes. Left, fragments reassemble at Pompeii. Right, the Temple of Concordia and its ridge. Animation stills.

Where Nature Remains is a heritage experience built around a journey back through time. It asks how nature and time shape a building, starting from the distance people can feel from the buildings they pass every day. A small sphere leads viewers from close details of stone, plants and ground to wider views of the site.

Pompeii begins on ground buried by the eruption. Vegetation returns and fragments reassemble as time runs backwards. At the Temple of Concordia, the route starts in a groove of a column and moves through water, earth, wheat and windblown grains before returning to the temple on the ridge. The dreamlike material changes compress slow weathering into a shot.

The two animations are the current visual prototypes. Close and wide views leave room to pause, drawing on the slower pace of cultural tourism. The exhibition proposal extends the route to a panoramic screen, a planned headset experience and a model table where children can reassemble a building.

## 3.2 Sites and Research Approach

Chapter 2 examined potential relational disconnection in the communication of historical sites through spatial, temporal, relational and sensory dimensions. It defined mediated perceptual reconnection as a short-term perceptual condition organised by VR: architecture, natural processes, human activities and temporal change are brought into a shared environment, allowing dispersed information to be observed and understood together. The concept concerns visual attention and site understanding.

This chapter moves from concepts to site evidence, analysing Pompeii, the Valley of the Temples in Agrigento, Yingxian Wooden Pagoda and the Hanging Temple.

The Sicilian case examines the terrain and environment of the Valley of the Temples, using the Temple of Concordia as its principal architectural subject. Pompeii's condition relates to the eruption of 79 CE, burial by volcanic deposits and continuing excavation since the mid-eighteenth century.

Agrigento's Doric temples form a site environment together with the sacred ridge, agriculture, stone condition and land stability. Yingxian's continuity involves a double-tube timber structure, mortise-and-tenon joints and bracket sets, alongside self-weight, earthquakes, wind, ageing and successive interventions. The cliff limits where the halls can stand. Cantilever beams embedded in the rock, timber frames and foundations at different heights let the buildings fit those limits.[1–6]

The comparison is limited to these four sites and the available evidence.

The four sites were selected to test whether the framework can identify relationships that can be shown in VR. Analysis examines how landforms, disasters, climate, structural movement, human activities and long-term maintenance jointly shape architecture, and why these relationships are difficult to observe together in static photographs or conventional descriptions.





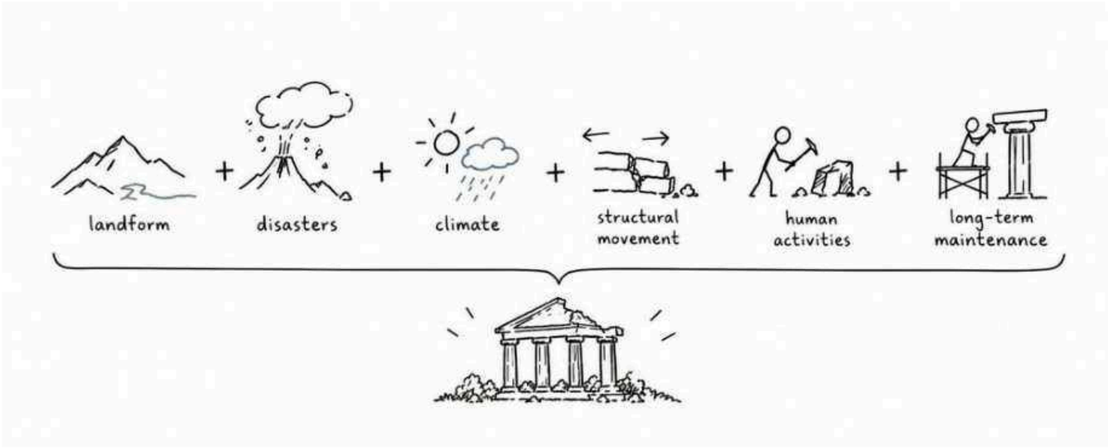



Figure 3.2-1. Landforms, climate, human activity and maintenance shape a site. Project illustration.

The chapter follows a common analytical sequence:

Site evidence → Relational processes → Perceptual issues → VR translation requirements

Perceptual issues here concern what the presentation or spatial layout makes difficult to notice.

Site evidence guides the layout, timing, movement and sound of each proposed scene. The chapter develops a proposal for what each scene could reveal and why. Chapter 4 follows this proposal into the visual methods and prototype work.

Chapter 5 organises testing feedback, research limitations and future development.

### Selection and analysis

#### Case selection

Five criteria guided the choice of sites: a documented relationship between architectural form and natural processes; temporal processes operating at different speeds and scales; comparable but distinct material and structural systems; relationships perceptible through bodily scale, direction, distance or movement; and sufficient evidence for analysis with potential for translation into immersive space.

## Case Selection and Comparative Logic

### Pompeii

BUILT SYSTEM

Masonry urban fabric, streets, houses and plastered interiors.

PRINCIPAL PROCESS

Eruption, burial, preservation and exposure.

HUMAN AND TEMPORAL DIMENSION

Ancient daily life, excavation and conservation.

COMPARATIVE FUNCTION

Connects sudden catastrophe with long-term archaeological exposure.

### Valley of the Temples

BUILT SYSTEM

Doric stone architecture situated on a ridge and sacred hill.

PRINCIPAL PROCESS

Material weathering, water ingress and land instability.

HUMAN AND TEMPORAL DIMENSION

Construction, reuse, restoration and landscape management.

COMPARATIVE FUNCTION

Connects architectural form with terrain and gradual material change.

### Yingxian Wooden Pagoda

BUILT SYSTEM

Double-barrel timber structure, mortise-and-tenon joints and dougong.

PRINCIPAL PROCESS

Structural movement, earthquakes, wind and material ageing.

HUMAN AND TEMPORAL DIMENSION

Religious use, repair, reinforcement and monitoring.

COMPARATIVE FUNCTION

Connects architectural continuity with flexibility and cumulative intervention.

### Hanging Temple

BUILT SYSTEM

Cliff-connected timber frames and cantilever beams.

PRINCIPAL PROCESS

Gravity, rock restraint and directional seismic response.

HUMAN AND TEMPORAL DIMENSION

Religious use, circulation, maintenance and structural monitoring.

COMPARATIVE FUNCTION

Connects architectural space with mountain conditions and bodily movement.

Author's synthesis based on the site evidence cited in Section 3.2.

Comparison emphasises differences. Pompeii shows how catastrophe can destroy a city and preserve its remains.

The Sicilian temples show how terrain and stone weathering shape a site. Yingxian shows structural movement, damage and long-term maintenance.





The Hanging Temple shows how the cliff, structural forces and human decisions shape the building. Conclusions must remain limited to whether the framework yields evidence-based VR requirements for these four cases given the available sources.

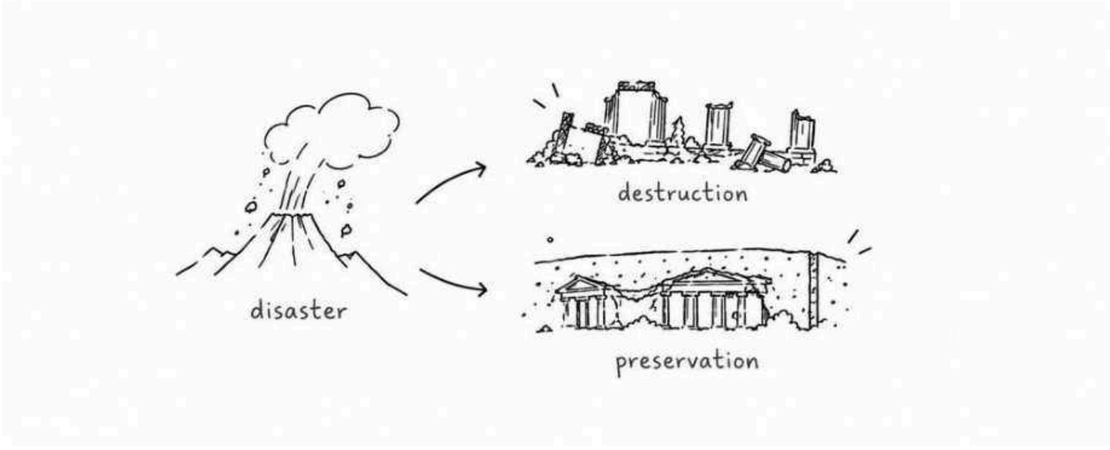



Figure 3.2-2. Catastrophe can cause destruction and preservation. Project illustration.

### Evidence and interpretation

The evidence includes official records from heritage and management bodies, published research in archaeology, architectural history, material conservation and structural engineering, and visual records such as site photographs, observations and historical images.

Sketches, three-dimensional experiments and revision records document how evidence informed design decisions. Official sources establish site boundaries, heritage status and basic history. Peer-reviewed studies support judgements about materials, structures and natural processes. Visual sources support analysis of visible forms and spatial relationships.

The analysis distinguishes three levels of engagement with historical material. Recorded evidence comes from official records, archaeology and peer-reviewed research. It supports judgements about history, materials and structures. Interpretation compares the records and suggests connections across sources. Design transformation introduces changes of scale, compressed time, fusion, growth and dissolution. These choices create the dreamlike experience, but do not provide archaeological evidence. Keeping the three levels visible allows imaginative work to remain accountable to its sources.





## Interpretive Levels for Visual Translation

### Documented evidence

DEFINITION

Information directly supported by official records, archaeological evidence or peer-reviewed research.

PERMITTED USE

May be presented as a historical, material or structural claim.

### Evidence-based interpretation

DEFINITION

A relational conclusion derived by comparing several documented observations.

PERMITTED USE

Must be presented as the researcher's interpretation rather than an uncontested fact.

### Design transformation

DEFINITION

Spatial reorganisation, temporal compression, scale change, fusion, growth or disappearance introduced for the immersive experience.

PERMITTED USE

Must be identified as a design decision and cannot be presented as historical reconstruction.

Author's framework, informed by the ICOMOS Interpretation Charter and the London Charter.

The ICOMOS Charter for the Interpretation and Presentation of Cultural Heritage Sites requires interpretation to draw on accepted scientific and scholarly methods, documenting sources and social, historical and natural contexts.[7] The London Charter requires digital visualisations to distinguish existing conditions, evidence-based restoration and hypothetical reconstruction.

The Charter also requires documentation of sources, reasoning, uncertainty and creative decisions.[8]

The project does not aim to completely reconstruct a particular historical moment. Production records should identify the evidence behind each imagined change. Each visual element retains at least this chain:

Source → Observed condition → Relational interpretation → Uncertainty → Design decision

The record lets readers trace a visual decision back to the evidence and interpretation that informed it.

### From evidence to design

Relational analysis begins with specific evidence. Each case addresses four questions: what material, structural, environmental or historical evidence can be verified; which factors jointly formed or changed the architecture; which relationships are difficult to observe in static images, descriptions or isolated models; and which spatial, temporal, movement or sound conditions could make those relationships more noticeable in VR.





## Site–Relationship–VR Translation Matrix

### Pompeii

SITE EVIDENCE

Ash and lapilli burial, preserved urban remains and archaeological exposure.

RELATIONAL PROCESS

Destruction, burial, preservation and excavation form a continuous sequence.

PERCEPTUAL ISSUE

The present ruin may appear detached from the processes that produced it.

PRELIMINARY VR REQUIREMENT

Layered temporal presentation of eruption, burial and excavation.

### Agrigento

SITE EVIDENCE

Doric temples, sacred-hill setting, porous stone and land instability.

RELATIONAL PROCESS

Terrain, material behaviour and conservation jointly affect the visible condition.

PERCEPTUAL ISSUE

Frontal images can detach the monument from its landscape and material change.

PRELIMINARY VR REQUIREMENT

Maintain topographical continuity and make gradual surface change perceptible.

### Yingxian

SITE EVIDENCE

Double-barrel structure, timber joints, residual deformation and repair evidence.

RELATIONAL PROCESS

Flexibility, accumulated damage and intervention contribute to continuity.

PERCEPTUAL ISSUE

A stable exterior can conceal movement, load transfer and maintenance.

PRELIMINARY VR REQUIREMENT

Reveal structural layers and use controlled motion without exaggeration.

### Hanging Temple

SITE EVIDENCE

Cliff-connected cantilever beams and unequal foundation levels.

RELATIONAL PROCESS

Rock restraint, gravity and timber framing produce directional structural behaviour.

PERCEPTUAL ISSUE

Exterior photographs provide limited access to the cliff–structure interface and bodily scale.

PRELIMINARY VR REQUIREMENT

Expose the rock–beam relationship; use viewpoint, parallax and routes to communicate height and constraint.

Chapter 2 sets four outcomes for the visual translation. Viewers should be able to notice overlooked relationships.

Viewers should also identify factors and their effects and describe how a building formed or changed.

The fourth outcome is to make connections among nature, architecture and history. These indicators capture immediate attention and understanding. Section 1.4.6 sets out the limits. The output is a traceable site–relationship–design requirements matrix, rather than the final scenes.

## 3.3 Pompeii: Eruption, Burial and Preservation

### Site context

Pompeii lies in Campania, Italy, approximately 9.5 kilometres southeast of Vesuvius. The site is a city that developed together with volcanic landforms, the Sarno plain, the ancient coast and regional transport. UNESCO describes a large commercial city whose approximately 44 excavated hectares preserve streets, forums, temples, baths, theatres, homes, shops and production spaces, allowing different layers of urban life to be observed together.[9] The analysis therefore begins with the city's longstanding relationships to regional resources, terrain, agriculture and risk, rather than treating the volcano solely as a backdrop to destruction.

Residents used their environment at several scales. Pompeii Archaeological Park's official Vesuvian displays document fishing around the Sarno estuary and coast, as well as cereals, fruit and vegetables.

The displays also document livestock, upland timber use, and urban and villa gardens. Pollen, plant remains and palaeogeographical research also identify the pre-79 CE floodplain as a cultural landscape shaped by sustained human activities.[10][11] Buildings adapted to the terrain and available resources. People also changed vegetation, land use and the movement of materials.

The city was already changing before the eruption. A major earthquake in 62 or 63 CE caused extensive damage; ancient texts and modern scholarship differ over its precise year. Repairs, structural deformation, dismantled materials and construction supplies show that some houses and public buildings were still being repaired, renovated or reorganised in 79 CE.[12][13] A pre-eruption model should therefore reflect an already dynamic state shaped by earthquake damage, repair, material reuse and social investment.

Pompeii's present condition combines at least three temporal layers: the evolving city and environment before the eruption; destruction, burial and exceptional preservation in 79 CE; and excavation, display, repair and renewed exposure since 1748.[9][18] These layers cannot substitute for one another. A complete pre-eruption city obscures disaster and archaeological processes; contemporary ruins alone disconnect architecture from life, deposits and excavation history.





The core relationship is the continuing interaction of resources, urban life, earthquake damage, eruption, burial conditions and heritage maintenance.

### Processes and evidence

Research on the 79 CE eruption has moved towards reconstructing multiple phases from stratigraphy. Recent synthesis indicates that the initial phreatomagmatic phase did not affect Pompeii. The subsequent Plinian column transported white and then grey pumice southeastward. Pumice fall deposits reached approximately 2.8–2.9 metres in maximum thickness, with thin interlayers from partial column collapse. Later column collapse and caldera formation generated pyroclastic density currents that crossed the city, followed by layered ash during waning activity.[12] Pompeii experienced successive events differing in direction, grain size, velocity and depositional mechanism.

Damage cannot be attributed entirely to pyroclastic flows. Accumulating pumice imposed vertical loads capable of destabilising roofs and walls; moving density currents exerted lateral impact and filled spaces. A 2024 multidisciplinary study proposes additional effects of syn-eruptive earthquakes on some wall collapses and deaths.[12] Its local excavation evidence does not attribute every collapse to earthquakes, but challenges a single-force explanation.

Earlier and later stratigraphic studies likewise distinguish pumice fall and density-current phases, while newer deposits reveal pauses or intervals of weaker activity.[14][15]

Pliny the Younger's Letters 6.16 and 6.20 describe the eruption column, darkness, ash and continuing earthquakes, providing important eyewitness material for volcanology.[16] However, he was in Misenum across the bay, wrote more than twenty years later and did not directly describe every phase inside Pompeii. The letters support regional phenomena and human responses, but cannot independently establish a minute-by-minute reconstruction within the city. Local processes require interpretation alongside stratigraphy, collapse locations and human remains.

Volcanic deposits caused death and architectural destruction while isolating, fixing and sealing some archaeological layers. Streets, decorations, building supplies, everyday objects and organic traces were preserved.[13] Hardened ash around some bodies retained cavities after soft tissue decomposed. From 1863, Giuseppe Fiorelli developed plaster casting of these cavities, producing the familiar casts of victims and organic objects.[17] Preservation was nevertheless selective.

CITY, VOLCANO AND SURVIVING SURFACES

# Pompeii

Figure 3.3-2. The Pompeii Forum with Vesuvius in the distance.

Photo: Heinz-Josef Lücking. Wikimedia Commons · CC BY-SA 3.0 DE.

Figure 3.3-5. Aerial view of Pompeii's Forum and excavated blocks.

3.5-5 · ElfQrin. Wikimedia Commons · CC BY-SA 4.0.

The panoramic view connects the excavated Forum to Vesuvius. At a smaller scale, street paving and surviving painted walls retain traces of daily life. Read together with Section 3.3, these photographs connect the present ruin to urban use, selective preservation and renewed exposure after excavation.

Figure 3.3-3. Paving, curbs and stepping stones on Via Stabiana.

Photo: Carole Raddato. Wikimedia Commons · CC BY-SA 2.0.

Figure 3.3-4. Interior walls and surviving frescoes at Pompeii.

Photo: Canadacow. Wikimedia Commons · CC BY 2.0.

Many roofs and upper structures collapsed, much timber and bodily material decomposed, and survival varied with location, deposits, temperature, moisture and later disturbance.

Modern excavation began in 1748 and gradually exposed approximately two-thirds of the ancient city.[9][18] Excavation made the city visible but removed protective deposits, exposing walls, plaster, paintings and floors to rain, temperature and humidity changes, plants, biological growth, earthquakes, visitors and earlier incompatible repairs. Management and conservation research emphasise drainage, roofs, slopes, structural reinforcement, wall maintenance and continuous monitoring.[18][19] Excavation brings the site into new environmental conditions and maintenance systems.

The eruption damaged the city, and some of the same deposits preserved its remains. Excavation made those layers visible and exposed them to weathering, creating a need for continued conservation.

Figure 3.3-1. Everyday Pompeii, pumice fall, burial and excavation. AI-generated sketches. The Roman arch is schematic.

### Design translation

The proposal for Pompeii links habitation, eruption, burial and renewed exposure. Static views of ruins often leave these relationships unseen, while disaster imagery concentrates attention on the moment of destruction. Showing the stages in sequence would help viewers see how the same deposits damaged buildings and later protected their remains. The sequence follows documented processes without claiming precise timings or inventing historical characters' actions.

The proposed scene begins at the scale of streets and homes, with Vesuvius and the surrounding plain establishing its wider setting.

The volcano's position matters more than an exaggerated size. The proposal distinguishes four states: everyday life and repair; accumulating white and grey pumice; burial by ash and pyroclastic density currents; and excavation followed by conservation. Sections, local dissolves and reversible transitions could connect these states while leaving the uncertain timing visible.

The two volcanic processes call for different movements. Falling pumice accumulates vertically, increasing depth and roof loads. Density currents move laterally through terrain and streets, carrying ash into enclosed spaces. Distinguishing them gives the material sequence a clearer spatial meaning.

Transparent sections or switchable layers can reveal relationships among buildings, deposits and cavities, providing a material explanation of preservation.

The proposed scene would evoke everyday life through gardens, crops, storage, shops, building materials and water, drawing on archaeological evidence. This offers a setting for habitation without relying on invented character stories. If body casts enter the scene, their role is to explain burial and the casting procedures developed in the nineteenth century. They are not bodies petrified during the eruption, and their interpretation requires that distinction.

The proposed sound design includes changes in direction, particle impacts, structural responses and the absorption of sound after burial. These are designed simulations informed by physical processes; no historical recording survives. Viewpoint height, movement and the occlusion of narrow streets would keep the experience close to bodily scale, with aerial views serving as occasional comparisons.

The scene aims to help viewers connect the city with the volcano and its surroundings and distinguish pumice fall, density currents, burial and exposure after excavation.

Viewers should also be able to explain how the eruption damaged and preserved remains, and how excavation exposed them to renewed deterioration. Observation, interviews and task testing are required to assess these outcomes. These are design requirements, not evidence that reconnection has occurred.

Interpretive boundaries must remain explicit. The eruption's exact date is disputed. A charcoal inscription discovered in 2018 supports an autumn date without resolving every debate.[20] Deposit thickness, flow intensity and damage vary locally. Characters, soundscapes, colours and compressed time involve design inference. The three interpretive levels in Section 3.2 allow dreamlike development in Chapter 4 without misidentification as archaeological reconstruction.

## 3.4 Valley of the Temples: Stone and Weathering

### Site context

The Valley of the Temples lies on Sicily's southern coast. Its World Heritage area extends across the ancient city, from Rupe Atenea and the acropolis through urban areas and the sacred hill to outlying cemeteries. UNESCO notes substantial remains beneath contemporary fields and orchards.





UNESCO's assessment of the site's integrity covers temples, roads, hydraulic works, housing, cemeteries, unexcavated areas and agricultural landscapes.[2] Terrain, stone, farming, religious architecture, later uses and heritage management have all shaped the landscape around the temples.

Ancient Akragas occupied elevated terrain near the coast, protected to the north by Rupe Atenea and Girgenti and defined to the south by the temple ridge, between the Akragas and Hypsas rivers. Official park information describes walls and gates exploiting rock and differences in elevation, with principal temples forming a directional sequence along the southern ridge.[21] The terrain shaped the city's boundaries, views, ritual sites and routes.

The Temple of Concordia is the main architectural subject. Its survival cannot be attributed directly to inherently strong stone. Park documentation dates it to the later fifth century BCE, recording a six-by-thirteen Doric colonnade and substantially preserved entablature and pediments. Around the late sixth century CE it became a Christian church; twelve arches cut into the cella walls and the new use contributed to continuity. Its original cult remains uncertain in archaeological and inscriptional evidence.[22] Preservation involves construction, religious reuse, restoration and management.

### Processes and evidence

The site's stone is predominantly local biocalcarenite. Taglieri et al.'s laboratory samples had approximately 26.1% mean porosity, with pores chiefly in the 10–100 micrometre range associated with strong capillary uptake. Untreated samples absorbed water rapidly.[3] These experiments inform mechanisms and treatment, but cannot establish uniform conditions for every temple or stone. Orientation, crusts, salts, cracks, repair materials and drainage alter actual deterioration.

Rain can run down the flutes and across the stone, while some water enters its pores by capillary action. Repeated wetting and drying may combine with salt migration, temperature changes and biological growth to alter the surface. Laboratory work supports the mechanism of uptake [3]. Without material tests at the temple, cracks and stains in photographs cannot establish the cause of decay in a particular area.

Human activities also contribute to long-term change. After antiquity, temples experienced religious conversion, abandonment, stone removal, agriculture, eighteenth- and nineteenth-century restoration and modern conservation.

RIDGE, COLONNADE AND REUSE

# Agrigento

Figure 3.4-2. The escarpment of the Valley of the Temples, looking towards Concordia.

Photo: Zde. Wikimedia Commons · CC BY-SA 4.0.

Figure 3.4-5. Aerial view of Concordia and its terrain (right), paired with Figure 3.4-2.

3.6-5 · Giacomo Palermo. Wikimedia Commons · CC BY-SA 4.0.

The ridge view places Concordia within a continuous terrain. The facade photograph brings columns and stone surfaces into view, while the cella openings support the discussion of later use. Visible surface differences provide a viewing reference; their causes require material evidence beyond these photographs.

Figure 3.4-3. Columns and entablature on Concordia's east façade.

Photo: Zde. Wikimedia Commons · CC BY-SA 4.0.

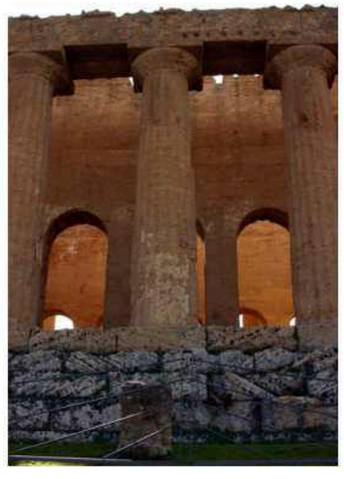


Figure 3.4-4. Arched openings on the south side of Concordia's cella.

Photo: José Luiz Bernardes Ribeiro. Wikimedia Commons · CC BY-SA 4.0.

Official sources record reuse of ancient blocks in Girgenti and Porto Empedocle's old port, while Concordia's conversion to a church contributed to structural survival. [21][22] UNESCO notes that some historical restorations conflict with later conservation principles and that land instability remains a concern.[2] Today's completeness and losses reflect materials, reuse, removal, restoration and geological conditions.

Park records describe olives, almonds, grapes, cereals, citrus and Mediterranean scrub [23]. Wheat offers a specific example. In 2024, the FAI documented the return of Chiattulidda, a traditional Sicilian durum wheat, to an arable plot in the valley's Kolymbethra Garden (FAI, 2024). Recorded in 1895, this variety matures early and has tall stems with comparatively short ears. The garden shows a farming tradition continuing today. The location and varieties of crops beside the ancient temple remain uncertain.

Figure 3.4-1. Ridge and farmland, porous stone and moisture, and religious reuse. AI-generated conceptual sketches.

### Design translation

At Agrigento, the proposal follows the ridge to connect the temple with the city, slopes, valleys and coast. Elevation, the rhythm of the columns and open views offer a way to understand the building through its terrain.

The animation opens close to a column flute. Raindrops, ripples and runoff make the path of water visible and draw attention to how porous stone takes up moisture. The stone turning into water compresses the slow effects of weathering into a dreamlike transition. The column's shape guides the depicted runoff.

The sequence illustrates possible weathering mechanisms. Identifying the causes of individual defects would require onsite tests.

The arches cut into the cella walls connect the temple's present form to its use as a church. The proposal uses reversible layers through which viewers could compare these periods and the modern heritage site. Keeping later alterations visible treats reuse as part of the building's history.

The sphere crosses bare ground into wheat, connecting the temple stone with cultivated land. The low view makes the height and density of the moving stalks and ears perceptible. The wider view brings the field and temple together in one landscape. Wind, insects and footsteps can also suggest distance and direction.

These sounds remain contemporary or designed simulations rather than reconstructed ancient soundscapes.

Immediate goals are to identify spatial relationships among temple, ridge, city and agriculture; understand how porous stone, exposure and maintenance jointly affect surfaces; and recognise religious reuse and restoration as parts of continuity. The case cannot prove a complete causal chain for specific deterioration or extrapolate present vegetation into ancient appearance.

## 3.5 Yingxian Wooden Pagoda: Structure and Movement

### Site context

Yingxian Wooden Pagoda, the Sakyamuni Pagoda of Fogong Temple in Shanxi, was built in the second year of the Liao dynasty's Qingning era. China's UNESCO Tentative List submission describes an octagonal timber structure 67.31 metres high with nine levels: five visible and four hidden. Inner and outer column rings form a double-tube structure, connected through mortise-and-tenon joints, bracket sets, beams and braces that organise vertical and lateral actions.[4] This source is a Tentative List submission and does not establish World Heritage inscription.

The pagoda comprises interacting components rather than a static stack. Alternating visible and hidden levels differ in stiffness and height. Tubes, columns, beams, brackets and eaves establish load paths from finial to platform. Timber joints permit rotation, sliding, compression and frictional dissipation under loads. Such deformation contributes to response but does not guarantee millennial survival. Continuity also depends on construction, material condition, damage, repairs, use restrictions and monitoring.

Religious use and maintenance form additional temporal layers. The submission records statues, murals, scriptures, inscriptions and repairs, including records left during later component replacement.[4] The pagoda combines a Liao structural core, religious remains and successive interventions.

### Processes and evidence

Mi et al. analyse residual deformation at component, storey and whole-building scales, identifying tilt, twist and local displacement.[5] Subsequent UAV imagery and terrestrial LiDAR research compares data across periods, identifying overall twisting and continuing displacement of some inner columns on the second visible level. The fused model's reported error is approximately six millimetres.[24] A stable appearance therefore does not establish complete immobility.





Deformation may accumulate with considerable differences between levels and directions.

Measured deformation does not establish a unique cause. Self-weight, joint relaxation, creep, historical earthquakes, wind, temperature and humidity, component damage and interventions may contribute. The point-cloud study explicitly analyses geometry without calculating each column's or level's capacity and stress.[24] This thesis can describe measurable spatial deformation, but cannot infer imminent failure from geometry alone or attribute all changes to one earthquake.

Engineering studies combine onsite dynamic tests and finite element models to assess seismic response. Deng et al. include nonlinear bracket and joint behaviour, initial inclination and measured dynamics in model updating.[25] Models support analysis of inter-storey displacement, joint rotation and potential weaknesses, but depend on material parameters, simplified joints, boundaries and seismic inputs. Outputs remain engineering inferences under stated assumptions, rather than directly visible historical facts.

Timber continues to be affected by humidity, temperature and biological factors. Orientation, eave shelter, ventilation and exposure to rain and snow produce differences in moisture and ageing, but this study has not obtained long-term microclimate and timber-testing data covering the whole pagoda. VR may therefore show directions and accumulation of environmental effects, without assigning untested causes to every crack, discolouration or bend.

The pagoda has remained standing through deformation, damage and successive repairs. Multiple structural levels and deformable joints respond to loads, while damage, residual deformation, repair and monitoring are continuously managed. Attributing survival entirely to ancient wisdom would erase modern conservation, while describing the tower as already severely unstable would also exceed the evidence.

Timber links the tower directly to trees. Huang et al. identified North China larch over a century old in the pagoda's horizontal timbers above the column heads (pupaifang). Most were cut to their full length from a single piece [Y2]. The length of a mature trunk helps explain how a tree could supply a continuous member across the column frame.

SCALE, EAVES AND TIMBER CONNECTIONS

# Yingxian Wooden Pagoda

Figure 3.5-2. The Pagoda of Fogong Temple from street level.

Photo: Zhangzhugang. Wikimedia Commons · CC BY-SA 3.0.

Liang Sicheng's survey records the Sanggan River plain north of town and the Yanmen mountains to the south [Y1]. In a 2011 view from the south, tree crowns partly hide the base while the layered eaves rise above them [Y6]. The trees give the tower a visible scale. Beyond the town, the Yingxian Nanshan Nature Reserve protects forests of North China larch. The photograph cannot identify the species at the base or establish what grew in the Liao courtyard.

Figure 3.5-5. Ceiling and beams on the pagoda's second level. See Figure 3.5-3 for the exterior brackets.

3.5-5 · Zeus1234. Wikimedia Commons · CC BY-SA 3.0.

Figure 3.5-3. Brackets and timber framing beneath the pagoda eaves.

Photo: Zhangzhugang. Wikimedia Commons · CC BY-SA 3.0.

Figure 3.5-4. The pagoda and its layered eaves from below.

Photo: Zhangzhugang. Wikimedia Commons · CC BY-SA 3.0.

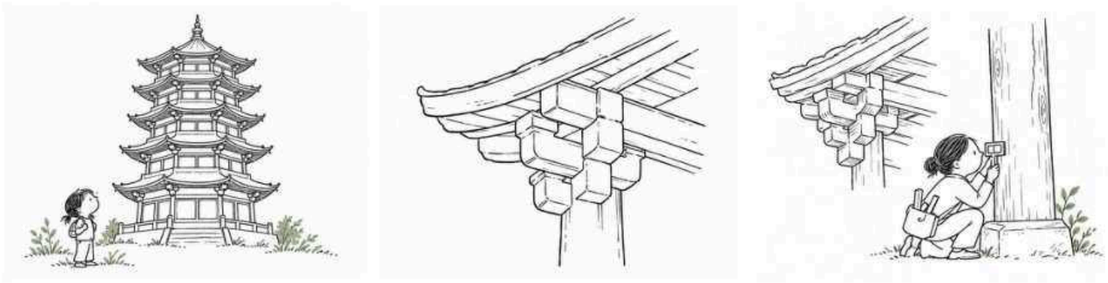



Figure 3.5-1. Scale, timber joints and inspection at Yingxian Wooden Pagoda. AI-generated conceptual sketches.

### Design translation

The timber source informs the opening of the pagoda sequence. A larch and the tower appear in turn on the same platform, so the trunk can be read against the columns and the branches against the projecting eaves. Chapter 4 uses this change to show where the building material comes from. Sections and changes in layer visibility could then reveal how the inner and outer column rings, hidden levels and brackets fit together.

To make small movements perceptible, the proposal calls for measured comparisons of tilt, twist and joint rotation, with any amplification clearly labelled.

Existing geometric deviations, movement under loads and simulated earthquake responses belong to different kinds of evidence. Showing them separately would avoid confusing accumulated deformation with the tower's motion during a strong earthquake.

Layered records of the Liao structure, religious use, repairs, surveys and monitoring could show how the pagoda has endured. In this proposal, repair traces give continuity a visible history rather than presenting survival as a property of the original structure alone.





A quiet soundscape could use timber friction, sounds responding to footsteps, and wind under the eaves and inside the tower. It would interpret the materials and spaces, without claiming to reproduce historical sounds.

Later testing could ask viewers to explain the relationships among the structural tubes, visible and hidden levels, and bracket joints. A further question is whether viewers distinguish present geometry from long-term deformation and recognise the role of maintenance. Impressions of grandeur or stability alone would not answer these questions.

## 3.6 Hanging Temple: Mountain, Gravity and Movement

### Site and topography

The Hanging Temple occupies a natural recess in Cuiping Peak on the west side of Jinlong Gorge, above the Tangyu River in Hunyuan, Shanxi. Rock ledges and changes in the cliff face shaped the size and position of its halls. Steps and walkways connect spaces at different levels. The rock overhang shelters parts of the complex from direct rainfall, while the cliff places it above the riverbed. The same cliff also exposes the building to weathering, runoff and falling rock.

Narrow timber walkways, successive steps and railings above the gorge make the distance between rock face and riverbed tangible. Figures 3.6-2 to 3.6-5 record the timber frame, entrance, route and setting.

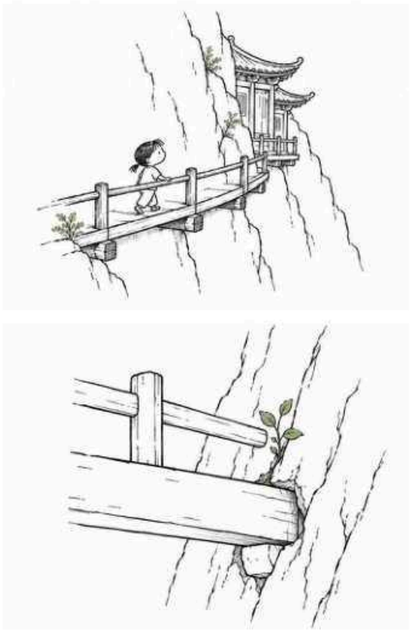


Figure 3.6-1. Above, walking beside the cliff. Below, the beam end at the rock face. AI-generated conceptual illustrations.

CLIFF, TIMBER FRAME AND MOVEMENT

# Hanging Temple

Figure 3.6-2. Thunder Hall and its timber frame against the cliff.

Photo: Gisling. Wikimedia Commons · CC BY 3.0.

Figure 3.6-3. The Hanging Temple entrance against the rock face.

Photo: Gisling. Wikimedia Commons · CC BY 3.0.

Figure 3.6-4. Internal steps and visitors at the Hanging Temple.

Photo: Zhangzhugang. Wikimedia Commons · CC BY-SA 3.0.

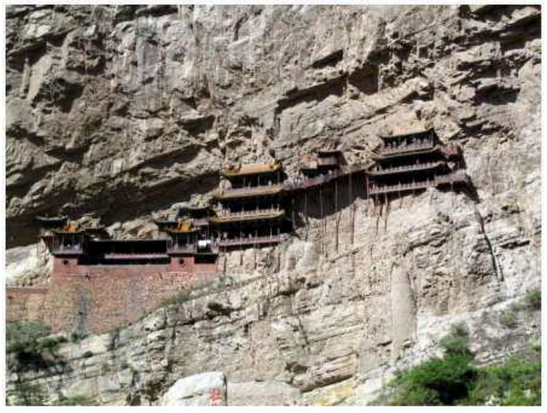


Figure 3.6-5. The Hanging Temple along the Cuiping cliff.

Photo: G41rn8, 2009. Wikimedia Commons · CC BY-SA 4.0.





## Timber and rock connections

Cantilever beams embedded in the cliff project out to carry floors or galleries, with some framing supported on ledges. Mortise-and-tenon joints connect beams and columns. The role of the outer posts varies with their position and loading. Under the visitor loads modelled for the walkway, posts carried a limited share of the main-beam stress but helped restrain movement. The archive's depth of about 1–1.2 m refers to two specific cantilever beams. The walkway study used 1 m as a model input.

Surveys and models make these connections easier to examine. In 2019, a Wuhan University team used 730 UAV images to reconstruct the site's existing condition in 3D. A later seismic study used field investigation and surveys to simulate the structural response under specified conditions. In the walkway model, concealed connections had to be inferred from their geometry.

Hengshan architectural archive · Zhang et al. (2026) · Zuo et al. (2018)

Yang et al. (2025) · Zhang et al. (2026) · Wuhan University (2019)

Figure 3.6-6. UAV survey of the Hanging Temple façade for 3D reconstruction.

Wuhan University, 2019

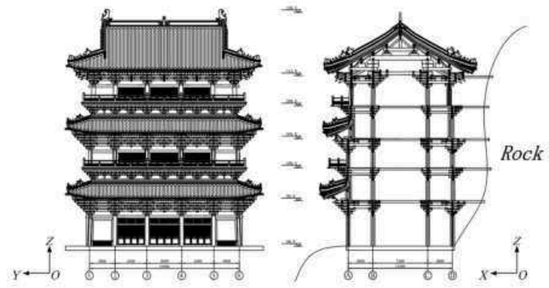



Figure 3.6-7. Elevation and section showing the halls, timber frame and cliff.

Yang et al., 2025, Fig. 2

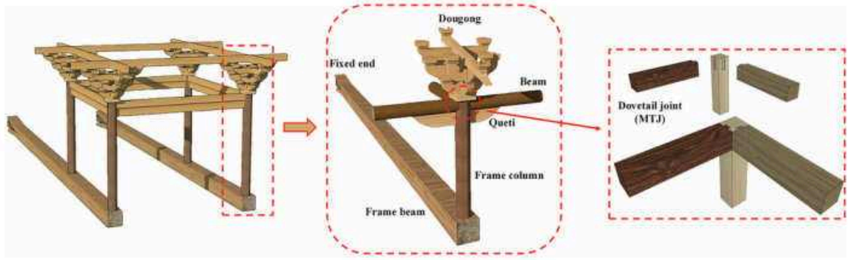



Figure 3.6-8. Mortise-and-tenon timber frame of the Hanging Temple.

Yang et al., 2025, Fig. 3

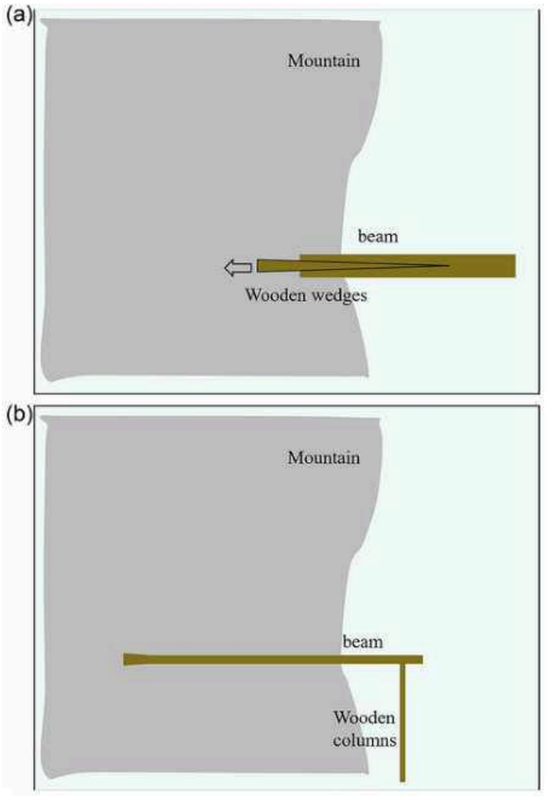



Figure 3.6-9. Walkway beam ends. (a) The inner end anchors into the rock. (b) The outer end meets a timber post.

Zhang et al., 2026, Fig. 2

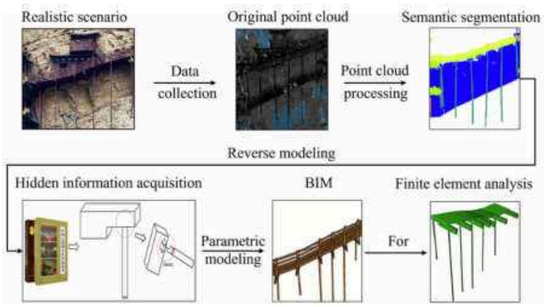



Figure 3.6-10. Point cloud to parametric walkway model. Concealed joints were inferred geometrically.

Zhang et al., 2026, Fig. 3

## Repair and changes to the site

Official accounts trace the foundation to the Northern Wei. Most surviving timber structures date from Ming and Qing rebuilding and have since undergone further repairs. Rockfall damaged the Drum Tower in 1973, leading to surveys and repairs in 1975. In 1978, the timber walkway linking the north and south buildings and its supporting posts were replaced. After rain in 2015, falling rock struck the Drum Tower roof again, prompting repairs to the cliff-top catch drain and work to stabilise loose rock. Changes to the walkways, posts and drainage record successive responses to the cliff environment.

Hengshan architectural archive · Rockfall protection report (2016)





## Construction and reconstruction

The original construction sequence remains uncertain. Temporary access, permanent floors and the walkway linking the north and south buildings serve different purposes. The available evidence leaves open where work began, where the first beam was fitted and whether workers were suspended throughout construction. Zhang Jing's diagrams explain rock cutting, beam anchoring and assembly. The halls follow Ming and Qing rebuilding styles, and the lower route is identified as a modern addition.

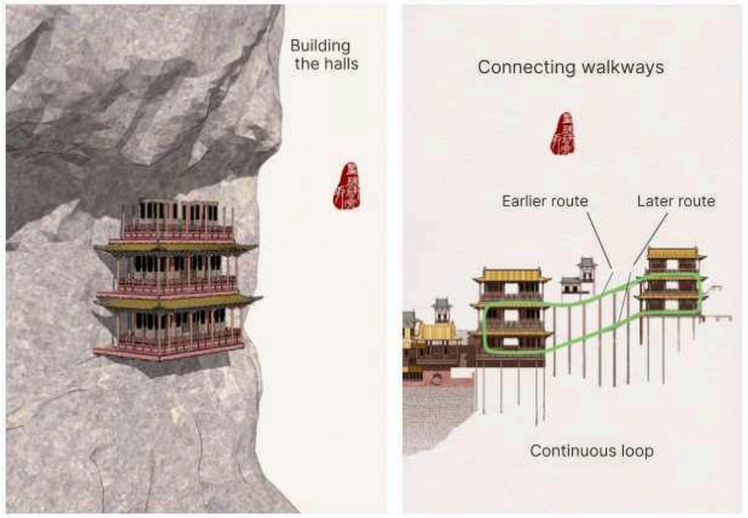



Figure 3.6-11. Construction and circulation diagrams. Left, the halls are drawn in their Ming and Qing rebuilding styles. Right, the lower walkway is a modern addition.

Drawings by Zhang Jing, Institute for Planets, 2019

Figure 3.6-12. Conjectural view of the late Northern Wei temple. AI illustration based on the cliff. The halls, roofs and approach are speculative.

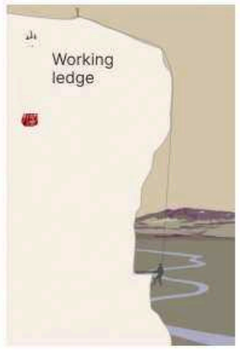
Project illustration generated with ImageGen

# Working ledges and cantilever beams

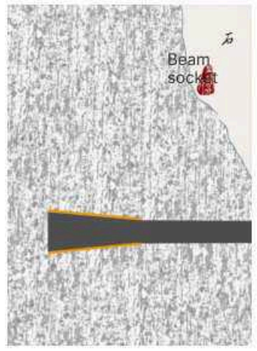
The four figure groups follow Zhang Jing's modern diagrams. The halls are drawn in the surviving Ming and Qing rebuilding styles.

The diagrams begin at a natural recess in Cuiping Peak. Panel (a) shows a narrow working ledge formed within the recess. Panel (b) takes the view into the cliff to show the horizontal beam socket. The ledge gives workers a place to stand, while the socket provides space for the beam inside the rock. Where the cliff can receive a beam helps determine where the halls can project into the gorge.

Panel (c) shows how a wedge tightens the beam end in the socket. Panel (d) puts the beam back into the wider structure. The rock restrains the embedded end, while the projecting end supports the floor and transfers its load into the cliff. These diagrams explain the principle of the connection. Embedment depths and loads depend on the individual beam.

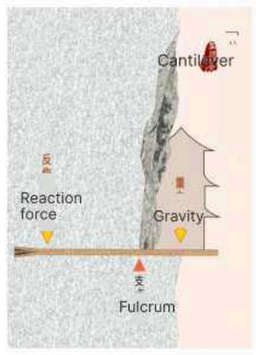



(a) Preparing the ledge




(b) Cutting the beam socket

(c) Wedging the beam end




(d) Cantilever action

Figure 3.6-13. Working ledge, beam socket, wedge anchoring and cantilever action.

Illustrations by Zhang Jing, Institute for Planets, 2019. View original illustrations and animations.

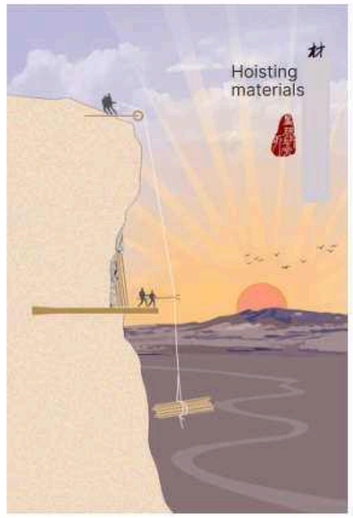




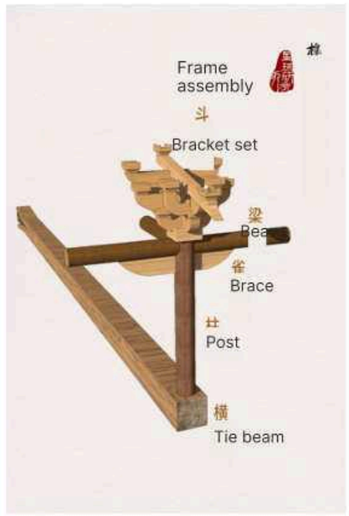
# Transport and timber assembly

The working area is narrow, so access for timber affects how the frame can be assembled. Panel (a) shows ropes lowering materials from above as one possible method. It also helps distinguish temporary access for workers and materials from the walkways used after construction. The available evidence leaves open where work began and whether ropes were used throughout.

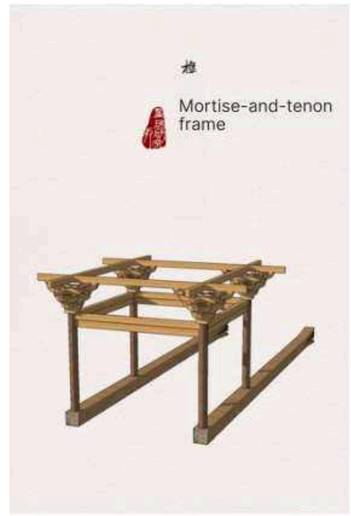
Panels (b) and (c) shift attention from the cliff to the components. Posts, beams, tie members and brackets occupy different positions in the frame. Mortise-and-tenon joints are formed as the members are fitted. A shot can follow one connection along the beam into the next bay, showing how individual joints become a continuous timber frame.




(a) Hoisting timber




(b) Assembling beams and posts




(c) Forming the frame

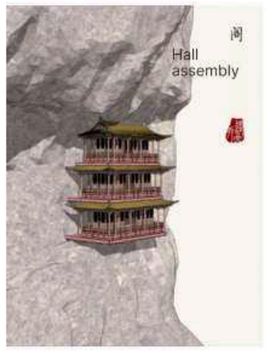
Figure 3.6-14. Hoisting timber, assembling joints and forming a frame.

Illustrations by Zhang Jing, Institute for Planets, 2019. View original illustrations and animations.

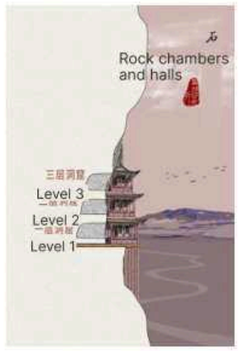
# Halls, rock chambers and vertical layout

The frame establishes the extent and levels of the floors. Roofs, doors, windows and railings then define interiors, galleries and exposed edges. Panel (a) shows the halls taking shape level by level against the cliff. Panel (b) places halls and rock chambers side by side. Some spaces extend into the mountain, while others project out on timber framing. Moving between them changes how close a visitor feels to the rock and the gorge.

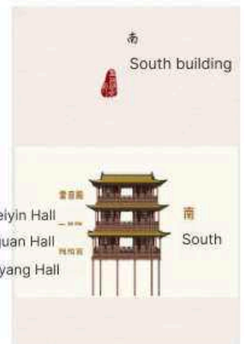
The diagrams of the south and north buildings make the vertical arrangement clearer. Halls occupy different levels, while stairs and galleries connect adjoining spaces. The illustrations follow the surviving Ming and Qing rebuilding styles, making them useful for studying the relationship between floors, halls and cliff. The number of levels at the time of foundation and the dates of the rock chambers still require direct evidence.



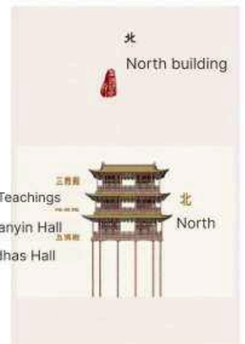

(a) Building the halls




(b) Rock chambers and halls




(c) South building




(d) North building

Figure 3.6-15. Hall construction, the rock-chamber section and the levels of the north and south buildings.

Illustrations by Zhang Jing, Institute for Planets, 2019. View original illustrations and animations.



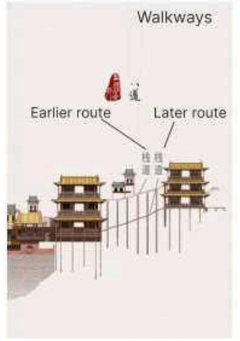


# Walkways and the completed ensemble

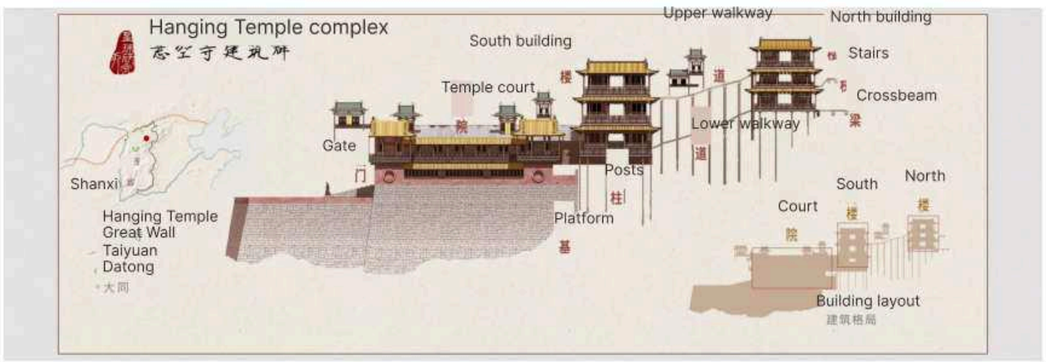
Once the individual frames and halls are in place, a continuous route allows spaces at different levels to function as a temple. Panel (a) pulls back to the full cliff. Panel (b) marks the walkway between the north and south buildings and the lower route added in modern times. Changes in level and direction determine when visitors see the valley floor and when they move close to the rock again. The present route reflects repairs and changes in access over time.

During the construction sequence, the camera moves from a beam socket to the hall framing, then follows the walkway to show how the halls connect. The loop through the mist can change the viewer's sense of direction, while the beam socket, railing and cliff remain recognisable between shots. When the valley floor is obscured, a familiar component can help the viewer regain their bearings.

(a) The ensemble along the cliff




(b) Connecting the halls




(c) Panorama of the completed complex

Figure 3.6-16. The building ensemble, connecting walkways and the completed complex.

Illustrations by Zhang Jing, Institute for Planets, 2019. View original illustrations and animations.

### Dreamlike visual translation

The scene opens on ripples as a clear drop rises from the water. The camera follows the reflection in the drop to an inverted view of the cliff. Mist hides the valley floor as the view moves closer to the rock. Beams extend from rock sockets, beams and columns join, and floors and roofs gradually appear. The view pauses at a beam end to show how the timber meets the rock, then continues along the walkway.

The rising drop, inverted cliff and growing timber frame are artistic devices. The connections draw on the surviving structure, while the order of assembly belongs to the animation. Later testing could examine whether changes in viewpoint help visitors recognise the relationship between cliff, beam ends and halls.

## 3.7 Comparative Synthesis and Design Criteria

### Cross-case comparison

Across the cases, visible appearance results from natural processes, human activities and conservation operating at different timescales. Their mechanisms differ. Pompeii links catastrophe, burial, selective preservation and renewed exposure; Agrigento links ridge terrain, porous stone, agriculture, reuse and slow weathering; Yingxian links layered timber construction, joint and residual deformation and maintenance; the Hanging Temple links rock support, directional constraints, gravity transmission and bodily movement.

Their temporal structures also differ. Pompeii combines an hours-long eruption, centuries of burial and continuing excavation and conservation. Agrigento principally combines slow material change, religious reuse and landscape management. Yingxian and the Hanging Temple combine instantaneous dynamic responses, long-term deformation, component ageing and repair. Identical visual rhythms would erase these differences. A shared framework can standardise questions and evidence labels, but not movement speeds, material effects or narrative structures.

Human activities belong within all four sites' formation. Habitation, agriculture, religious use, stone removal, excavation, repair, monitoring and visitor management participate in change. Dividing nature and human-made elements into fixed opposing categories would reproduce the relational disconnection criticised in Chapter 2.





### Design strategies

Across the four proposals, routes connect architecture to its setting: Pompeii's streets to interiors, plain and volcano; Agrigento's columns to ridge and cultivation; Yingxian's silhouette to its hidden levels and joints; and the Hanging Temple's galleries to beam ends and rock.

Each process calls for a different timescale. Eruption, deposition, weathering, structural movement and repair cannot be understood through the same accelerated sequence. The proposed approach uses local comparisons, compressed intervals and changes of state, with labels distinguishing recorded conditions, simulation and design interpretation.

Each scene is designed around a relationship that viewers could be asked to describe. Pompeii returns from volcanic burial to vegetation and architecture. Concordia moves from stone, water and windblown grains to the temple on its ridge. Yingxian and the Hanging Temple provide comparative studies of structure and terrain. A later viewing test can check what viewers notice in the two animations.

Sound, scale, parallax, movement and enclosure offer information that static images cannot convey in the same way. Heritage VR reviews discuss spatial exploration, multimedia, immersion and presence alongside navigation, participant backgrounds and evaluation. [27] The viewing test could ask what people understood about the site and note whether they report novelty, discomfort or difficulty moving through the scene.

### Design requirements

The comparison produces the following requirements for the VR project:

1. Organise transitions through site relationships rather than country or date, preserving each case's distinctive terrain and materials.

2. Label timescales for every dynamic process. Give measured states, historical records, engineering models and design animations distinct visual codes to prevent simulations being mistaken for direct evidence.

3. Support overall orientation and close observation. Show volcano and city, temple and ridge, tower and internal structure, temple and rock, while allowing approaches to strata, pores, joints and beam interfaces.

4. Prioritise comparison, switching and observation over complex operations. Core functions should include time-state switching, local sections, evidence labels and comfortable movement.

5. Use sound to indicate direction, distance, materials and stages. Identify unrecorded soundscapes as design simulations rather than authentic historical recordings.

6. Maintain distinctions between recorded evidence, evidence-based interpretation and design transformation. Make citations, sources and uncertainties available on request without permanently obscuring scenes.

7. Align evaluation with Chapter 2's attention, recognition, understanding and association outcomes. Include at least one task requiring participants to describe a relational process; liking, surprise and immersion cannot replace evidence of understanding.

Viewing tests could examine whether visitors can recognise changes, find their bearings and pause comfortably.

## Chapter summary

The four cases show how present forms carry different histories of natural processes, construction, occupation and care. Their appearances cannot be assigned to a single original moment. Comparing these histories identifies the relationships that each proposed scene could make perceptible.

The chapter also establishes boundaries between recorded evidence, evidence-based interpretation and design transformation.

Chapter 4 develops this analysis through spatial reorganisation, changes of scale, fusion, growth and dissolution. Planned headset studies and visitor feedback will examine how these choices affect viewing, comfort and understanding.

Figure 3.7-1. Design criteria in one scene: landscape, rainfall, long-term weathering and bodily scale.

# 04

# Implementation

# Pompeii:

# A return through time

## Architecture, landscape and the traces of life

*The route moves forward.*

*The city returns to an earlier time.*

Historical references · Form studies · Material tests · Animation

In this chapter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## 4.1 Design development

Pompeii grew near Vesuvius, connected to the surrounding plain, rivers, cultivation and gardens. The eruption of AD 79 damaged and buried the city, while its deposits preserved buildings, paintings and traces of daily life. Section 3.3 describes how excavation exposed them to weather and continuing care [9, 10, 18]. Today's ruins hold the histories of occupation, disaster, burial and renewed exposure.

The research made me notice how much of the life around surviving columns, walls and objects is difficult to imagine. The sequence invites viewers to approach slowly and reconnect architecture with ground, plants and light. A journey back through time became the starting point.

I chose to change the surroundings before revealing the building. Reddish grass and porous volcanic stones recall the eruption, then green returns along the sphere's path. The red grass is an imagined colour used to evoke the buried landscape. Scattered pieces reassemble, and a dark skin splits into particles to reveal the colonnade and interior.

The chosen setting leads through grass, water and architectural fragments into the temple. Reflections help viewers follow pieces rising into place. Inside, branches, fruit and oil lamps give viewers details to pause over.

The sphere accompanies viewers and slows before the steps. The Forum's axis, historical column forms, painted figures and documented lamps informed the layout and details of the imagined scene.

Figure 4.1-1. Integrated Pompeii scene. Author's render.

## 4.1.1 Reading Pompeii through images

(a) Forum and urban context

(b) Temple, porticoes and square

(c) Mosaic at a house entrance

Figure 4.1-2. Contemporary reconstruction of the Forum, temple and porticoes. Source: Lost in Time [31].

I studied wall paintings alongside views of the Forum. The paintings preserve distinctive faces, gestures, headwear and objects even where much of the original surface is missing. These recognisable details became starting points for the figures.

The Forum offered another kind of reference: an open centre, a temple axis and the repeated rhythm of the porticoes.

Reconstructed views helped me read the relationships between roofs, columns and surrounding buildings and imagine movement through the space, while the official site guide anchored the Forum's organisation and its relationship with Vesuvius [30–31].

Painted imagery also belonged to Pompeii's religious, commercial and everyday life [28]. The figure designs aim to retain this connection to the city and give the figures a role beyond ornament.





## 4.1.2 From fragments to figures

The first figure studies focused on Mercury, Jupiter, Apollo/Sol and Diana/Luna. The façade paintings at IX.7.1–2 supplied a small set of clear visual clues: Mercury's winged headwear and staff, Jupiter's beard and sceptre, and the solar and lunar signs associated with Apollo and Diana [28–29].

I kept these attributes visible while developing bodies, clothing and poses beyond the surviving fragments. A full figure needs decisions that a painted head cannot supply: where the weight rests, how a cloak falls, and how an object is held.

GPT-assisted image generation offered possible completions. I compared them with the source images and selected versions whose posture and silhouette could work at architectural scale.

A successful front view could still leave an implausible arm, an unclear grip or a costume that did not resolve from behind.

The development sheet places fresco fragments, full-figure sketches and front and rear white-model references together. The sheet shows which details came from the fresco and which were added.

The figures are interpretations developed from Pompeian imagery. Their completed bodies, costumes and positions in the scene are design decisions; the paintings do not provide a complete three-dimensional statue to reproduce.

Fresco fragment → full-figure proposal → white-model reference → scene model

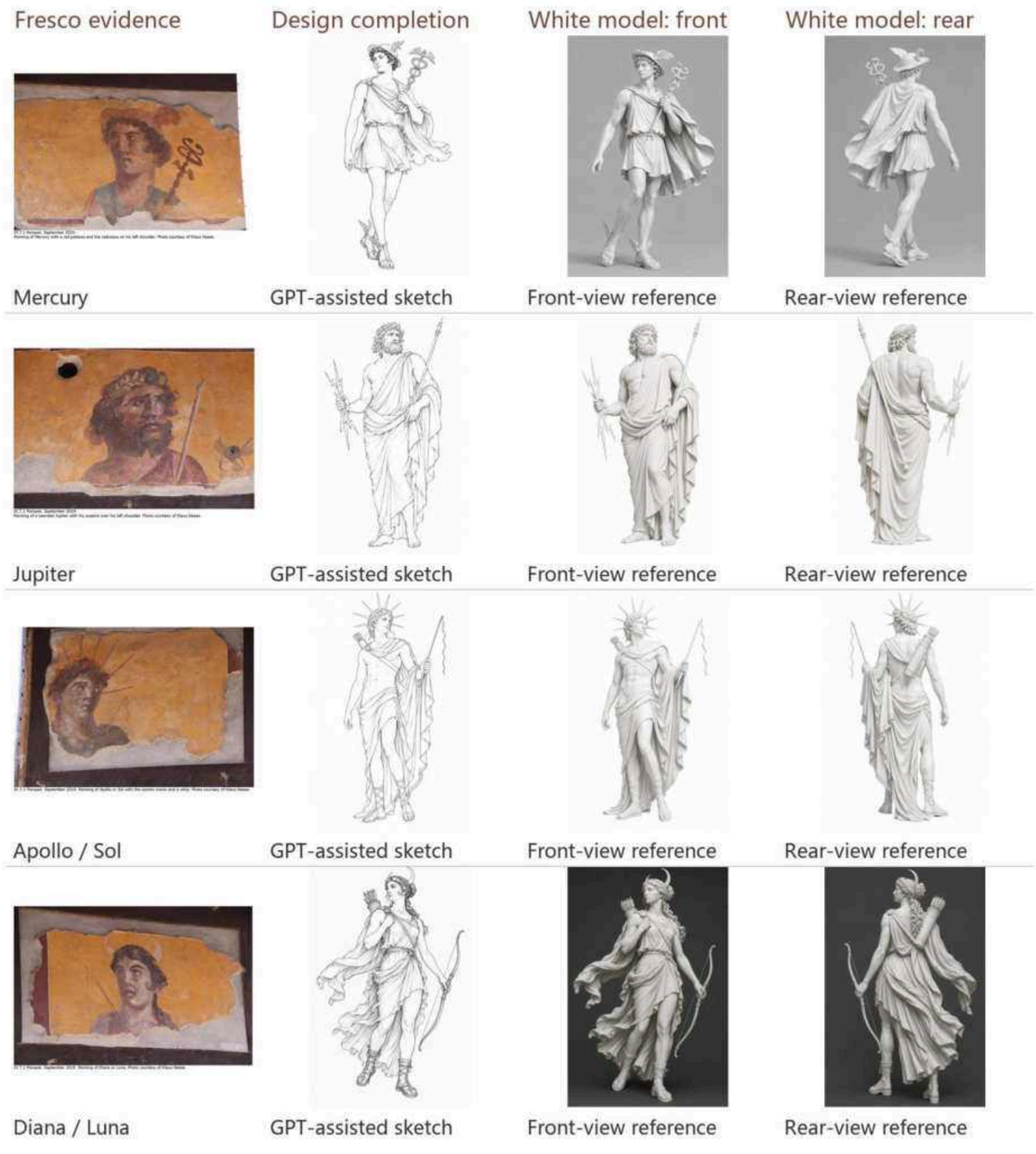



Figure 4.1-3. Fresco fragments, figure studies and white models of four deities. Fresco photographs: Klaus Heese / Pompeii in Pictures [29].





## 4.1.3 Testing form through white models

Pale white-model studies bring the figure's proportions, the volume of its drapery and the connection between hand and attribute into focus. With colour reduced, the forms that support the pose become easier to see.

I checked the proportions in these views before generating the 3D model.

The main questions were practical. Could the figure stand convincingly? Did the arm connect to the object it carried? Would the silhouette remain clear against a row of columns?

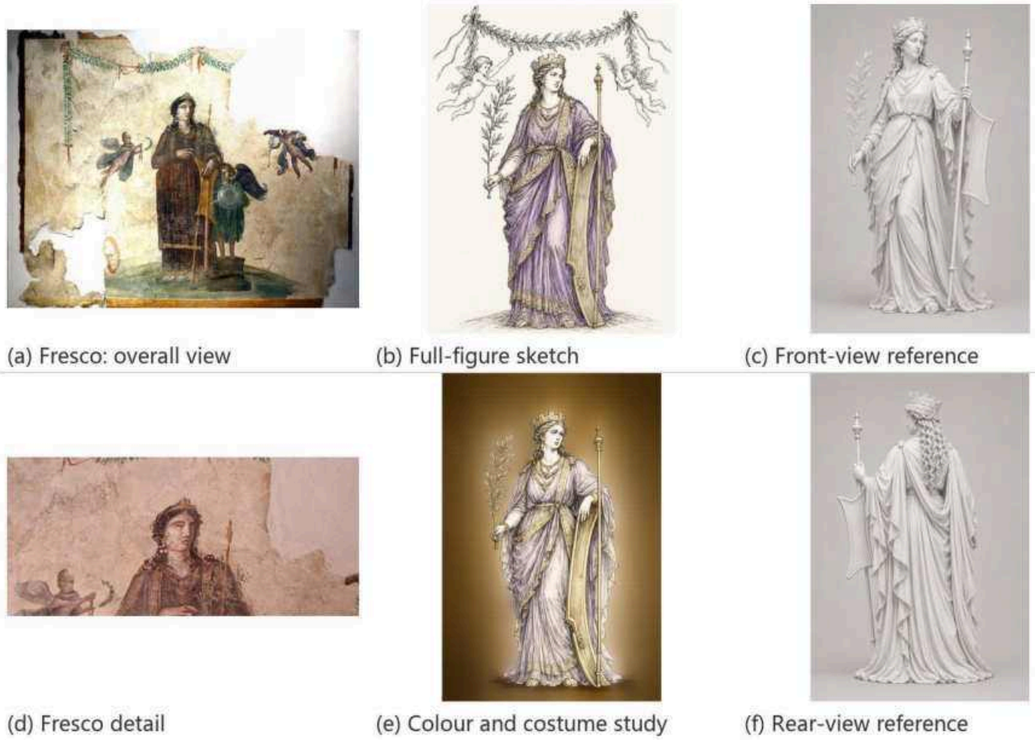
Hair, enclosed arms and hanging cloth need to make sense from both the front and the back. A held object also needs a direction that belongs to the pose.

I used the front and rear views to adjust clothing and silhouette together, developing a figure that could hold its form from different angles in the scene.

The fresco provides the figure's identifying features. Drapery, body proportions and the relation to held objects continue to develop at the intended viewing distance.

White-model views make proportion, drapery and silhouette easier to compare.

## Venus: costume, silhouette and volume



(a) Fresco: overall view
(b) Full-figure sketch
(c) Front-view reference
(d) Fresco detail
(e) Colour and costume study
(f) Rear-view reference

Figure 4.1-4. Venus across frescoes, costume sketches and white models.

Venus required a different balance. The study focused on the long robe, headdress and standing posture. I then compared the density of folds and ornament against the figure's overall silhouette.

The white-model stage made it possible to judge whether the costume gave the body a coherent form without relying on colour.

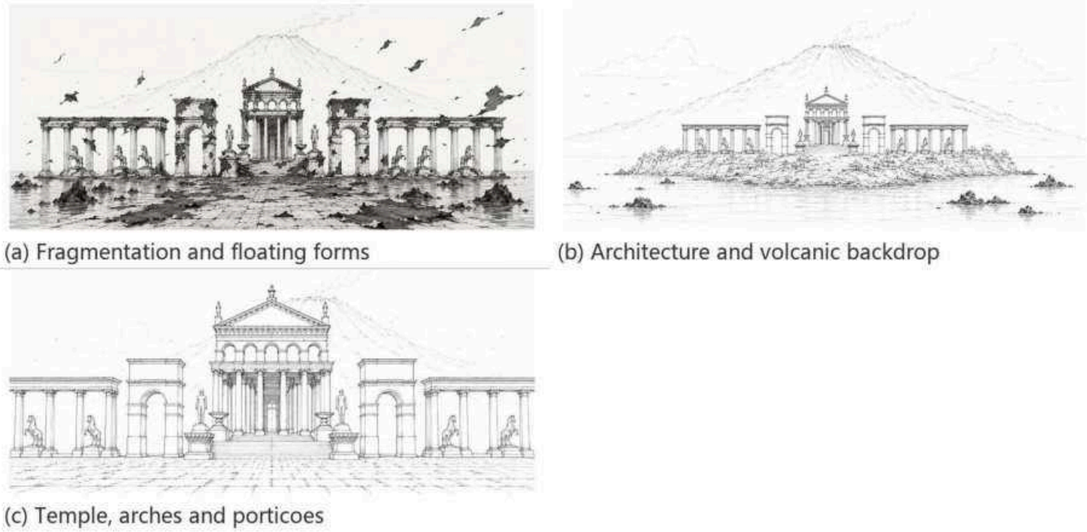
The robed Venus imagery provided an iconographic reference, including the façade image discussed in relation to Venus Pompeiana [28]. It does not establish that the completed figure occupied this invented interior.

I placed the figure at the end of the interior axis so that robe, pedestal and light could form a steady destination for the moving camera.





## 4.1.4 Reimagining the Forum



(a) Fragmentation and floating forms
(b) Architecture and volcanic backdrop
(c) Temple, arches and porticoes

Figure 4.1-5. Scene sketches of fragmentation, volcanic terrain and architectural layout.

The Forum's open centre, raised temple and repeated colonnades gave the scene a clear structure. Vesuvius placed this architecture within a larger landscape [30].

I worked through three spatial studies: one broke the architectural edge into floating pieces; another strengthened the mountain in the distance; a third clarified the temple, arches and surrounding porticoes.

These drawings tested how much a familiar classical form could change while remaining recognisable. Symmetry made the return to order readable. Displacement, suspension and changes of scale opened a less stable, dreamlike space.

The final arrangement follows the needs of the route and the view.

## 4.1.5 Building with separate forms

(a) Temple: front view

(b) Temple: side view

(c) Roof module

(d) Portico module

(e) Arch module

(f) Altar module

Figure 4.1-6. Temple, roof, portico, arch and altar studies.

I separated the architecture into modules so that the temple entrance, steps, column spacing and roof could be adjusted independently. Repetition could then be tested across the whole composition.

This also prepared the buildings for animation. A column, roof section or wall could become a fragment while still retaining a known place in the complete form.

The same parts were checked in two states, scattered across the water and assembled into a building. The original positions linked these two states.

These modules became the parts used in the later scattering, return and skin-reveal tests.





## A shared space

(a) Horse sketch

(b) Horse white-model reference

(c) Separate pedestal

(d) Branch forms

(e) Leaf forms

(f) Door-frame module

Figure 4.1-7. Horse, pedestal, branches, leaves and door-frame studies.

The horse and pedestal added scale and a second kind of silhouette beside the buildings. Branch and leaf studies introduced more flexible forms that could cross the architectural rhythm.

In the integrated scene, I checked where the temple drew the eye, how the porticoes repeated, and how the figures changed in scale across the view.

The early renders established the arrangement of buildings and figures. I then tested changes in the grass, the return of fragments and the opening skin along the same route.

The horse serves the scene's scale, rhythm and surface animation. A specific archaeological source for this model has not yet been identified.

## 4.1.6 Sources within the exterior

| Element and setting | Reference | Design choice |
|---|---|---|
| Temple, arches and porticoes | Pompeii Forum; official guide [30]; contemporary reconstruction [31]. | The central axis and repeated columns organise the route. Spacing and fragmentation are adjusted for the scene. |
| Mercury and Jupiter | Façade paintings at IX.7.1–2; recognisable attributes [28–29]. | Headwear, staff, beard and sceptre preserve identity while complete bodies and poses are developed. |
| Apollo/Sol and Diana/ Luna | Solar and lunar imagery in the same façade group [28–29]. | The paired identities give time a visual reference. Diana also becomes a close-up for the opening skin. |
| Fragments and horse | Parts cut from the complete scene models; project form studies. | Fragments return to their original transforms. The horse supplies scale and a second moving surface. |

The route determines the spacing, scale and viewing distance of the forms shown in the table.





## Sources within the interior

| Element and setting | Reference | Design choice |
|---|---|---|
| Venus and the end of the axis | Pompeian Venus imagery; project costume and white-model studies [28]. | Robe, headdress and standing pose form the interior focus. |
| Column rhythm | House of the Faun peristyle photograph, c. 1925–1949 [32]. | The Ionic profile informs volutes, fluting and a moulded base. Repeated columns frame the pool; a shorter pair marks the entrance. |
| Oil lamps and stands | Pompeii bronze lamp/stand, image record 2545b [33]; Louvre ED 2777 [34]. | The nozzle, curved handle, support tray and feet provide a structural reference. Height and placement are adapted. |
| Olive and fig | Regional evidence from Boscoreale/ Herculaneum and Oplontis [35–36]. | Plants connect architecture with cultivation and everyday life. Rapid growth and winding branches are dreamlike transformations. |
| Red grass, water and dark skin | Project colour, space and material experiments. | These elements carry time reversal through colour, reflection and revelation. |

## 4.1.7 Entering the interior

Figure 4.1-8. Ionic peristyle, House of the Faun. Scrocchi collection, c. 1925–1949 [32].

The interior continues the exterior's axis. A long pool leads from the entrance towards the figure, while paired rows of columns give the movement a measured rhythm.

The House of the Faun photograph provided a reference for the spacing of columns, the mouldings and the relationship between shaft and surrounding opening.

The model follows an Ionic vocabulary: scrolls at the capital, a fluted shaft and a moulded base. I adapted the proportions and repetition to the interior; the following column study shows the actual scene mesh.

Open space around the pool and the final figure keeps the route legible. Plants and lamps sit within this structure, close enough to affect the view without closing the central passage.





## An object reference for the lamps

Figure 4.1-9. Bronze oil lamp and freestanding stand, Archaeological Park of Pompeii. Image record 2545b. Photograph: Amedeo Benestante / Pompeii Commitment [33].

I replaced the small wall- and column-mounted cups that appeared in some early generated concepts with oil lamps carried on separate stands.

The photographed object offered a more concrete basis: a lamp body with a nozzle and curved handle, held above a slender shaft and feet.

The Louvre's bronze stand ED 2777 provides a related Pompeian comparison for the three-footed base, narrow stem and upper tray [34].

The references guided the lamp's form. I placed a pair beside the figure.

# The interior column

Figure 4.1-8a. The actual interior column, isolated from the Blender scene with its existing material.

## An Ionic profile

The House of the Faun photograph is catalogued as an Ionic peristyle [32]. The scene model uses the same vocabulary: scrolls at the capital, a fluted shaft and a moulded base [38]. Its proportions and arrangement are adapted to this interior.

### CAPITAL AND SHAFT

The paired volutes widen the narrow shaft towards the square abacus above. Below them, the vertical flutes produce alternating bands of light and shade. These changes in depth keep the column readable as the camera moves beside it.

### BASE AND THRESHOLD

Rounded mouldings step out towards a square plinth, giving the slender shaft a clear contact with the floor. A shorter pair marks the entrance, while taller versions repeat along the side and rear rows.

### REPEATING THE UNIT

The repeated profile frames the pool and measures the approach to the statue. Olive branches pass in front of and behind the shafts, so their growth can be read against a steady architectural rhythm.





# Volutes, flutes and a moulded base

Figure 4.1-8b. Volute capital, square abacus and upper shaft. Actual model detail under neutral studio lighting.

Figure 4.1-8c. Ring mouldings and square plinth at the base of the same column.

### LIGHT ON THE PROFILE

The pale surface has small variations in tone and relief. Side light catches the flutes and the rounded base, leaving the recesses darker. The main profile comes from the mesh; textures add finer surface detail.

These views retain the scene mesh and material. Isolating the model makes its profile easier to compare with the House of the Faun reference.

## 4.1.8 Developing the lamp

Figure 4.1-10. AI-assisted front, side and top references used to compare the lamp body.

I began by separating the lamp body from the stand. The shallow bowl, recessed top, central filling hole, projecting nozzle and bird-head handle could then be checked individually.

The support tray forms the transition to the slender shaft and three-footed base.

The three views made it easier to compare the nozzle's alignment, the return of the curved handle and the distinction between the filling hole and the wick opening.

These were design references. The following close-up shows the model actually placed in the Blender scene.





## The lamp in the Blender scene

Figure 4.1-11. My lamp model, shown with its wick and flame. Blender production render, October 2026.

I checked the bowl's curved wall, the rims of the openings, the turn of the bird-head handle and the layers of the support tray. A continuous highlight helped reveal whether these surfaces joined cleanly.

## Structure and proportion

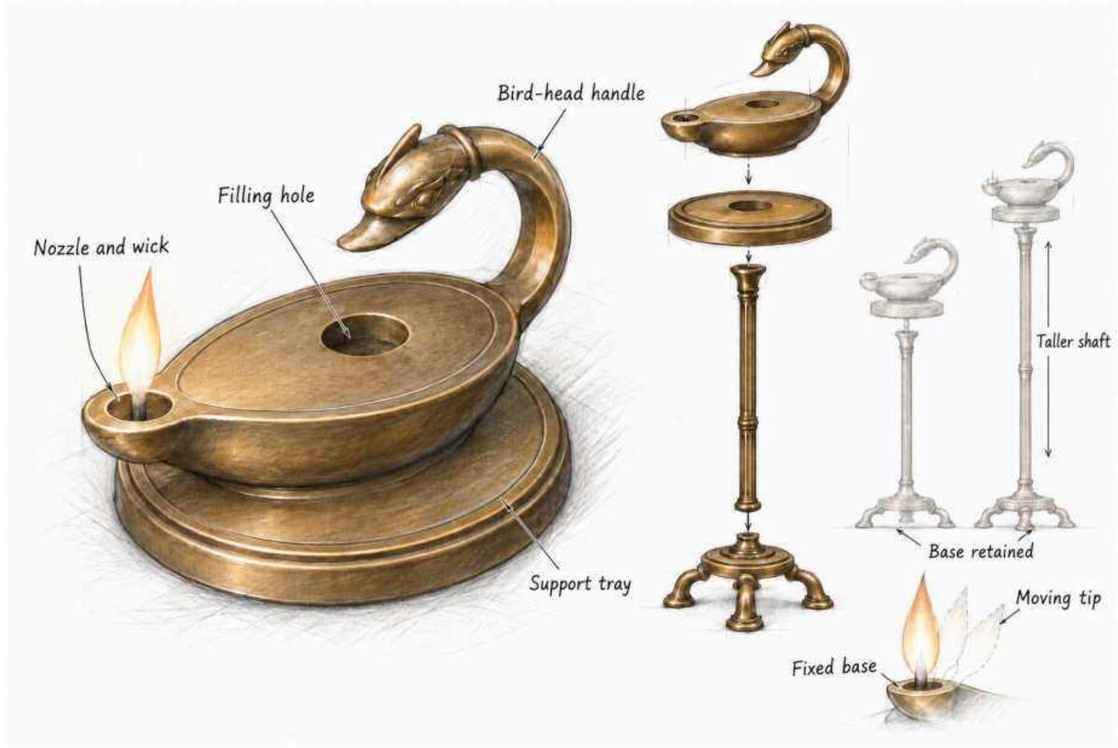



Figure 4.1-12. Retrospective process drawing: lamp parts, the extended shaft and the fixed flame base. AI-assisted drawing based on the project records.

The lamp was still too short after it entered the interior. I kept the feet on the floor and lengthened the middle shaft.

The tray, lamp, wick and flame moved upwards together, preserving their assembly and bringing the flame higher beside the figure.

The drawing gathers these decisions into one view: the object is built in layers, while the change in height is concentrated in the stem.

Retaining the base avoided changing the lamp's footprint or allowing the upper body to separate from its support.





## Metal, wick and reflected light

The historical object constrained the main outline. The warm gold colour and small surface variations were developed for the project's lighting.

Highlights describe the bowl and handle, while holes and recesses remain darker. Small variations in reflection break up an otherwise uniform polished surface.

At close range, the edge around the filling hole needed to remain distinct from the flatter top. The support tray also had to read as a separate layer beneath the lamp.

A short wick sits inside the nozzle. The flame starts at that point, so the light appears to come from the lamp's working end rather than float above the bowl.

The setup used separate controls for the visible flame and the warm point light that illuminates nearby surfaces. The visible flame could stay small while the point light illuminated the metal and surrounding surfaces.

The separation was especially useful when reducing reflections on the water: the lamp could stay readable without creating a broad, distracting glow.

## Lighting the interior

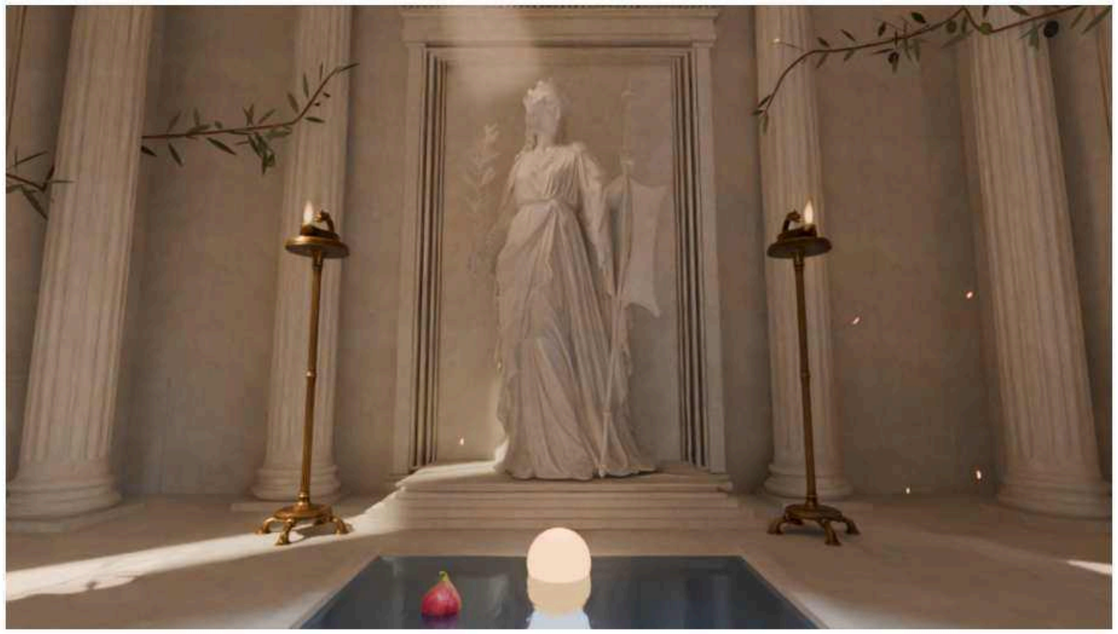


Figure 4.1-13. The extended lamp stands beside the interior figure. Actual Blender scene render.

I placed freestanding lamps close to the colonnade and on either side of the final figure. Their repeated verticals reinforce the destination while leaving the pool and the ball's path open.

The scene view was the deciding test for height. The flame had to belong to the lamp, but also remain visible beside the much larger figure and columns.

The exterior sparks retain the memory of disaster. Inside, fire is held within an object people use. Its small movement, together with leaves, fruit and daylight, returns a sense of activity to the architecture.

The lamp retains the reference object's outline while its height, position and light are adjusted to the room.





## Small movements

The six flames do not sway together. Each follows an offset rhythm and moves slightly in two directions, with small changes in length and brightness.

The bottom stays anchored to the wick while the tip moves. This keeps the flame attached to the nozzle and avoids the impression of a rigid object rocking on its base.

At the scale of the interior view, these changes need to be restrained. Their role is to make the room feel active without competing with the ball and the growing branches.

The skylight fabric introduces a slower movement. Continuous deformation lets it suggest air passing through the opening, while its transparency and fine weave preserve a sense of thin material.

Later in the shot, the fabric moves away from the area around the figure. The movement remains visible, but the destination is no longer covered at the point where the camera settles.

Leaves, fire and fabric use different rhythms. Together they make light and air perceptible within the otherwise heavy stone setting.

# 4.2 Reverse transformation

The visitor advances through the scene while the city is read backwards in time.

## 4.2.1 Red ground, returning green

Figure 4.2-1. Actual production tests: red ground and the persistent green passage.

Red-brown grass and porous red stones recall volcanic burial. The stones stand for volcanic rock before the complete architecture appears.

Volcanic burial is the starting point for the animation's reversal of time.

As the ball passes, green begins near its route, spreads sideways and stays behind. Each stretch of movement returns another part of the environment.

The green refers to the landscape before the eruption, including cultivated fields, orchards and gardens. This part of the route introduces vegetation before the complete buildings appear.

I kept the same grass clumps and distribution in both states. The change occurs through the material, so the viewer can follow a continuous ground instead of seeing one landscape replaced by another.





## How the green passage develops

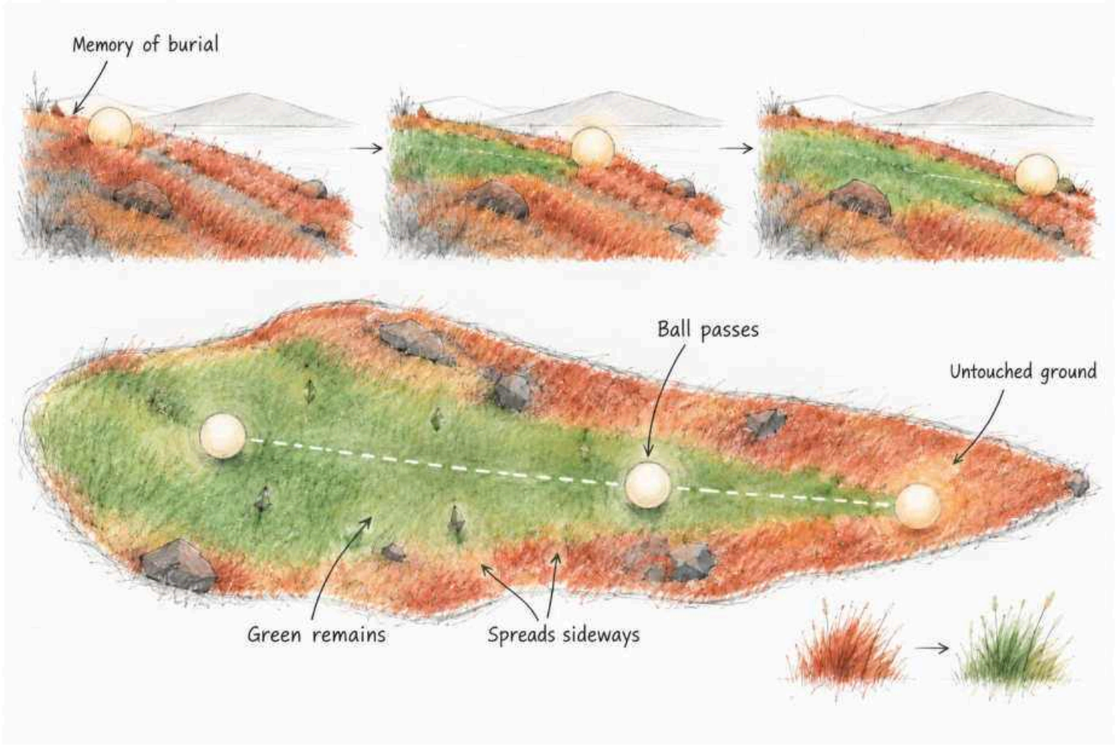



Figure 4.2-2. Retrospective storyboard of the accumulated green passage. AI-assisted process drawing.

The material mixes red and green through a mask driven by position and time. Sampled points along the ball's route determine where it has passed and how long each area has been recovering.

Earlier sections keep widening, so the width records how long it has been since the ball passed.

A softened, slightly noisy edge prevents a perfectly even stripe. Low grass, taller clumps, scattered seed heads and flowers also break up the surface.

The storyboard separates the untouched red ground, the first green area and the wider passage left behind, making the timing visible in a single sheet.

## 4.2.2 Water as a continuous route

Figure 4.2-3. Production tests of scattered architectural parts and the assembled building.

The ball continues from the grass onto reflective water. Reflections help show where fragments meet and leave the surface along this uninterrupted route.

I kept the ball close to the surface and replaced an earlier second island and landing sequence with a direct approach to the building steps. It slows gradually before stopping.

Dynamic Paint supplies the ball's contact ripples. A trailing disturbance spreads behind it and fades, while a small warm reflection helps the viewer follow its movement.

Lowering the water's overall brightness preserved the reflections of the ball and the building. In the imagined Forum, water links the grass to the steps and reflects the fragments as they rise.





## 4.2.3 From fragments to architecture

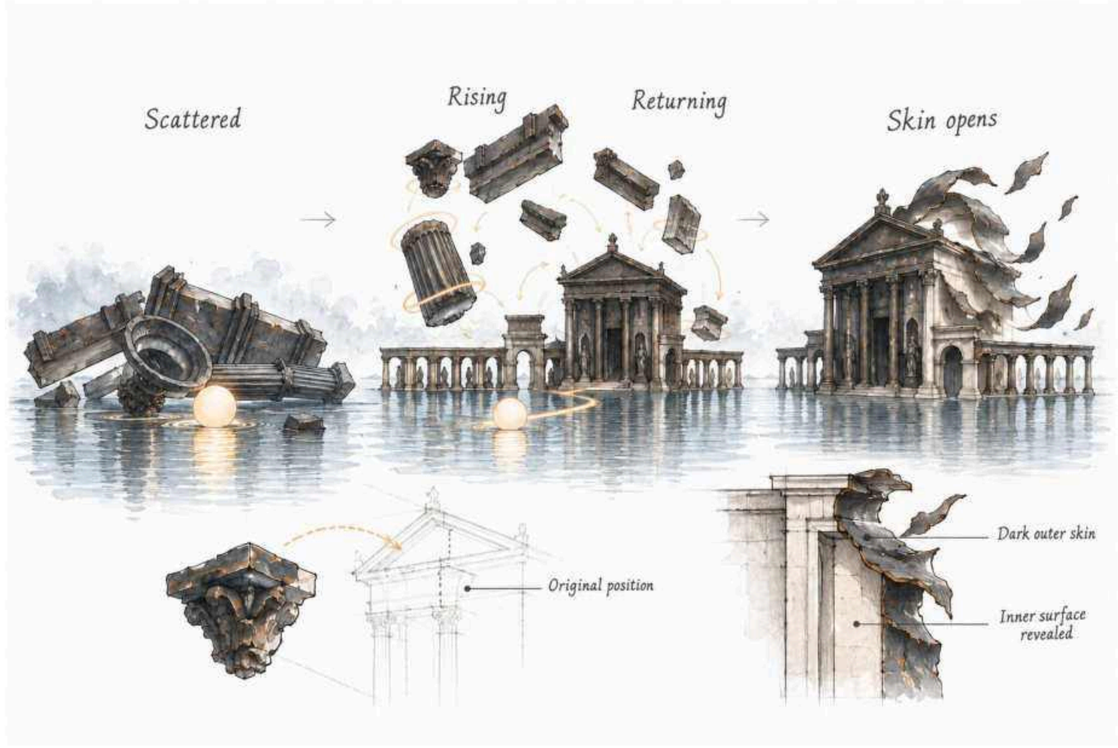



Figure 4.2-4. Scattering, rising, returning and opening the outer skin. Retrospective AI-assisted process drawing.

I cut columns, walls, roofs, arches, bases and figure-related parts from the complete models, retaining each fragment's original transform.

The scattered state and the assembled state therefore share the same geometry. A recognisable edge can be followed back to its place in the building.

Reassembly restores the structural relationships first. The surface then opens while the completed form remains stable.

The fragments settle into place before the outer layer begins to pull away.

## Returning to the original position

I did not distribute the fragments evenly across the water. I adjusted their depth, sideways displacement, orientation and distance from the camera while leaving a passage for the ball.

Some pieces remain close to their original ruined positions. Others begin with small movements, rise and rotate, then ease towards their saved transforms at different times.

The final alignment needed more attention than the initial scatter. Surface continuity and shadows at the joints had to settle so that the viewer could recognise a complete building before the next transformation.

Each part also needed a visible connection to the water. The animation uses the part's scattered position and the moment it leaves the surface to generate an expanding ring at the contact point.

The ring remains centred on the original contact. It widens, lowers and fades using the water material, rather than following the fragment upwards.

These fragment ripples use Geometry Nodes so their timing and radius can be adjusted individually. They work alongside the ball's Dynamic Paint contact waves.

Scatter → lift → rotate → settle → pause → reveal





## The pause before the surface opens

Early tests moved too quickly from the ball's arrival to tearing. I delayed the whole revelation sequence so that the ball could stop at the steps and the assembled building could be seen as a stable form.

The pause lets the complete outline remain visible before the skin starts to tear.

The outer layer withdraws as the inner architecture appears. Their shared timing makes the change in surface continuous, while the building’s outline gives the viewer a steady form to follow.

The sequence places two directions of time beside one another. Architectural parts move back towards a whole, while plants later follow the familiar order of growth.

This difference makes the return of history visible without reversing every element's motion.

The sequence reveals the setting in stages: green ground, an assembled building, exposed surfaces and signs of activity inside.

The fragments settle into place before the covering starts to open.

## 4.3 Growth and spreading

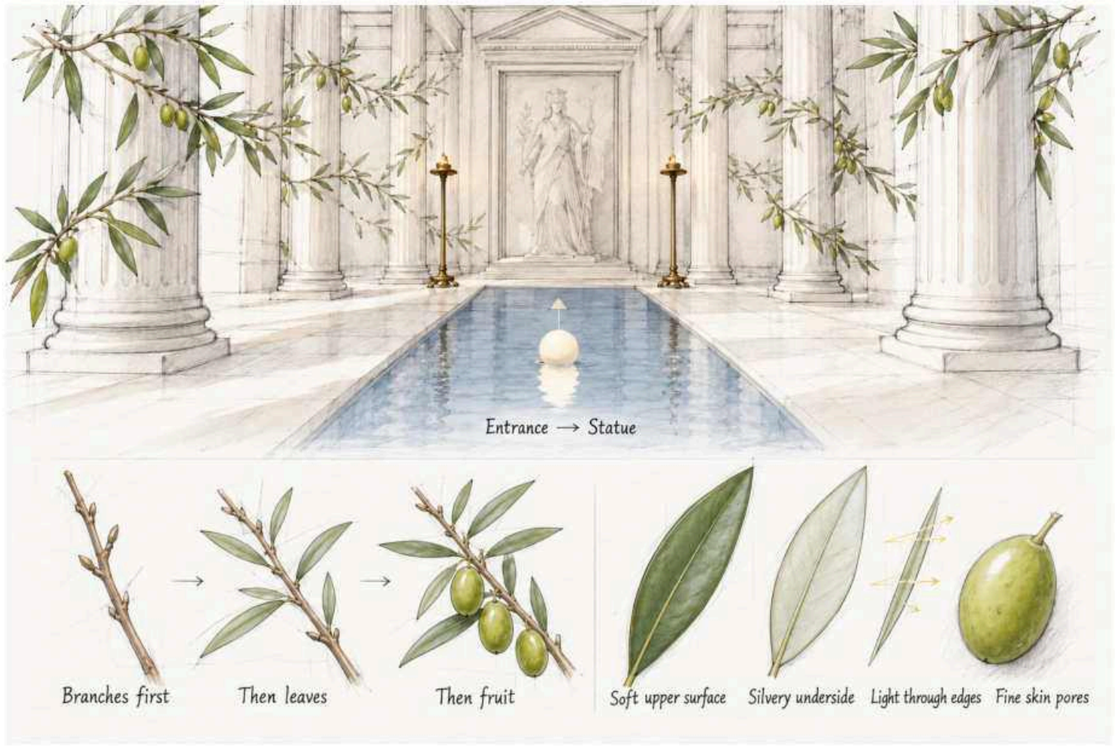



Figure 4.3-1. Interior direction, branch–leaf–fruit sequence and surface details. Retrospective AI-assisted drawing.

Olive and fig connect the interior to the cultivated landscapes and everyday life of the Vesuvian region [35–36].

The Boscoreale Antiquarium describes botanical and food evidence, including fig specimens from Herculaneum. The account of Oplontis Villa A places olive trees within a garden of plants, architecture and sculpture.

I chose the olive's narrow leaves, silvery underside and oval fruit, with the softer, ridged fig as a contrast.

These regional sources support the plant choices. The flexible branches winding through the columns and the compressed speed of growth are part of the dream.





### 4.3.1 Branches, then leaves, then fruit

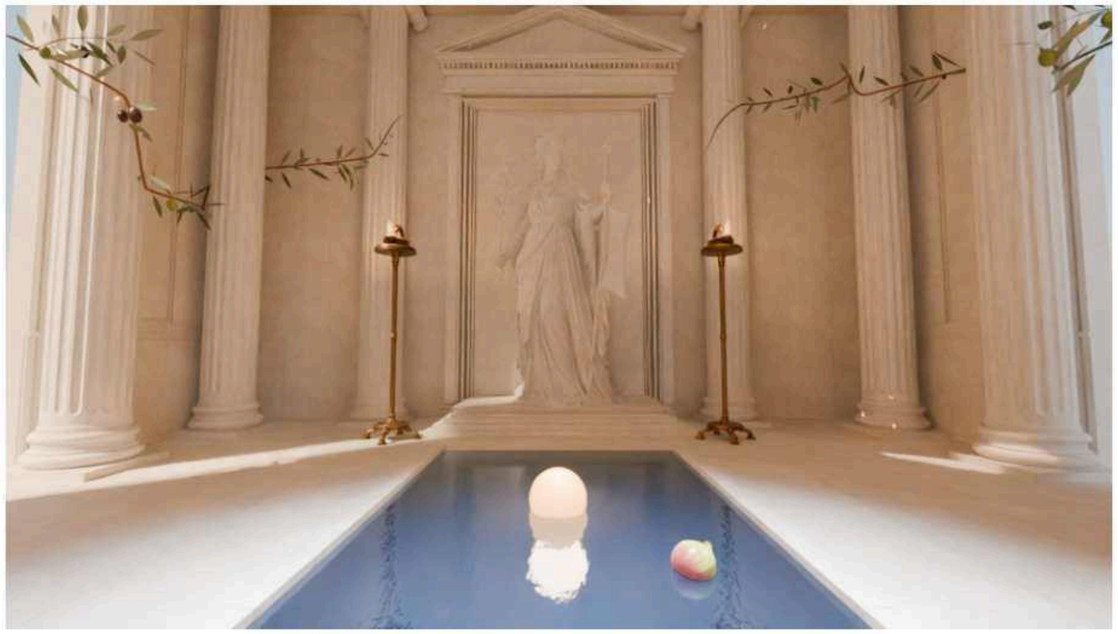


Figure 4.3-2. Interior test render: branches, pool, lamps and the figure at the end of the route.

The main branches extend from the entrance towards the statue. I adjusted their curves and leaf spacing so that they pass in front of and behind the columns, while leaving the central pool and the final figure relatively open.

The animation begins with the main branch, then the smaller branches, leaves and fruit.

Growth follows distance along the actual branch curve. A smaller branch appears after the main branch reaches its junction; a leaf grows from its attachment point, extending before it opens.

I used the supporting branches to place the leaves and fruit. This reduced floating leaves and fruit appearing before the wood beneath them.

## 4.3.2 Making a leaf respond to light

Figure 4.3-3. Olive-leaf material close-up. Actual Blender test render.

The first leaf surface was too coarse. Strong granularity broke the highlight into small, noisy patches and made the leaf feel thick.

I reduced the larger mottling and rough bumps, refined the veins and controlled the colour and roughness of the two sides separately.

The upper surface has a softer green sheen; the underside is paler and more silvery.

Thin-surface scattering lets light through the edge, while the veins retain more weight. The leaf remains a solid thin surface rather than a transparent cut-out.





## Fruit skin and the pace of growth

Figure 4.3-4. Fig and olive material tests. Actual Blender renders.

The fig material retains its colour and normal maps, with additional fine pores, longitudinal microtexture and varying roughness.

The olive uses a smoother skin, small pores and subtle colour differences between stages of ripeness. Both retain a waxy highlight, interrupted enough to avoid a painted-plastic finish.

Fine texture and roughness produce different reflections on leaves, fruit and stone.

I also brought forward the growth near the entrance so that the change is already visible when the ball enters. Further growth continues towards the statue, sharing the direction of the camera.

## 4.4 Opening the dark skin

Figure 4.4-1. Diana close-up: the dark skin curls away from the inner figure. Actual Blender production test.

After the architecture returns, its dark outer skin tears, breaks into particles and disperses, revealing lighter buildings and figures beneath.

A separate cloth layer covers the architecture and figures. The tearing simulation lets it curl and pull away along their contours before dispersing.

The dark skin evokes volcanic burial. It hides the buildings until it tears away.

The light inner surface is also a choice of form and lighting. Pompeii's surviving paintings and coloured decoration remain an important reminder of the city's colour.





## Before and after the reveal

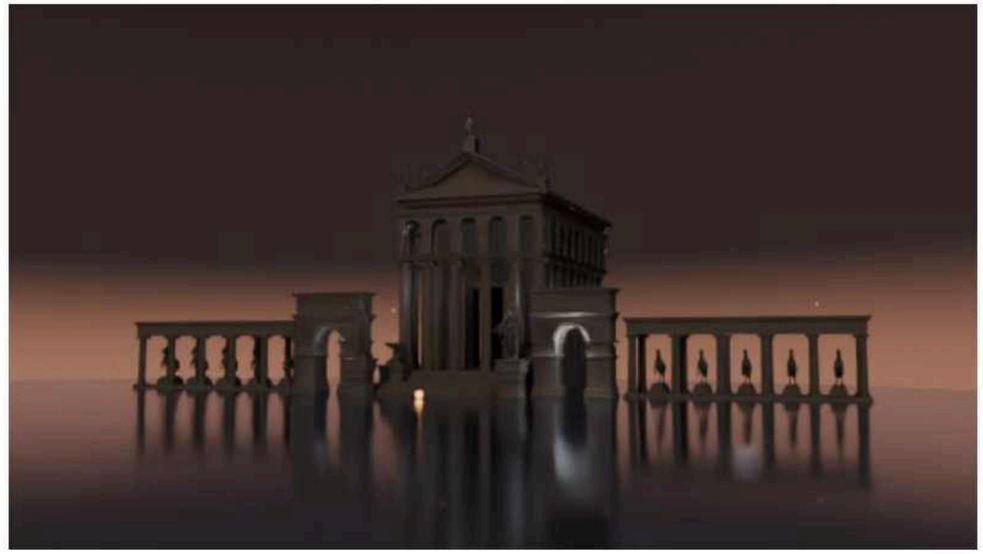


Figure 4.4-2a. Dark skin still covers the reassembled architecture. Blender render.

Figure 4.4-2b. The skin withdraws and the lighter architecture appears. Blender render.

The paired views show the change in surface and atmosphere, including the sky and light. The Diana close-up brings the curling and separation of the skin into closer view.

## 4.4.1 Texture scale across the building

A dark, rough stone surface gives the outer layer and the architecture before revelation a heavy, covered appearance. Warmer patches echo the light around the sphere.

Colour supplies the broad value and mottling. Roughness controls the width and strength of highlights. Normal and bump detail make smaller changes in the surface visible under side light.

The complementary fragment boundaries and the curling skin are actual geometry and animation. Surface maps can describe texture, but they cannot replace these changes in shape.

The same texture initially produced different grain sizes on columns, figures, bases and roofs, weakening their sense of belonging to one building.

I added missing UVs and adjusted texture density to bring those surfaces to a common scale.

A similar grain size across fragments and the complete architecture helps the material stay continuous through reassembly.

### Colour · Roughness · Normal detail · Geometry

The comparison uses the same lighting for the colour, roughness, normal and bump maps.





## 4.4.2 Reflection, transparency and air

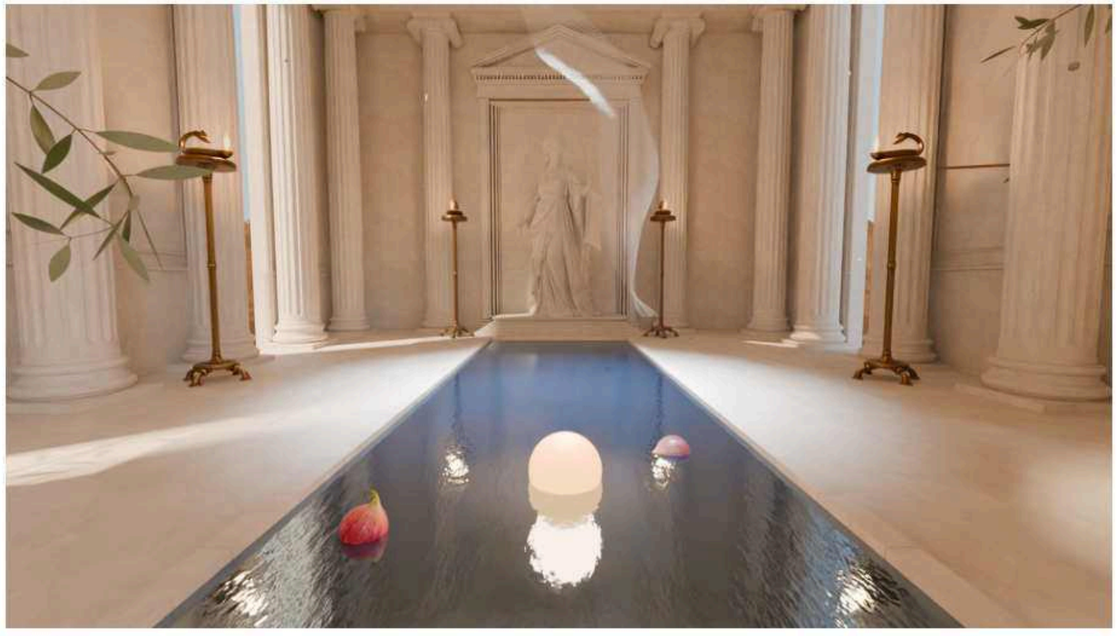


Figure 4.4-3. Interior skylight and gauze material check. Actual Blender test render.

I darkened the water while retaining a warm area close to the ball. Strong lamp reflections were reduced so the water could support movement and reflections without dominating the composition.

The ball's glow was also lowered to preserve shading across its surface. Atmosphere and brightness could then be tuned to each shot.

Moon Rock 01 and 03 from Poly Haven supplied irregular scanned forms for the grass and shore. I adjusted them towards charcoal grey to connect them visually to the landscape; the asset names do not identify Pompeian stone [37].

Gauze uses transparency, fine weave and continuous movement to show its thinness and the air passing through the opening. Leaves use thin-surface scattering instead.

A distant haze separates mountains, sky and water, but its density is kept low enough for the foreground action to remain clear. Warm interior daylight marks a gentler stage after the red ground and dark covering.

## 4.5 Organising the view

The camera moves through three scales. The landscape begins with grass, rocks, mist and the ball. The exterior opens from fragments to a complete architectural view, then returns to the figure and its surface. The interior follows the pool towards a final destination.

The opening focus begins on nearby grass, then moves to the ball as it reaches the rise. This shifts attention from the ground's texture to the route through it.

A low following view gives fragments a scale close to the body. Side views show them leaving the water; a wider view makes the completed relationships visible.

During the revelation, the sequence includes closer views of Diana and the horse before returning to the full building. Diana is framed from the knees upwards, slightly to the side. The horse view focuses on one of the existing horses in the scene.

Explicit camera targets keep the figures and the moving skin readable. The return to the wide shot reconnects these local changes to the whole space.

Inside, the camera follows the ball from the entrance along the pool and gradually slows near the statue. Branch growth shares this direction. I moved the slanting light onto the front folds of the robe so that light and movement arrive at the same place.

Landscape → fragments → whole building → surface detail → inhabited interior





## 4.6 Production and iteration

| Observed in the render | Adjustment | Result checked |
|---|---|---|
| Lamp shafts felt too short beside the figure. | Retain the feet, lengthen the stem and lift the upper assembly. | A clearer relationship between flame, tray, figure and columns. |
| Leaf highlights looked noisy and grainy. | Reduce coarse bump; refine veins; separate upper and lower surfaces. | A smoother sheen with light visible at the thinner edges. |
| The same rock material changed scale between objects. | Add missing UVs and align texture density. | Columns, bases, roofs and fragments read at a more consistent scale. |
| Tearing began as soon as the ball arrived. | Delay the cache and synchronise skin fading and inner reveal. | A pause to read the complete building before the surface opens. |
| The horse close-up faced the wrong direction or cut through the interior. | Set a clear look target and check it against the wide shot. | A readable side view that returns coherently to the full composition. |

## 4.6.1 From images to scenes

The work moved from source images and figure completion to white models, architectural modules, landscape, vegetation, materials, movement and cameras. Generated images remained proposals that had to be tested in the models and actual renders.

The landscape, exterior architecture and interior each need a different balance of light and movement. Darker exterior light picks out the fragments and opening surfaces, while side light inside falls on leaves, lamps and the statue.

The water and sphere connect these passages, allowing changes of setting to follow one viewing route.

Green remains behind the sphere, ripples mark fragments leaving the water, and scattered parts recover a complete form.

The inner architecture appears as its covering withdraws. Leaves unfold after the branch reaches them, so each movement grows from something already visible.

Inside, the slanting light and the movement of the gauze leave the statue at the end of the route, giving viewers a place to pause.

## 4.6.2 Current outcome and the VR stage

The Pompeii animation moves from changing grass, through reassembly and surface revelation, into an interior with branches, fruit and lamplight. Materials, movement and camera direction follow this route, bringing the view from a distant landscape towards architectural and everyday details.





## A return to relationships

The scene began as a study of how a ruined building might come back together. The returning grass and the movement of branches and lamplight extended that idea to the landscape and interior.

The returning green makes nature present before architecture. Reassembly shows how separate parts depend on one another. The opening skin turns covering and revelation into a visible surface.

Moving branches, fruit, flames and fabric share the view with the still columns and statue. Their different movements bring activity back into the interior.

In an immersive space, viewers may look away or want to linger beside a column or the water. The animation needs to allow time for that attention to return to the unfolding scene.

Viewing trials could show whether the reassembly and opening skin are clear, where the pace should slow, and how the scale and movement feel in a headset.

## 4.6.3 Research and production records

References [28–38] connect the scene's figures, architectural forms, lamps and plants to their sources. Contemporary reconstruction imagery informs the spatial arrangement, while the rock scans provide material for the landscape. The image credits record how historical references, generated studies and the author's renders contribute to each figure.

## 4.7 Sicily: Temple of Concordia

The Temple of Concordia was built in the later fifth century BCE on the ridge at Agrigento. Its later use as a church helped preserve it. Section 3.4 places the temple among roads, slopes, fields and orchards, tracing its continuity through changing uses and surroundings [2, 21–23].

From a distance, the colonnade stands out. Local biocalcarenite absorbs water, and cultivation continues around the site [3, 23]. The animation begins close to the stone and ground so viewers have time to notice them before the temple comes into view.

A column flute became the starting point. This recognisable Doric feature becomes a small valley for the sphere to cross. Following the groove lets viewers compare stone, water and plants, with the sphere offering a familiar guide through the changes.

Water and rain bring weather's effects on stone into view. Wheat recalls Sicily's agriculture and the cultivation of land around the ruins. Windblown grains draw attention to loosening surfaces and moving material. The animation enlarges these processes into an imagined passage, with grains gathering into a temple as its turning point.

The chosen sequence moves from the flute through water, earth and wheat, then returns to stone. The surface disperses into grains that gather into a column, a colonnade and the temple. The final wide shot reveals the ridge, slopes and field together.

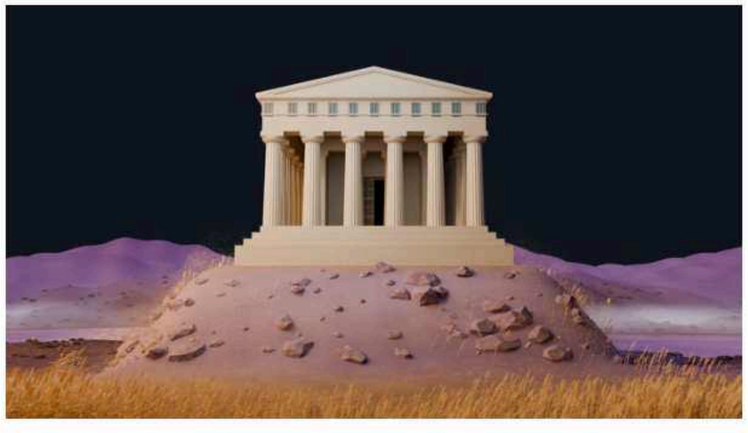


Figure 4.7-1. The temple, ridge and wheat in the closing shot. Still from the author's animation.





### 4.7.1 From research to sketches

The model retains the temple's stepped platform, fluted Doric colonnade and pediment [21–22]. Figure 4.7-1a pairs a surviving column at Concordia with the one reassembled in the animation. Both show vertical flutes below an outward-curving capital and its square top slab. The flute enlarged into a route for the sphere at the start returns as part of the complete column.

Figure 4.7-1a. Concordia's Doric capital and shaft (left) and the column reassembled in the author's animation (right). Photo by Zde, Wikimedia Commons, CC BY-SA 4.0 (cropped).

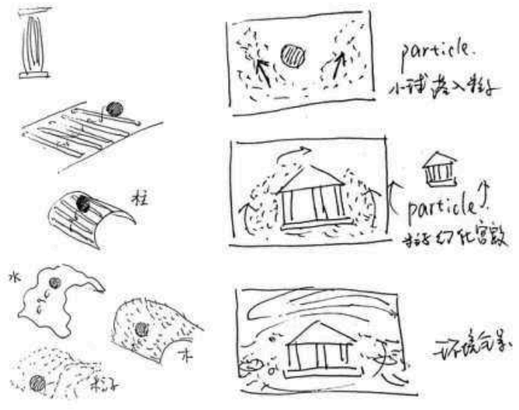



Figure 4.7-2. Author's sketches, cleaned from the original photographs. Left, column grooves and material changes. Right, particle movement and assembly.

The first drawing separates the opening into studies of the column, the sphere and the changing surface.

The second traces how loose grains might move around the sphere and gather around the building. I used the two sheets to work out a change of scale, from a narrow groove the sphere can travel along to a full temple that can come into view in stages. Early ideas tested enlarged wheat and other dreamlike changes. The rendered sequence keeps the wheat at the scale of a field and uses the sphere and camera to make it feel expansive.

## 4.7.2 Rain, water and exposed ground

Figure 4.7-3. Water first crosses the stone grooves, then fills the view with ripples.

The stone study in Section 3.4 describes the local biocalcarenite as porous and able to draw in water by capillary action [3]. Rain and runoff bring attention to how weather affects the stone surface. The sequence begins with pale stone. Shallow relief and roughness keep its grooves visible at close range.

A darker, reflective layer crosses the grooves as the sphere advances. Rain falls from above and sends rings across the surface, while water already on the ground continues to flow.

Falling rain and flowing water remain visible as distinct movements. Blue reflections then give way to brown earth and a narrow gold edge. Relief and colour change beneath the same moving sphere as the camera continues along its route.

The shift from stone into water is a dream transition. It brings a gradual process into the time of one shot and leads into the exposed earth that follows.





## 4.7.3 Wheat and the guide sphere

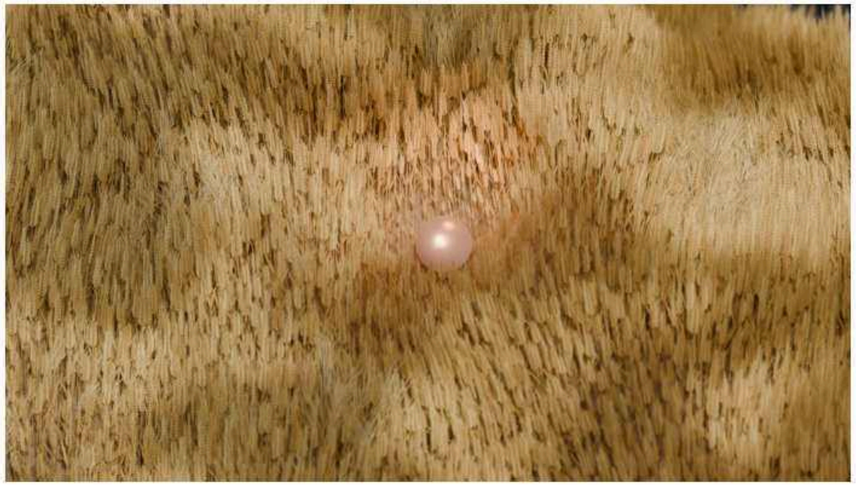


Figure 4.7-4. The guide sphere among the wheat heads.

Figure 4.7-5. Blender lighting test, before (left) and after (right) lighting the sphere.

Wheat carries the route from exposed ground into a cultivated landscape. The park's landscape account records cereal cultivation in the Valley of the Temples [23], and finds from Greek Selinunte document wheat in ancient Sicily. I used wheat as a crop grounded in that wider context and designed the field's position and density for the shot.

The camera remains low enough for the ears to fill the foreground.

When the ball first entered the dense wheat, its colour was too close to the surrounding gold. A light that follows the ball was adjusted for this passage and linked to the ball alone. Comparing the same test frame before and after shows a clearer highlight on the sphere. That small reflection lets the viewer pick up its movement again without lifting the brightness of the whole field.

## 4.7.4 Windblown grains

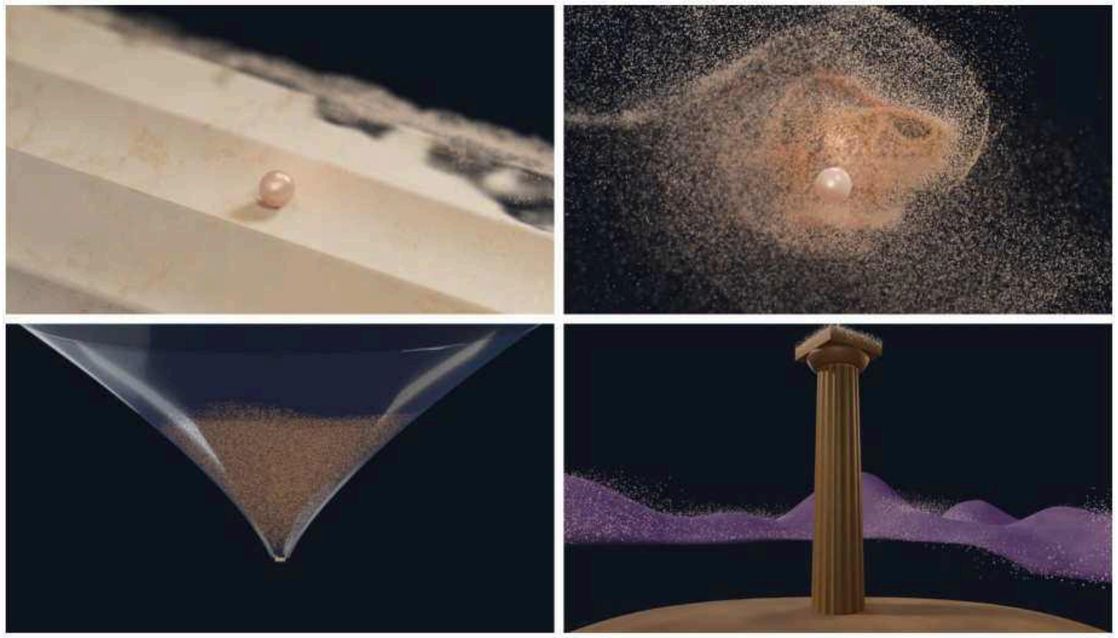


Figure 4.7-6. Upper left, stone loosens. Upper right, grains whirl. Lower left, the translucent funnel symbolises time. Lower right, a Doric column forms.

After the wheat, the path comes back to the pale stone. The surface begins to loosen at the edges, and fine grains cross the frame and curl around the sphere. The motion draws on windblown sand. A few grains detach first, so the viewer can follow their movement before the denser turn encloses the sphere. The second hand drawing was used to work out the curve and where the ball would remain visible within it.

The translucent funnel symbolises time. Its narrowing shape recalls an hourglass and brings dispersed grains back together. As the particles gather into a Doric column, the weathering seen just before seems to run backwards. The flutes and capital recall the opening close-up, taking the view from a grain to a column and then to the full row.





## 4.7.5 Reassembling the temple

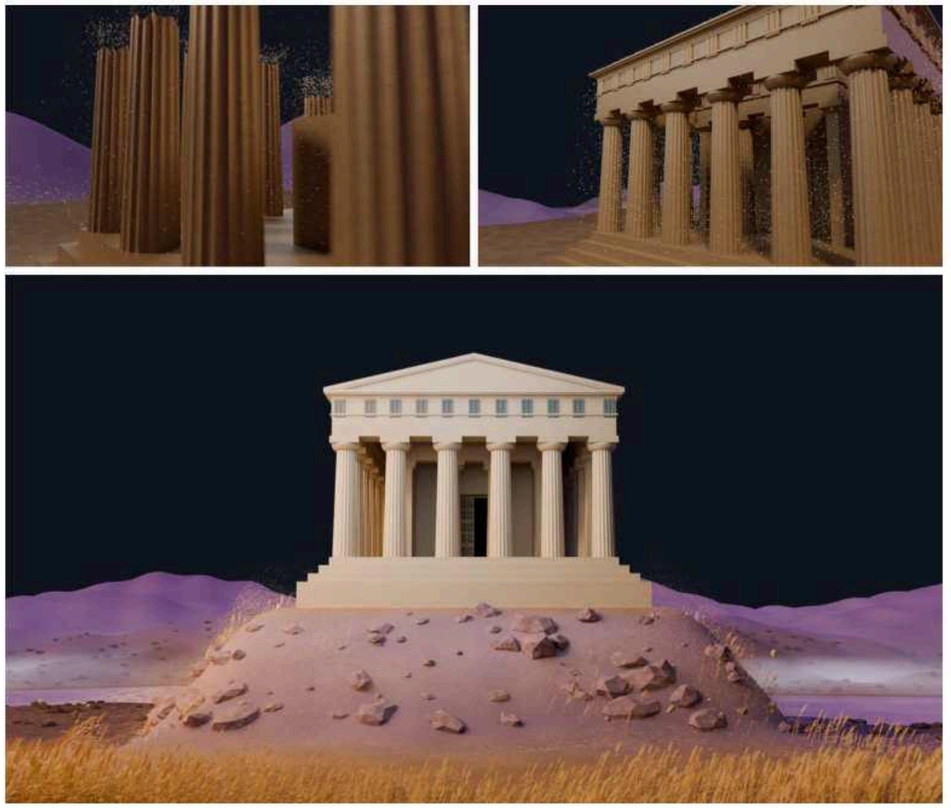


Figure 4.7-7. Columns, upper structure and the temple on the ridge.

Reassembly begins with a single column and the stepped ground beneath it. The camera passes the emerging row before revealing the entablature and pediment. The gathering grains answer the surface's earlier dispersal.

The closing view places the finished form back on a raised ridge. The frame leaves space for the uneven slope, scattered stones, wheat at the bottom edge and the distant terrain. The temple keeps the form established by the site research, while the colours and positions of the landscape belong to the dream scene. The last pullback lets the relationship between the building and its surroundings come into view after the material close-ups.

## Comparing the site and the reconstructed temple

Figure 4.7-8. The standing east façade (left) and the author's colonnade assembly (right). Photograph by Zde, Wikimedia Commons, CC BY-SA 4.0, resized for layout.

The photograph shows the standing east façade at an angle. Grooves repeat along the shafts, the capitals meet the horizontal stonework, and the columns rise from a stepped platform. These details give the new column row its order in the animation. A groove first appears at the scale of the sphere. The same detail returns on the full column, then becomes part of the rhythm of the row.

The photographed stone has irregular surfaces. The model uses continuous edges and a more even surface so the shafts, capitals and upper structure can be read as they come together. The site evidence establishes the main architectural features. The movement of the particles, the colour of the ground and the timing of the assembly come from the animation design. In the final wide view, the ridge and the wheat place the building back in its surroundings.





## Keeping the sphere visible through material changes

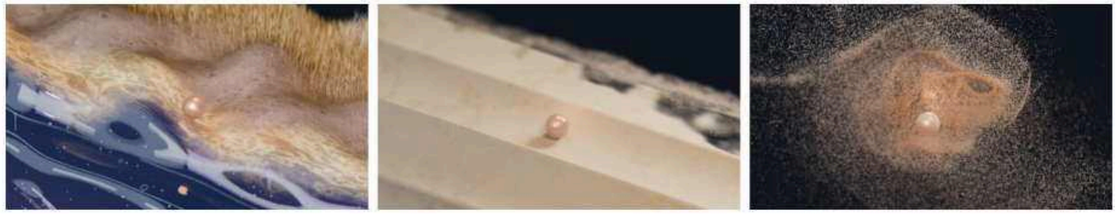


Figure 4.7-9. The sphere crosses water and earth, stone and particles. Stills from the author's animation.

The edge where water meets exposed earth and wheat makes the sphere's direction visible. As the pale stone returns, its grooves continue that route. Fine grains then curl around the sphere, and a glint on its surface keeps it visible through the cloud.

The wheat lighting test addressed the same problem in a different material. The first render let the gold of the wheat and sphere merge. Lighting the sphere separately recovered a small reflection while retaining the field's overall brightness. Later, the funnel shifts attention from the moving sphere to the gathering grains. A single column gives the viewer a form to recognise before the camera reveals the whole row. A future viewing test can ask where the route was easy to follow and where the particles obscured it.

## Animation and viewing pace

Figure 4.7-10. The close-range route through stone, water, exposed earth, wheat, returning stone and windblown grains.

The animation follows the sphere close to the ground, across stone, water and wheat. As the temple comes back together, attention shifts to the particles gathering into a single column and then to the full colonnade. The last wide shot shows where the temple sits on the ridge.

In an immersive space, viewers may linger over the water or the wheat. They may also be looking elsewhere as particles gather into a column. A viewing trial could show whether they have time to notice those changes and where a slower transition or a longer pause might help.





## Sicily sources and image credits

Site and architecture. Section 3.4 and references [2, 21–22] provide the site context, ridge and Doric form. The finished temple is a visual model developed for the animation.

Stone and landscape. Reference [3] informs the choice to show water on porous stone. Reference [23] and the archaeobotanical study at Selinunte inform the choice of wheat.

The movement of windblown sand was used as a reference for the particle choreography.

Sketches and renders. Figure 4.7-1a pairs Zde's photograph of the Doric column with a still from the author's film. Figure 4.7-2 shows two original sketches. Figure 4.7-5 compares two Blender lighting passes. In Figure 4.7-8, the photograph on the left is by Zde, and the still on the right is from the author's film. The other Sicily animation plates come from the author's finished animation.

UNESCO World Heritage Centre · Archaeological Area of Agrigento [2]

Taglieri et al. · Porous biocalcarenite from Agrigento [3]

Parco Valle dei Templi · Archaeological area [21]

Parco Valle dei Templi · Temple of Concordia [22]

Parco Valle dei Templi · Landscape of the Valley [23]

Stika, Heiss and Zach · Archaeobotanical finds from Selinunte

USGS · Eolian processes

## 4.8 Yingxian Wooden Pagoda · A Night Around the Pagoda

The Yingxian Wooden Pagoda stands in Fogong Temple in Shanxi. Formally named the Sakyamuni Pagoda, it was built in 1056 during the Liao dynasty. Its octagonal structure has five visible storeys and four hidden levels. Inner and outer rings of columns, beams and bracket sets connect individual timbers into a tall Buddhist space [4].

The 1933 survey photograph places the raised platform, layered roofs and courtyard in one view. These records drew my attention from the tower's silhouette to the wood that supports it, and to the change in scale when someone looks up from its base. The scene starts low and close to the ground, giving viewers time to notice the setting and the timber before they see the whole tower.

Figure 4.8-6. South view of the pagoda, photographed during the Society for the Study of Chinese Architecture's 1933 survey. Reproduced in a 2025 Sohu article introducing the reissue of Chen Mingda's Yingxian Wooden Pagoda. No open licence stated. Rights remain with the respective holders.





## 4.8.1 The setting and the timber

Section 3.5 discusses the pagoda's inner and outer column rings, hidden levels and joints. Here, the research turns to details a viewer can recognise on screen. From the top of the tower, Liang Sicheng recorded the Sanggan River plain to the north and the Yanmen mountains to the south [Y1]. A low wall, open ground and distant ridges establish the setting. The courtyard is an imagined arrangement. Later surveys and present-day vegetation provide regional context, while the Liao temple planting remains undocumented in the sources used here.

Figure 4.8-2. Front render of the author's pagoda model, showing the octagonal form, five visible storeys and double eave.

North China larch is a principal timber in the pagoda. A field study identified larch over a century old in the pu-pai fang, horizontal members that carry the bracket assemblies [Y2]. Liang also recorded the masonry platform and the upturned lotus and crescent forms of the finial [Y1]. Placing a larch on the same platform brings the origin of the timber into view. The species and use of the wood are documented. The location of the tree and its passage into architecture belong to the dream.

With Codex, I developed Yingxian Photo Structure, a Blender add-on based on photographs and timber research. It adjusts storey count, brackets between columns, proportions and storey spacing. Hiding roofs and enclosure reveals the bracing in the hidden levels. The editable model uses proportions inferred from photographs and simplifies unseen joints. A later Eevee add-on tested ink lines and paper grain on the same geometry.

Figure 4.8-7. Blender studies from the project. Left, framing with roofs and enclosure hidden. Centre, eaves, brackets and latticework. Right, the Eevee ink and paper study.

## From timber to a dream

My first thought was to show the whole pagoda at once, but that revealed its main form before viewers had time to become curious about it. They could see the tower without noticing its material or what lies beneath the eaves. I moved the complete building towards the end and gave the opening to a shadow, the ground and a tree. The final reveal lets viewers match the tree and the eave shadow to the completed tower.

North China larch gave me a concrete way into the idea. Timber begins as a growing tree before it is worked into columns, beams and bracket sets. I placed a large tree and the later pagoda on the same platform so that this relationship could unfold in view. Spreading branches echo the projecting eaves, allowing viewers to compare the two silhouettes in one place. The disappearing tree and opening tower express my understanding of the material's origin through a dream.

Paper and lotus imagery grew from the pagoda's Buddhist setting. Liao scriptures found inside the tower suggested the sheet of paper, while the lotus form of the finial and the layered timberwork suggested opening petals [Y1, Y3]. Sunlight turns the ink lotus gold before the tower opens with a similar movement. Swallows rise from their shadows, and the moon appears at the end.





## The chosen setting

The courtyard sequence moves from late sunlight into a starry night. A low wall, stone paving, steps and distant hills make a quiet setting, with the site of the tower always easy to locate. The scene begins with the shadow of an eave. An amber sphere rolls along a paving joint. Bird shadows lift from the ground as swallows, leading the camera around to reveal a larch where the pagoda should stand.

A sheet bearing a lotus drifts in front of the viewer, briefly hiding the platform. As it falls, the tree has disappeared and the pagoda opens in rotating tiers. The sphere stops at the steps, leaving time to watch the timberwork unfold and the night settle. The slow approach lets the shadow and the larch appear before the tower fills the view.

Figure 4.8-1. Moonlit pagoda at the end of the sequence. AI concept image.

## 4.8.2 From separate studies to one route

The shadow gives an outline of the building, the sheet of paper brings the view closer, and the lotus suggests how the timberwork might open. Liao printed scriptures found in the pagoda informed the paper motif [Y3]. The plaque ‘A Lotus Opens a Hundred Feet High’ and its reading through the layered brackets also connect petals with eaves [Y4]. The images unfold along one route, gradually introducing the tower.

The sphere follows a paving joint towards the steps, linking the shadow, tree, empty platform and pagoda along one route. The camera first follows from the side, then moves behind the sphere with the birds. While the paper briefly covers the view, the sphere keeps moving towards the platform. That continuous direction connects the views before and after the change. The whole tower appears at the end as evening turns to night.

The paving joint, steps and sphere would remain visible through the camera turn so that viewers can recognise the route. A path in Blender would guide the sphere, with its rotation tied to the distance travelled. A separate projected shadow would introduce the tower before it appears. Bird shadows would meet the swallows at takeoff, and the birds would rise in succession to lead the camera towards the larch. The dream changes a place viewers already recognise.

The larch and pagoda would share an axis and a similar height, aligning the trunk and branches with columns, beams and eaves. Geometry Nodes would distribute smaller branches and needle clusters with slight variations in density, while wind moves mainly the tips. Close views would distinguish fissured bark, grain running along the worked timber, rings at cut ends and cooler grey tiles. The same raking light would make the differences between these surfaces visible.





## 4.8.3 The wordless storyboard

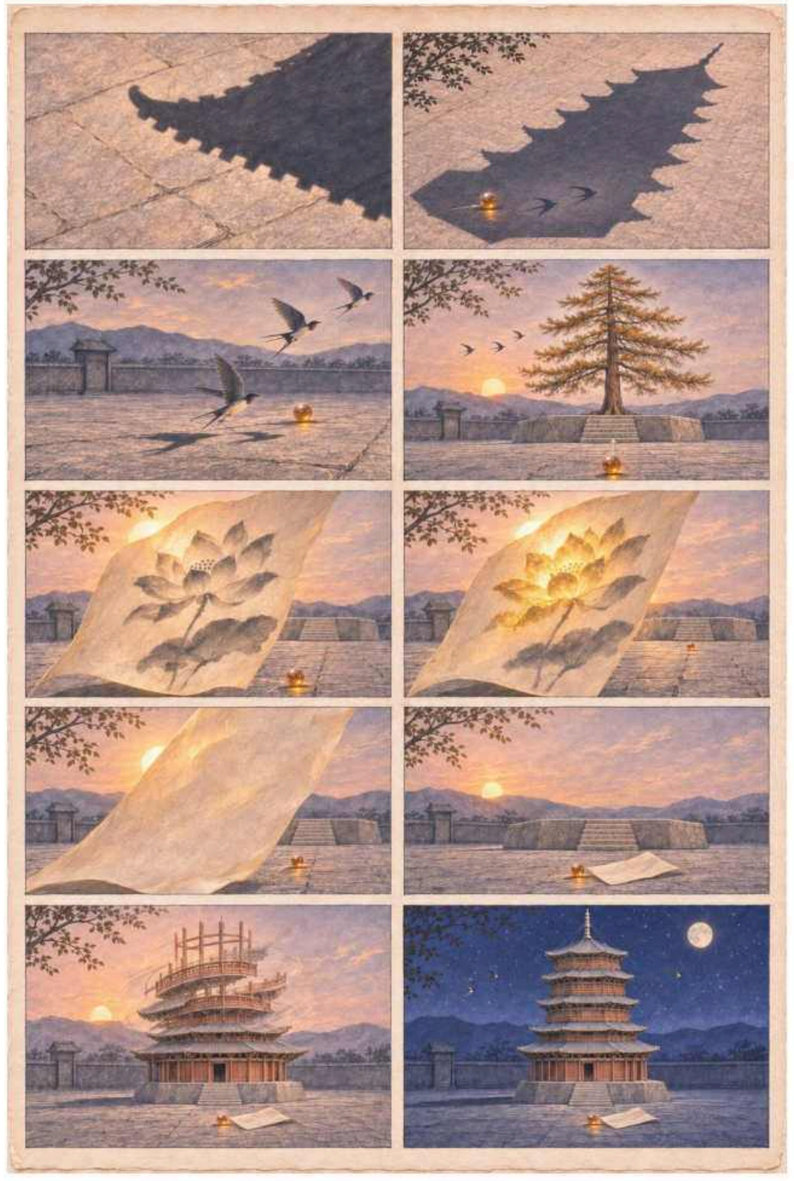


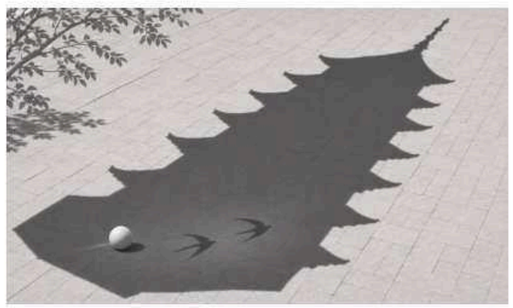
Figure 4.8-4. Final ten-panel wordless storyboard for Yingxian.

Paired views make the changes easier to read. The lotus brightens and fades within closely matched compositions. The steps align the tree, the empty platform and the completed tower. The sphere and the paving seam help the viewer find the route again after the camera turns.

## 4.8.4 Shots 01–04 · Shadow, swallows and larch

01 The shadow of an eave

A fixed shot close to the paving shows a single upturned eave in shadow. The tile edge gives the viewer a fragment of the architecture before the building appears.



02 The sphere enters the shadow

The view widens to show the complete pagoda shadow. The sphere rolls in from the left along a seam, with two bird shadows crossing ahead. Both shadows remain on the paving so their positions are easy to read.

03 Swallows lift from the paving

As the sphere reaches the shadows, the swallows lift from the ground in succession. One is still close to its shadow while the other two are already higher and moving ahead. Their direction leads the turn of the camera. Swallows can be seen circling the real pagoda today [Y5]. Their emergence from the shadows is part of the dream.

04 The turn reveals the larch

The camera comes behind the sphere, with the seam still leading towards the steps. A large North China larch stands where the tower will rise. Its upright trunk and layered branches give an early sense of the tower's height.

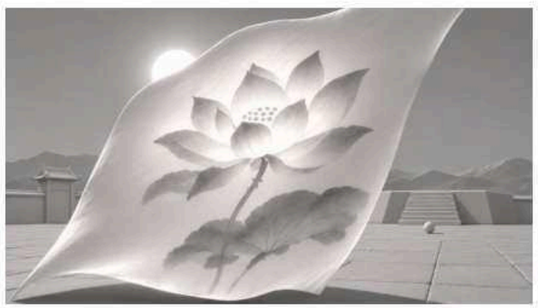




## 4.8.5 Shots 05–07 · Paper, light and the ink lotus

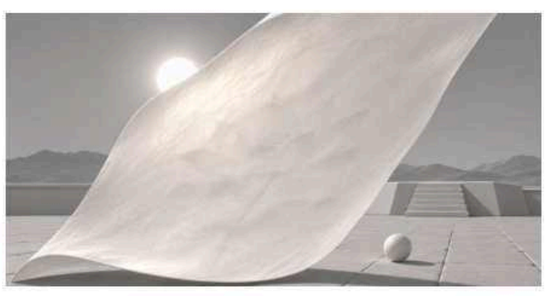
05 The sheet crosses the lens

Wind brings the sheet close to the lens, with the low sun behind it. The lotus is still dark ink and the sphere continues towards the steps below the sheet. The paper fills enough of the frame to conceal the change that follows.



06 Light passes through the lotus

As the sphere rolls beneath the sheet, gold light gradually follows the painted petals. Traces of grey ink keep their edges visible, allowing viewers to follow the flower's opening form as the light moves.



07 The lotus fades

The illuminated flower fades into blank paper as the sheet begins to fall. It still masks the centre of the platform, leaving the disappearance of the larch to the next shot.

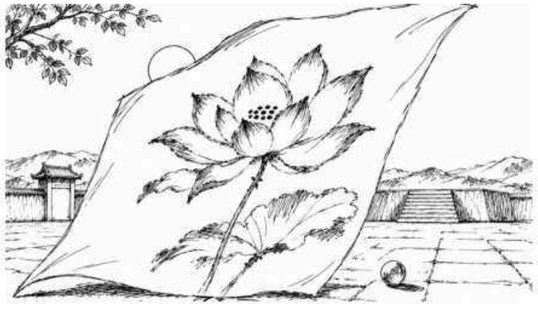
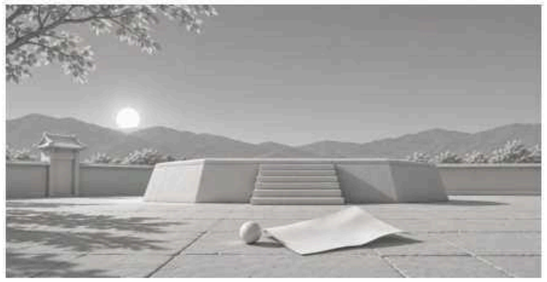


Figure 4.8-5. Shot 05. Black-ink study of the sheet, lotus petals and steps.

Keyframes would bring the sheet to the lens. Cloth and a light wind field would make its edges flutter. Fine fibres would texture the paper, while the ink would transmit less sunlight so the lotus stays legible. As the sphere passes, an animated mask would carry a gold highlight along the strokes before the image fades. The larch would disappear while the paper covers the platform. Once the sheet falls, the camera would hold on the empty base before the tower forms.

## 4.8.6 Shots 08–10 · The tower opens and night falls



08 The sheet lands before the empty platform

The sheet lands and the sphere stops before the first step. A brief hold on the platform lets the viewer register that the tree has disappeared.

09 The timber tiers unfold

Posts and beams rise first. Frames, bracket sets and eaves then open from bottom to top around the same axis, turning into place one tier at a time. Their rhythm follows the lotus petals from the preceding sequence. The octagonal outline and lower double eave follow the author's model [Y1, Y3].

10 The completed pagoda under one moon

Night falls on the completed pagoda in the same place. Swallows return to the eaves, the sphere rests on the paving and a single moon lights the tower. The view holds here, giving viewers time to connect the earlier tree and shadow with the timberwork that has just opened, then look at the eaves and distant hills.

In the animation plan, the existing model's posts, beams, brackets, eaves and roofs would be grouped by level. Keyframes would stagger their arrival from the base upwards. Each group rotates slightly around one axis before settling into place. The eaves move first and the remaining pieces close around them, echoing the opening lotus. The camera and steps stay aligned across the cut. After the tower forms, warm evening light would fade into cooler moonlight. A small warm highlight on the sphere marks where it has stopped.





## 4.8.7 Visual studies and sources

The pagoda model, storyboard and concept images share one platform. Grey-white concept stills make it easier to compare camera position, the scale of the paper, and the silhouettes of tree and tower. The pen drawing develops the lotus lines and their overlap. Successive views of larch, empty platform and pagoda show one place changing. The birds' return to the eaves extends the idea from timber supplied by nature to a building imagined as shelter for life.

The dream borrows the layering of the timber structure for the tower's rotating growth. The real pagoda's long history also involves weathering, sustained loads, wind and earthquakes, damage and successive repairs [Y6]. Chapter 3 discusses these processes separately.

Image credits. Figure 4.8-2 is a render of the author's Blender model. Figures 4.8-1, 4.8-4 and 4.8-5 and the ten clay-style studies were developed from the author's design and references using OpenAI's built-in image generation tool, image_gen. The tool record does not identify the image model version.

[Y1] Liang Sicheng, Survey of the Liao Sakyamuni Pagoda at Fogong Temple

[Y2] Huang et al., 2024. Timber and damage in the pu-pai fang of Yingxian Wooden Pagoda

[Y3] Wang Nan and Zhang Xiaoqin, Yingxian Wooden Pagoda, Tsinghua University

[Y4] On compiling the Yingxian Wooden Pagoda gazetteer, Chinese Social Sciences Net

[Y5] Ge Shuiping, 2025. A firsthand account of swallows at the pagoda, People's Daily

[Y6] Wu Yuhua and Yong Xinqun, Past, Present and Future Deformation of Yingxian Wooden Pagoda

# 4.9 Hanging Temple

The Hanging Temple occupies a natural recess on Cuiping Peak above the Tangyu River in Hunyuan, Shanxi. Its foundation is traced to the Northern Wei period, while its surviving timberwork largely reflects Ming and Qing rebuilding. Section 3.6 shows how the cliff shelters and supports the halls while restricting their position and size. Repairs record a continuing response to rain and rockfall.

In sections and construction diagrams, I noticed where beams enter the rock (Figure 4.9-7). These connections explain how projecting floors are supported and tied to the mountain. As visitors move between the narrow galleries and platforms, they see the cliff and the gorge from different heights.

Figure 4.9-7. Walkway beams where they meet the cliff. See Figure 3.6-7 for the section. Zhangzhugang, 2013. Wikimedia Commons, CC BY-SA 3.0 (cropped).

## 4.9.1 From visual research to the scene

I delayed the full view of the temple to draw attention to how it is supported. At the cliff, the camera approaches a socket in the uneven rock and stays close as the first beam enters. The halls come into view after this connection is established.

The platform and gallery first appear upside down. As the view rotates upright, the gallery stays in frame to help viewers regain their bearings before the camera moves along it.

Hengshan architectural archive · Institute for Planets construction diagrams · Yang et al., 2025, elevation and section

The chosen dream enters a mountain through a reflection in a water drop. Rock, sockets, beams and halls gradually appear. A gallery leads deeper into the building, a water mirror exchanges it with its reflection, and the final wide shot returns it to the gorge. The twelve-panel storyboard develops this route.

Hengshan architectural archive





## 4.9.2 Construction sequence and viewing scale

The structural research in Chapter 3 shows beams embedded in the cliff supporting the projecting platform. The camera begins close to a socket so that viewers can see how the timber enters the rock, then draws back as crossbeams, joists and flooring form a platform. The shot shows how the beams carry the platform and how the halls sit above it.

The storyboard moves between close views of the beam end, a middle distance on the platform and a wide view of the temple. The close view reveals a connection hidden inside the rock. The middle view shows timber becoming a place to walk, and the wide view relates it to the cliff. Grey-white concept stills emphasise volume, joints and occlusion, helping me decide which structures should stay visible as the inverted scene turns upright.

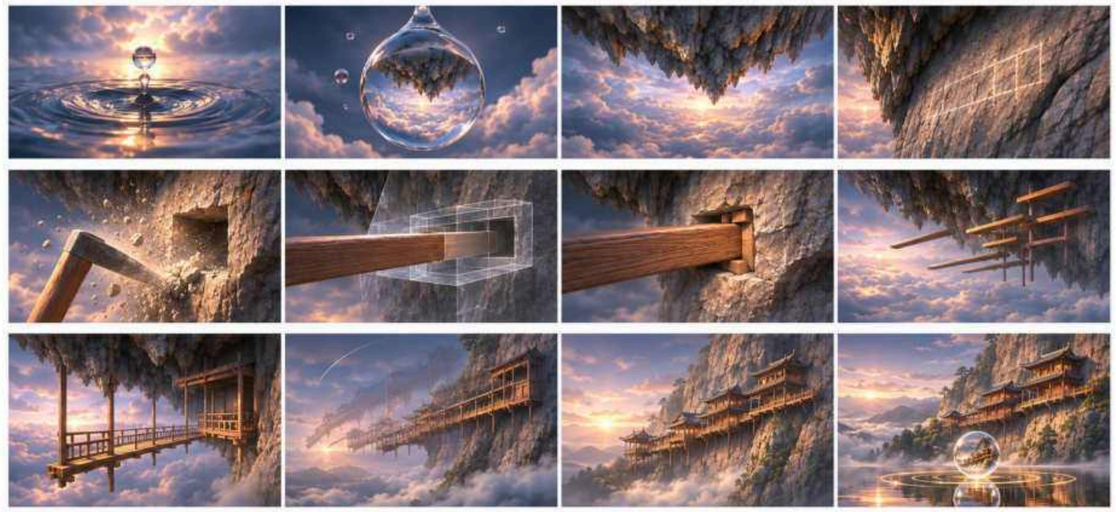


Figure 4.9-1. Final twelve-panel storyboard for the Hanging Temple.

Image credits. The final storyboard and the selected clay-style stills were generated in ChatGPT for concept previsualisation. The source conversation does not expose the image model version.

## 4.9.3 The inverted space and the growing timber frame

A square socket opens in the rock face and stone chips scatter. After the first cantilever enters the rock, more beams project from the same face. Secondary beams and joists connect and the platform begins to form. The cliff remains inverted throughout this sequence. The world starts its slow rotation upright only after the first suspended gallery frame has appeared. The gallery stays visible on either side of the rotation, giving viewers a continuous reference for the cliff, platform and walkway.

Once the scene is upright, posts, tie beams, mortise-and-tenon connections, roof frames and halls gradually appear. Galleries link the neighbouring volumes. These connections make the growth of the structure visible. Beams project from the cliff, the platform extends from them, and the roof frames and halls acquire a place to stand. The first beam and order of appearance are arranged for the viewer. The connections refer to surviving timber work and modern diagrams. The original Northern Wei construction sequence remains open to investigation.

Institute for Planets construction diagrams · Yang et al., 2025, mortise-and-tenon frame

Figure 4.9-2. Shot 01. A drop rises above the ripples.

Figure 4.9-3. Shot 06. Cutaway of the first cantilever entering the cliff.

The direction in which the beam enters the cliff is central to the close view. In Blender, keyframes would guide it along its axis, while a local cutaway reveals the embedded end and wedge. A few rigid-body stone chips and finer dust would draw attention to the socket. Once the platform and gallery form, a slow camera roll would turn the scene upright while keeping the first stretch of gallery visible as a point of reference.





## 4.9.4 Beam-end cutaway and material studies

From outside the cliff, the projecting beam is easy to see, but its embedded end is hidden. The shot uses a local cutaway as the first beam enters the socket. A small area around the opening reveals the internal connection, while the surrounding cliff retains its mass. The cutaway shows the wedge tightening the beam end inside the rock. The socket and wedge draw on a construction diagram. The image adjusts their scale for composition and uses the anchoring to show how the components connect.

The grey-white concept images clarify the outlines of cliff, frame and floor, showing connections through volume and occlusion. The reflection in the drop leads the eye towards the mountain, while a local cutaway reveals the beam end inside the rock. Mist conceals the valley floor and distance, leaving nearby timberwork visible.

Zhang et al., 2026, walkway beam-end details

Figure 4.9-4. Shot 08. Beams and floor framing project from the inverted cliff.

Figure 4.9-5. Shot 09. Framing study for the first suspended gallery.

For the wide shots, the cliff mesh would carry the recesses, strata and larger fractures. Closer in, colour, roughness and bump maps would follow the rock layers, with lighter fresh stone around the socket. Wood grain would run along the beams and posts, with separate end grain and a slight sheen on worn edges and floorboards. In the cutaway, these textures help the viewer distinguish the cliff, the cut surface and the timber inside it. From a distance, broader areas of light and shade keep the halls legible against the rock.

### 4.9.5 Dream logic and the viewing route

Ripples and a rising transparent drop open the sequence. As the camera enters the drop, the reflected clouds and inverted mountain expand into a wide view. The camera then approaches a recess, strata and fissures. This establishes the mountain's scale and direction before drawing attention to where the socket will open. Once the temple appears, the camera moves along the newly formed gallery that the beams support.

Shot ten enters a timber gallery that seems to extend without end. Repeated columns, drifting mist and changing light give it an impossible depth. The camera crosses a water mirror and the temple exchanges places with its reflection, recalling the passage through the drop at the opening. The final view pulls back to show the complete temple in its relationship to the cliff. Clouds and water remain part of the dream. Putting the frames, railings and galleries back into the gorge invites viewers to notice how terrain shaped the architecture.

Figure 4.9-6. Shot 12. The temple against the cliff, framed by water and clouds.





### 4.9.6 Current outcomes and next steps

The storyboard takes viewers from a beam socket into the platform and halls, then back to the whole temple. AI-assisted concept stills establish the framing. The proposed pacing would leave time at the joints and through the rotation so viewers can follow how the timberwork connects to the cliff.

The drop turns a small reflection into a mountain space the viewer can enter. Keyframes would control its rise and slight deformation, while a low-roughness transmission material keeps the reflected mountain recognisable. A Wave modifier could spread concentric ripples with diminishing amplitude. As the camera approaches, a highlight would pass across the view before revealing clouds and an inverted cliff in the same orientation. The changing scale would retain the scene first seen inside the drop.

The fog treatment would use a low layer in the valley, a thinner veil along the cliff and drifting bands at the edge of the gallery. Slow-moving noise in the volume shader would vary the density while keeping the socket and gallery entrance clear. Repeating bays would extend the apparently endless gallery, with mist and changes in light hiding the joins. At the water mirror, the upright and inverted temple would align across one horizontal plane. Ripples would cover the seam as the camera passes through it. The foreground mist would thin in the final pullback, exposing the point where the halls meet the cliff and placing the gallery back within the whole temple.

In an immersive space, viewers may linger at the socket or turn back towards the cliff as the gallery extends. The rotation upright and passage through the water mirror need to allow time for these movements. A viewing trial could show where a slower pace is needed, whether the gallery helps viewers find their bearings, and whether the final wide view lets them locate the timberwork they have just passed within the cliff.

## 4.10 From four animations to Vision Pro

The four animations will come together in one Apple Vision Pro experience. Pompeii rewinds from volcanic cover to the reappearance of vegetation and buildings. In Sicily, rain, wheat and windblown grains bring attention to changes in the stone surface. At Yingxian, a larch gives way to the unfolding timber pagoda. At the Hanging Temple, viewers see how the cliff takes the first beam before moving along the gallery. Seen together, the four scenes trace how the natural setting alters buildings over time and how it shapes where they stand, what they are made of and how they are built.

From the opening screen, viewers can play the four scenes in order or choose a site directly. A short pause between scenes gives a change in wind, water or light time to signal the next place. The sphere, the shadow of the pagoda and the water droplet give viewers a point to look toward before leading their attention to the ground, the structure or the surrounding landscape. Viewers can pause or return to the scene list whenever they want to spend longer in one place.

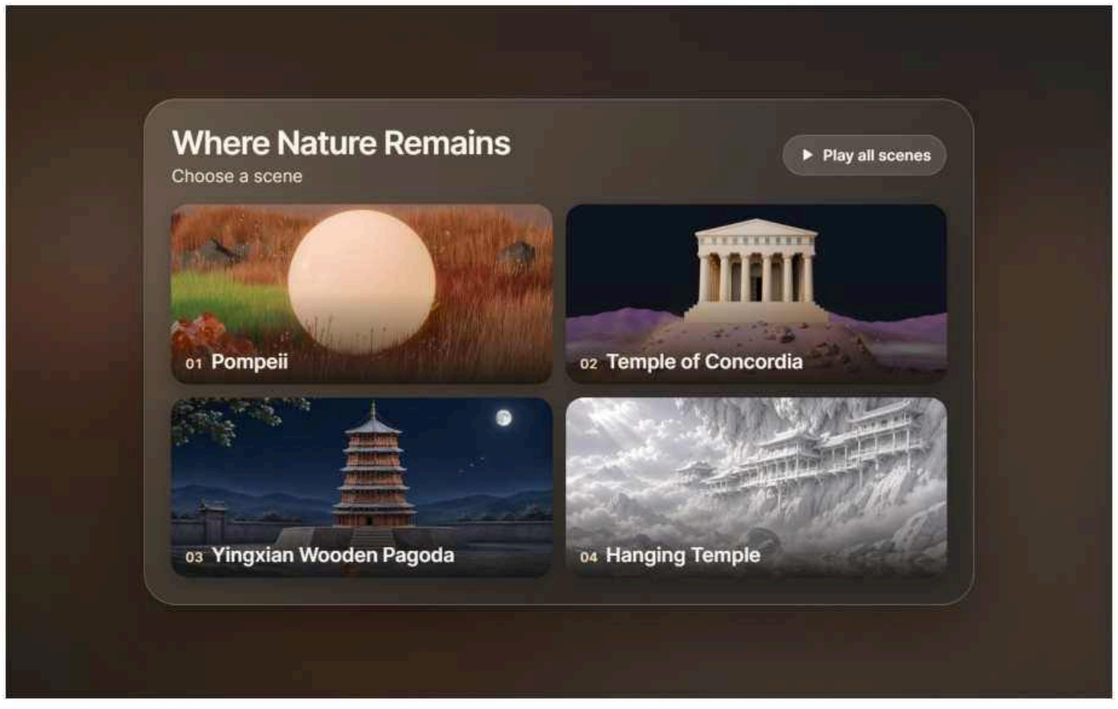



Figure 4.10-1. Vision Pro scene selection. Project frames and concept images.





The Vision Pro proposal combines window playback with spatial viewing. A window preserves the existing compositions and their sequences of material change and reassembly. Spatial viewing suits details understood through distance, such as a beam embedded in the Hanging Temple's cliff and the gallery beyond it. Placing rock, timber and landscape at different depths could make their supporting relationships easier to perceive. Keeping the gallery in view during the rotation would give viewers a reference point as they find their bearings again.

Immersive viewing needs time for people to turn, orient themselves and pause over a detail. When the inverted cliff turns upright or the view crosses the water mirror, someone may still be studying a joint. A rapid transition could interrupt their understanding of how it is supported. The pace therefore needs to respond to how people actually look around. Chapter 5 considers this through visitors' pauses, movement through the exhibition and feedback.

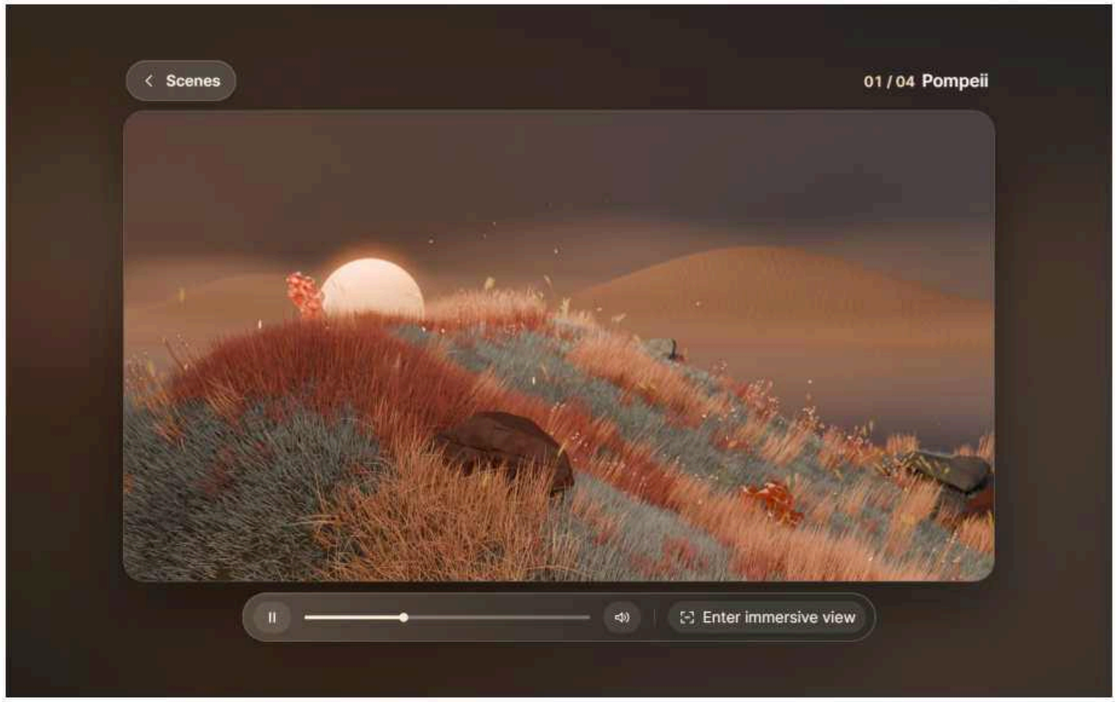



Figure 4.10-2. Vision Pro window playback using a frame from the Pompeii animation.

Apple Developer · Immersive experiences · Apple Developer · Destination Video

## Window playback

The viewer watches the Pompeii film while the room stays visible around the window.

Figure 4.10-3. Window playback at home. AI concept image based on the Pompeii animation.



WHERE NATURE REMAINS / VISION PRO / PRACTICE

## Immersive viewing

In immersive viewing, the grass slope, volcanic stone and distant terrain surround the viewer. The sphere provides a point of orientation in both.

Figure 4.10-4. A girl on the virtual Pompeii slope. AI concept image based on the project.

## References

[28] Potts, Charlotte R. “The Art of Piety and Profit at Pompeii.” Greece & Rome 56, no. 1 (2009): 55–70. DOI: 10.1017/S0017383508000697.

[29] Pompeii in Pictures. IX.7.1–2. Façade paintings including Mercury, Jupiter, Apollo/Sol, Diana/Luna and Venus imagery. Fresco photographs used in the development sheet: Klaus Heese.

[30] Parco Archeologico di Pompei. Pompeii: Guide to the Excavations. French edition, p. 84, “Forum.”

[31] Lost in Time. The Advanced City of Pompeii. YouTube, 2025. Forum segment from 03:43. Contemporary interpretive reconstruction; screenshots collected by the author.

[32] Lombardia Beni Culturali. “Sito archeologico – Pompei – Casa del fauno – Colonne ioniche del peristilio.” Scrocchi collection, c. 1925–1949. Record IMM-LMD40-0000104. Photographer unconfirmed.

[33] Pompeii Commitment / Parco Archeologico di Pompei. “Candelabra.” Image record 2545b; photograph Amedeo Benestante. The related “Tablinum, Triclinium, Convivium” entry identifies the ensemble as 2545.

[34] Musée du Louvre. “Candélabre.” ED 2777. Bronze, 25–75 CE; findspot Pompeii. collections.louvre.fr/en/ark:/53355/cl010258066

[35] Parco Archeologico di Pompei. “Antiquarium, Room I.” Boscoreale. Botanical and everyday-life evidence from the Vesuvian region; fig specimens described are from Herculaneum.

[36] Parco Archeologico di Pompei. “Villa A, also known as Villa Poppaea.” Oplontis. Garden plants and their architectural setting.

[37] Poly Haven. Moon Rock 01; Moon Rock 03; Asset License. CC0 production assets. Their irregular forms support the landscape; they are not archaeological records of Pompeian stone.

[38] Hemingway, Colette. “Architecture in Ancient Greece.” Heilbrunn Timeline of Art History. The Metropolitan Museum of Art, October 2003. metmuseum.org/essays/architecture-in-ancient-greece





## Image credits and process records

Figures 4.1-1, 4.1-5–7: project scene render and design studies. Parts of the architectural arrangement draw on Lost in Time's reconstruction [31].

Figure 4.1-2: screenshots collected from Lost in Time, The Advanced City of Pompeii [31].

Figure 4.1-3: fresco photographs by Klaus Heese / Pompeii in Pictures [29]; GPT-assisted full-figure proposals and white-model references developed for this project.

Figure 4.1-4: Venus reference images collected during research; some photographer and licence details remain unconfirmed. Costume and white-model studies are project design material.

Figure 4.1-8: Scrocchi collection, Lombardia Beni Culturali, IMM-LMD40-0000104 [32].

Figure 4.1-9: Archaeological Park of Pompeii, image record 2545b; Amedeo Benestante / Pompeii Commitment [33]. An open reuse licence has not been established.

Figures 4.1-8a–c: the author’s interior column model, shown separately with studio cameras and lighting.

Figures 4.1-10 and 4.1-12: AI-assisted design reference and retrospective process drawing. Figures 4.2-2, 4.2-4 and 4.3-1 are retrospective AI-assisted drawings based on the recorded production decisions; English editions prepared for this chapter.

Figures 4.1-11 and 4.1-13: the author’s Blender lamp model and interior renders.

Figures 4.2-1 and 4.2-3: the author’s grass and architectural reassembly studies. Figures 4.3-2–4: Blender renders of the interior, olive leaves, figs and olives.

Figures 4.4-1–3: the author’s Blender renders of surface revelation, exterior atmosphere, skylight and gauze.

Source details still to resolve: the horse model's specific archaeological reference, some early Venus photographs and the original external asset page for Rock Dark. No object number or author has been inferred for these items.

# 05

# FEEDBACK, TESTING AND FUTURE STEPS

## 5.1 Proposed exhibition and visitor experience

Figure 5-1. Proposed Pompeii exhibition with panoramic screen, headset area, seating and model table.

The exhibition begins with the Pompeii animation. Visitors first encounter a site changing over time on the panoramic screen. They can then choose the proposed headset area or try putting loose architectural parts back together at the model table. Each option draws on the same scene. Visitors can watch it together, choose where to look, or use their hands to work out how the parts fit.





### 5.1.1 VR headset experience

Section 4.10 proposes a Vision Pro experience for all four sites. This exhibition pilot focuses on Pompeii and sets out how visitors enter the scene and what the first headset trial will examine.

In the proposed headset experience, visitors start where they can see the sphere and the ground around it. The sphere follows the animation route and slows where grass begins to return or fragments start to move. This gives visitors time to turn or pause. They can look down at stone, plants and ground covered by ash, then look up to see where those changes sit in relation to the building.

Looking freely around the scene means visitors may briefly miss the sphere or a change in the building. The proposed short trial will focus on how they find the route again and whether returning grass and moving fragments need more time in view. Their responses will inform the direction cues and pace.

Figure 5-2. Visitors using VR headsets at a technology exhibition. Reference photograph by Nenad Stojković (Shixart1985), Wikimedia Commons, CC BY 2.0, resized for layout.

Image source · CC BY 2.0

## 5.1.2 Panoramic screen experience

The panoramic screen faces a shared viewing area. Visitors can take a seat or stop to watch while waiting or passing through. In the Pompeii film, the sphere first directs attention towards the ground. A wider view then shows returning vegetation and reassembling fragments in the same site.

Visitors may arrive partway through the film. A short cue at the start of each loop could identify Pompeii and the change about to appear. The wider shot also gives people an image to recall at the model table. The plan shows how the screen, seating, proposed headset area and model table relate to one another.

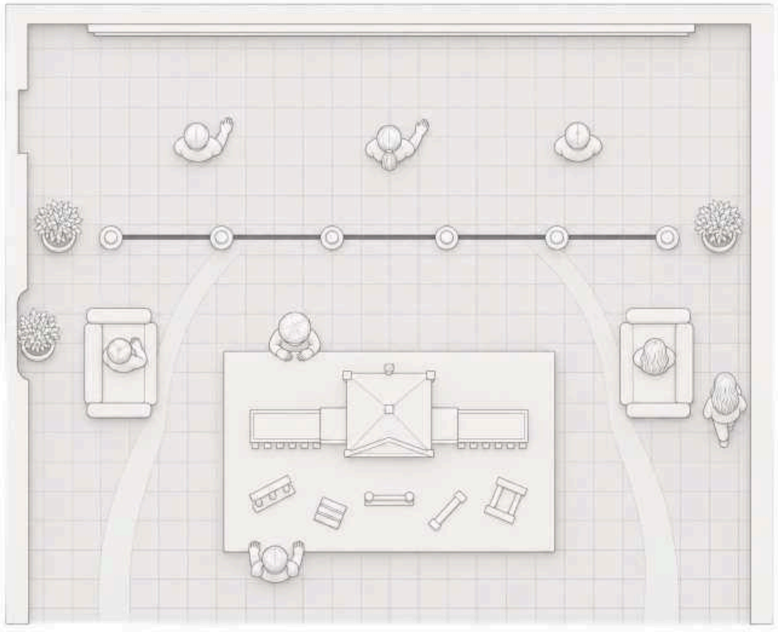


Figure 5-3. Exhibition plan showing the screen, seating, headset area and model table.





## 5.1.3 Exhibition toy and tabletop play

The model table turns the film's reassembly into an activity. Children select a loose column, step or wall section, look at its shape and work out where it belongs. A step can mark an entrance, and several columns placed together reveal a colonnade. After assembling the parts, children can compare the animation still with the model and point out the element they found.

The table can ask two short questions: where did this piece sit, and what happened to the city before it was put back together? The tactile model in Trier offers a reference for reading a building through touch. For the proposed Pompeii table, the size, grip and connections of the loose pieces will be checked with a physical model before a visitor trial.

Figure 5-4. Pompeii fragments (left) and the proposed tabletop model (right). Author's animation still and render.

Figure 5-5. Tactile model of the Aula Palatina in Trier. Reference photograph by Virtual-Pano, Wikimedia Commons, CC BY 4.0, resized for layout.

Image source · CC BY 4.0

## Feedback, testing and future steps

### 5.2 Prototype review and feedback

In the review on 7 November 2026, the tutor found the project easier to understand after seeing the Sicily animation. He asked for a clearer connection between the research, the prototypes and the exhibition, and suggested playing the films separately from the presentation. Production checks included comparing the wheat-field shot before and after lighting the sphere. These observations informed the presentation and visual route. Formal visitor testing, including the headset trial, has yet to take place.

### 5.3 Visitor testing and next steps

Visitor testing will begin with a screening. After each film, people new to the project will describe the changes they saw and how they understood the building's relationship to its surroundings. Where they lose sight of the sphere or miss a transition can help guide changes to pauses, viewing distance and captions.

Exhibition trials will look at how visitors use each format. Can headset viewers find their bearings after turning and follow changes comfortably? Can people joining the panoramic screening partway through recognise the place and follow what is happening? At the table, can children connect the pieces in their hands with the building on screen? Their descriptions, pauses and choices at the table will inform the exhibition and the animation.





# 06

# CONCLUSION

Through these animations, I wanted to give viewers a little more time in places as they once were. It is easy to miss how a beam meets the cliff or how rain and wind change the surface of a stone when you are looking at a building from a distance. I used changes in the ground, the materials and the buildings to let viewers move back through time. As the scenes unfold, they can begin with one detail and gradually get to know the building again, along with the place around it.

When I first thought of Pompeii, I felt a sense of loss at the idea of a whole city buried by a volcano. During my research, the surviving wall paintings and the life they depict let me see how beautiful the city had once been. That changed how I wanted to open the animation. I used red grass and porous volcanic rock for the ground after burial. As the sphere passes, green grass grows along its path and gradually covers the red. I wanted that quiet change to leave viewers some room to imagine the city as it might have been.

At the Temple of Concordia in Sicily, I felt the weight of time. The surfaces of the columns have changed, yet the building still feels imposing. I began close to the fluting and followed rainwater over the stone, through the wheat field and into the windblown grains. The grains gather in a translucent funnel that recalls an hourglass and gives time a visible shape. When the colonnade comes back into view, I hope that having seen the stone up close gives viewers something new to look for in the whole temple.

What drew me to the Yingxian Wooden Pagoda was the intricacy of its timberwork and the symmetry of the tower. Thinking about how long it had stood made me wonder how the builders took trees and made them into a structure that tall, one storey at a time. In the storyboard, a larch appears where the pagoda will stand. When the sheet of paper falls away, the tree has gone and the tower turns as it rises, storey by storey, from the same spot. I wanted that brief change to let viewers think about the wood before they see the tower.

At the Hanging Temple, I wondered how anyone had found a place to build and walk on a cliff so steep. After looking at photographs and sections of the beams where they meet the rock, I wanted to show that easily missed detail in the storyboard. The camera approaches a socket as a beam enters the rock, then pulls back as the platforms and halls take shape. I wanted the completed temple to feel a little like a palace suspended in the mountains. Remembering that first beam, viewers might also sense the judgement and courage it took to build there.





## Personal reflection

At the start, I was thinking about how to make four sites into a set of animations. Over time, I kept coming back to things I had barely noticed at first: the life in a wall painting, a weathered stone surface, the point where a beam meets the cliff. The more I read and the more I adjusted the scenes, the more my idea of each building changed. Now those small details are often what move me most. If I get the chance, I would like to walk through all four sites and see how it feels to stand in the places I got to know through images and shots.

Finally, I would like to thank Professor Luca Fois. This was my first attempt at making a series of animations of this length. Whenever I got caught up in the images and details, he helped me work through my ideas and return to a practical question: what can someone see at this moment, and what can they understand from it? His encouragement helped when I was unsure and made it clearer what I wanted people to notice. This project changed how I look at architecture and nature. It also made me more willing to spend time with things I would once have passed by.

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# APPENDIX A / AI IMAGE CREDITS

### Tool and model record

OpenAI's built-in image generation tool (image_gen), accessed through ChatGPT/Codex, was used for the verified generation and editing calls. The saved tool outputs do not expose the underlying image-model identifier. The exact GPT Image version is therefore recorded as unavailable.

### Verified process illustrations

Figures 4.1-12, 4.2-2, 4.2-4 and 4.3-1 are retrospective process illustrations based on the author's existing project images and design decisions. Generation/editing and English-annotation edits are recorded on 31 October 2026.

### Design references

Figure 4.1-10 is an AI-assisted lamp view study. Figure 4.1-3 includes GPT-assisted figure proposals and white-model references. Their original model and version are not retained in the available records.

### Earlier conceptual sketches

Figures 3.3-1, 3.4-1, 3.5-1 and 3.6-1 are AI-generated concept sketches. Figures 1.3-1 and 3.7-1 were made with image_gen from the author’s brief and visual references. Figure 3.6-12 is a conjectural view of the Northern Wei Hanging Temple generated with image_gen. The exact model versions are unavailable.

### Author's contribution

The author supplied project renders, historical references and design decisions and reviewed the images and labels. These illustrations explain the project; they are not photographs or measured archaeological reconstructions.

### Scope of this note

Historical photographs and actual Blender production renders retain their existing credits. No AI image model is attributed to them. Original project drawings are not relabelled as AI-generated unless the retained records identify them as such.

### Yingxian · Practice studies

Figures 4.8-1, 4.8-4 and 4.8-5 and shots 01–10 are AI-assisted concept and storyboard images based on the author’s design and references. They were generated or edited with OpenAI’s built-in image generation tool (image_gen). The exact image model version is unavailable in the tool record. Figure 4.8-2 is a render of the author’s Blender model.

### Hanging Temple and Vision Pro

Figures 4.9-1–6 are Hanging Temple storyboard and clay studies. Figures 4.10-3 and 4.10-4 are viewing concepts based on the Pompeii animation. They were made with ChatGPT’s built-in image generation tool. The exact model version is unavailable.